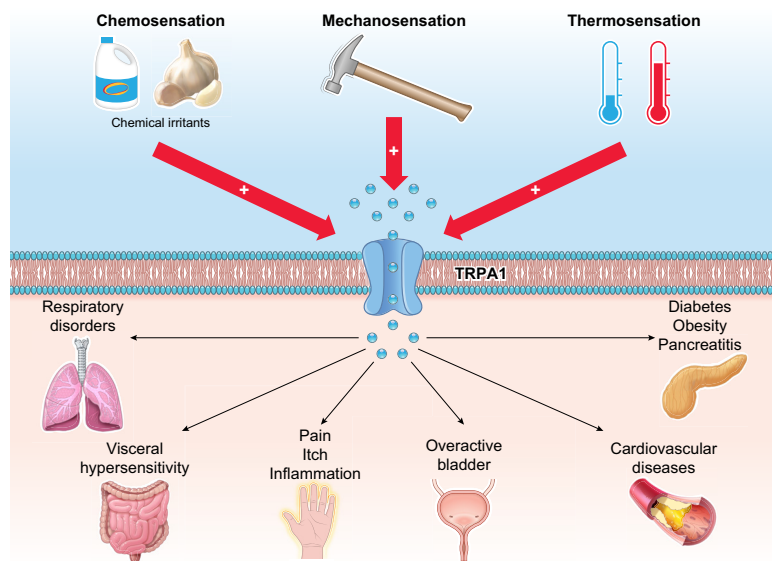


MAMMALIAN TRANSIENT RECEPTOR POTENTIAL TRPA1 CHANNELS: FROM STRUCTURE TO DISEASE

GRAPHICAL ABSTRACT



AUTHORS

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KEYWORDS

Ca²⁺ channel; channelopathies; chemosensation; inflammation; mechanosensation; nociception; pain; thermosensation

CLINICAL HIGHLIGHTS

TRPA1 is a Ca²⁺-permeable cation channel activated by a wide spectrum of noxious external stimuli, such as intense cold, pungent compounds, and reactive chemical species, as well as by endogenous signals associated with cell damage.

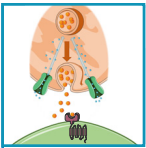
Animal experiments indicate that TRPA1 is expressed in sensory neurons and epithelial cells and is involved in acute and chronic pain and inflammation, playing key roles in the pathophysiology of nearly all organ systems.

Further clinical research is required to evaluate the suitability of TRPA1 as therapeutic target for the treatment of peripheral and visceral hypersensitivity, as well as of pain and inflammatory conditions arising from the exposure to environmental pollutants and tissue injury.

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Talavera K, Startek JB, Alvarez-Collazo J, Boonen B, Alpizar YA, Sanchez A, Naert R, Nilius B. Mammalian Transient Receptor Potential TRPA1 Channels: From Structure to Disease. *Physiol Rev* 100: 725–803, 2020. First published October 31, 2019; doi:10.1152/physrev.00005.2019.—The transient receptor potential ankyrin (TRPA) channels are Ca^{2+} -permeable nonselective cation channels remarkably con-

served through the animal kingdom. Mammals have only one member, TRPA1, which is widely expressed in sensory neurons and in non-neuronal cells (such as epithelial cells and hair cells). TRPA1 owes its name to the presence of 14 ankyrin repeats located in the NH_2 terminus of the channel, an unusual structural feature that may be relevant to its interactions with intracellular components. TRPA1 is primarily involved in the detection of an extremely wide variety of exogenous stimuli that may produce cellular damage. This includes a plethora of electrophilic compounds that interact with nucleophilic amino acid residues in the channel and many other chemically unrelated compounds whose only common feature seems to be their ability to partition in the plasma membrane. TRPA1 has been reported to be activated by cold, heat, and mechanical stimuli, and its function is modulated by multiple factors, including Ca^{2+} , trace metals, pH, and reactive oxygen, nitrogen, and carbonyl species. TRPA1 is involved in acute and chronic pain as well as inflammation, plays key roles in the pathophysiology of nearly all organ systems, and is an attractive target for the treatment of related diseases. Here we review the current knowledge about the mammalian TRPA1 channel, linking its unique structure, widely tuned sensory properties, and complex regulation to its roles in multiple pathophysiological conditions.

Ca^{2+} channel; channelopathies; chemosensation; inflammation; mechanosensation; nociception; pain; thermosensation

I.	INTRODUCTION	725
II.	THE TRPA1 GENE	726
III.	STRUCTURE	726
IV.	TRAFFICKING AND MEMBRANE...	730
V.	BIOPHYSICAL PROPERTIES OF TRPA1	731
VI.	MODULATION, REGULATION, AND...	734
VII.	PATHOPHYSIOLOGY OF TRPA1	760
VIII.	CONCLUSIONS	776

I. INTRODUCTION

The superfamily of transient receptor potential (TRP) cation channels is composed by unique proteins that are expressed in almost every cell type and that play key roles in diverse homeostatic functions. According to their amino acid sequence homology, TRP channels are divided into seven subfamilies: TRPC (Canonical), TRPV (Vanilloid), TRPM (Melastatin), TRPP (Polycystin), TRPML (Mucolipin), TRPA (Ankyrin), and TRPN (NOMP-C) (226, 253,

TRPA1 is a Ca^{2+} -permeable cation channel activated by a wide spectrum of noxious external stimuli, such as intense cold, pungent compounds, and reactive chemical species, as well as by endogenous signals associated with cell damage. Animal experiments indicate that TRPA1 is expressed in sensory neurons and epithelial cells and is involved in acute and chronic pain and inflammation, playing key roles in the pathophysiology of nearly all organ systems. Further clinical research is required to evaluate the suitability of TRPA1 as therapeutic target for the treatment of peripheral and visceral hypersensitivity, as well as of pain and inflammatory conditions arising from the exposure to environmental pollutants and tissue injury.

595). One of the members of this family, TRPA1, is of special interest for being a sensor of a wide variety of noxious external stimuli such as intense cold, pungent compounds, reactive chemical species and of endogenous signals associated to cell damage. This functional diversity and its expression in nociceptive nerve fibers, epithelia, and a

wide variety of other cells implicate this channel in multiple diseases and make it an attractive therapeutic target.

Here we provide a comprehensive review of our current knowledge on this ion channel. Given the increasing volume of publications in the field, currently more than 200 papers per year, we here restricted our focus to mammalian TRPA1 channels, but made pertinent allusions to the extremely illuminating literature on many other species, including *Caenorhabditis elegans*, *Drosophila*, zebrafish, etc. We may also warn the reader about the high rate of reports on newly-described TRPA1 natural and synthetic agonists and antagonists, which makes every revision on the field relatively obsolete after a few years. Finally, we would like to acknowledge the contribution of recent reviews on TRPA1, which are, because of their specialization in specific fields or because of their completeness on specific aspects of TRPA1 properties, excellent complements to this work (60, 80, 90, 116, 210, 321, 351, 360, 410, 487, 547, 550, 557, 586, 618, 687, 691, 704, 726, 741, 746, 774, 805, 855, 899, 957, 959).

II. THE TRPA1 GENE

In humans, the *Trpa1* gene is located in chromosome 8, band q21.11, and comprises 73,635 bases and 29 exons. Homologous genes have been cloned or identified in several mammal species, including 10 non-human primates, rodents, dogs, cattle, pigs, etc. (TABLE 1). On the basis of structure and function, the TRPA family has only one member in mammals. The *Trpa1* gene has also been described and cloned from birds, fishes, amphibians, insects, and nematodes (TABLE 2). Unlike mammals, some of these contain more than one gene homolog, such as *Drosophila melanogaster* (fruit fly, 4 homologues) and *Danio rerio* (zebrafish, 2 homologues).

III. STRUCTURE

The *trpa1* gene encodes a large protein, consisting of ~1100 amino acids (aa) (e.g., 1119 aa in human, 1125 aa in rat, 1115 aa in mouse, 1120 aa in zebrafish, 1197 aa in fruit fly, 1193 aa in *C. elegans*), with an estimated molecular mass between 120 and 130 kDa.

The functional channel protein assembles in homotetramers, through “domain-swap” interactions (178, 651). TRPA1 contains a transmembrane core conserved among the members of the TRP family, consisting of six transmembrane α -helices (TM1–6) with a reentrant pore loop between TM5 and TM6. These two TM domains converge and form the central cavity of the channel, with two gates or restriction points. The upper gate involves diagonal interactions of opposed D915 residues, which have a functional role in Ca^{2+} permeation. The lower gate consists of two hydrophobic seals, formed by residues I957 and V961, that constrain the permeation of rehydrated cations by narrow-

ing the funnel to ~6 Å. At the mouth of the channel, TRPA1 has two pore helices, where the negative charges of the second helix may act as the first gatekeeper by exclusively repelling anions from the gate entrance (651).

TRPA1 has distinctively large intracellular NH_2 and COOH termini, which together account for ~80% of its molecular mass (651). The long NH_2 terminus contains between 14 and 18 ankyrin repeat domains (ARDs), each consisting of 33 amino acids (349, 358, 596, 935). The recent high-resolution three-dimensional (3D) reconstruction of TRPA1 (at ~4 Å resolution) also provided a molecular scaffold to understand channel function (FIGURE 1). It revealed unexpected structural features, such as a TRP-like domain in the COOH terminus, directly after the TM6. Spatially, this α -helical TRP-like domain is in close apposition to the pre-TM1 helix, the linker region in the NH_2 terminus, and other noncontiguous domains such as the TM4-TM5 linker, serving as a node for allosteric regulation of the channel (651). TRP-domain helices, which act as a structural nexus between the channel gates and other domains, may be a feature conserved across the entire TRP family and, possibly, other allosterically-gated channels.

Another distinctive feature of TRPA1 is the presence of a tetrameric coiled coil in the center of the channel, beneath the permeation pore. This stalk-like structure is stabilized by the interaction of positively charged residues in the exterior surface of the coiled coil with inositol polyphosphates (126, 651). These interactions are essential for TRPA1 channel activity (388, 596, 651), suggesting that polyphosphate unbinding could act as a molecular kill-switch responsible for TRPA1 inactivation.

Human TRPA1 features 16 ARDs, spanning for at least 150 Å. Of these, the distal 11 ARDs adopt a propeller arrangement that facilitate side-chain interactions with the COOH -terminal coiled-coil region, stabilizing the ARD proximal to the plasma membrane and possibly contributing to channel assembly (651). Indeed, ARD deletion results in a nonfunctional channel protein with disturbed trafficking to the plasma membrane (596). In addition, the high-resolution 3D reconstruction of human TRPA1 revealed steric interactions between ARDs 12–16 and the helix-turn-helix moiety of the linker region, and the regulatory TRP-like domain in the COOH terminus. This web of interactions possibly accounts for the regulatory effect of ARD-embedded chemical- and thermal-sensitive regions on the gating properties of the channel (651).

Notably, the *Trpa1* gene from mouse, but not from human and rat, has two splice variants (949). The shorter variant (named TRPA1b) lacks exon 20, which encodes for 30 amino acids (from 777 to 807 in longer variant, TRPA1a) that span from the second transmembrane domain and the first intracellular loop. Mouse sensory neurons express both gene

Table 1. Mammalian TRPA1

Common Name	Binomial Name	Gene ID	Chromosome (Map Location)	GNAv	Exon Count
Human	<i>Homo sapiens</i>	8989	8 (q21.11)	NC_000008.11	29
Common chimpanzee	<i>Pan troglodytes</i>	464230	8	NC_036887.1	27
Sumatran orangutan	<i>Pongo abelii</i>	100460642	8	NC_036911.1	27
Western gorilla	<i>Gorilla gorilla</i>	101150478	8	NC_018432.2	27
Crab-eating macaque	<i>Macaca fascicularis</i>	102139057	8	NC_022279.1	28
Rhesus macaque	<i>Macaca mulatta</i>	694623	8	NC_027900.1	27
Northern white-cheeked gibbon	<i>Nomascus leucogenys</i>	100607287	16	NC_019831.1	27
Common marmoset	<i>Callithrix jacchus</i>	100414472	16	NC_013911.1	27
Bonobo	<i>Pan paniscus</i>	100973158	8	NC_027876.1	27
Olive baboon	<i>Papio anubis</i>	101016452	8	NC_018159.2	27
Gelada baboon	<i>Theropithecus gelada</i>	112630413	8	NC_037676.1	27
House mouse	<i>Mus musculus</i>	277328	1 (A3)	NC_000067.6	27
Gairdner's shrewmouse	<i>Mus pahari</i>	110338816	22	NC_034611.1	27
Ryukyu mouse	<i>Mus caroli</i>	110298275	1	NC_034570.1	27
Common rat	<i>Rattus norvegicus</i>	312896	5 (q11)	NC_005104.4	27
Prairie vole	<i>Microtus ochrogaster</i>	101984403	LG5	NC_022031.1	27
Domestic dog	<i>Canis lupus familiaris</i>	486994	29	NC_006611.3	27
Domestic cat	<i>Felis catus</i>	101080611	F2	NC_018740.3	27
Domestic goat	<i>Capra hircus</i>	102170065	14	NC_030821.1	27
Domestic sheep	<i>Ovis aries</i>	101115717	9	NC_019466.2	29
Cattle	<i>Bos taurus</i>	505317	14	NC_037341.1	27
Horse	<i>Equus caballus</i>	100061564	9	NC_009152.3	27
Przewalski's horse	<i>Equus przewalskii</i>	103548063	Un	NW_007673276.1	26
European rabbit	<i>Oryctolagus cuniculus</i>	100341337	3	NC_013671.1	27
Wild boar	<i>Sus scrofa</i>	100152934	4	NC_010446.5	29
Water buffalo	<i>Bubalus bubalis</i>	102397027	15	NC_037559.1	27
Tibetan antelope	<i>Pantholops hodgsonii</i>	102315761	Un	NW_005812652.1	28
Polar bear	<i>Ursus maritimus</i>	103681282	Un	NW_007927247.1	28
Weddell seal	<i>Leptonychotes weddellii</i>	102730954	Un	NW_006383700.1	27
Minke whale	<i>Balaenoptera acutorostrata scammoni</i>	103012702	Un	NW_006728019.1	28
Cape golden mole	<i>Chrysochloris asiatica</i>	102826219	Un	NW_006408554.1	29
Aardvark	<i>Orycteropus afer afer</i>	103202460	Un	NW_006921768.1	27
Cape elephant shrew	<i>Elephantulus edwardii</i>	102862466	Un	NW_006399758.1	27
Gray short-tailed opossum	<i>Monodelphis domestica</i>	100028386	3	NC_008803.1	29
Tasmanian devil	<i>Sarcophilus harrisii</i>	100918272	2	N/A	Unk
Sunda flying lemur	<i>Galeopterus variegatus</i>	103585496	Un	NW_007726355.1	27
Big brown bat	<i>Eptesicus fuscus</i>	103293988	Un	NW_007370710.1	27

GNAv, genomic nucleotide accession version.

variants, but only TRPA1a was found to be a functional channel according to the responses to allyl isothiocyanate (AITC), 2-aminoethoxydiphenyl borate (2-APB), thymol, or carvacrol. The expression of TRPA1b enhances the level of TRPA1a at the plasma membrane and therefore the efficacy of chemical agonists. Interestingly, the expression of *Trpa1b* mRNAs correlates with the role of TRPA1 in the late phase of mechanical hyperalgesia induced by the complete Freund's adjuvant or by partial sciatic nerve ligation. These findings suggest that mouse TRPA1 may be regulated through alternative splicing in pathological conditions.

The structural analysis of TRPA1 revealed the spatial distribution of critical cysteine residues within the pre-TM1 region. These residues, namely, C621, C641, and C665, are involved in the channel activation by electrophilic compounds, and its location in solvent-accessible regions of the pre-TM1 suggests that covalent modifications in the pre-TM1 may provide the driving force for conformational changes (178), involving the neighboring subunits (i.e., TRP-like domain) and regulating the gating properties of the channel. In addition, the electron microscopy structure predicts that several NH₂-terminal cysteines form a binding pocket allowing disulfide bonding be-

Table 2. *TRPA1* in nonmammalian species

Common Name	Binomial Name	Gene ID	Chromosome (Map Location)	GNAv	Exon Count
<i>Birds</i>					
Red junglefowl	<i>Gallus gallus</i>	420180	2	NC_006089.5	27
Eurasian blue tit	<i>Cyanistes caeruleus</i>	111924651	2	N/A	
Japanese quail	<i>Coturnix japonica</i>	107310278	2	NC_029517.1	27
Great tit	<i>Parus major</i>	107214741	2	NC_031769.1	27
Anna's hummingbird	<i>Calypte anna</i>	103527146	Un	NW_007619513.1	27
Collared flycatcher	<i>Ficedula albicollis</i>	101813018	2	NC_021673.1	28
Zebra finch	<i>Taeniopygia guttata</i>	100221097	2	NC_011465.1	27
Domesticated turkey	<i>Meleagris gallopavo</i>	100545876	3	NC_015013.2	14
Helmeted guineafowl	<i>Numida meleagris</i>	110394756	2	NC_034410.1	29
<i>Reptiles</i>					
Green anole	<i>Anolis carolinensis</i>	100556580	4	NC_014779.1	30
Green sea turtle	<i>Chelonia mydas</i>	102944221	Un	NW_006642402.1	29
<i>Amphibians</i>					
Western clawed frog	<i>Xenopus tropicalis</i>	100158526	6	NC_030682.1	27
African clawed frog	<i>Xenopus laevis</i>	108695342	6S	NC_030735.1	27
<i>Fish</i>					
Japanese rice fish	<i>Oryzias latipes</i>	101174541	20	NC_019878.2	28
Northern pike	<i>Esox lucius</i>	105019660	LG21	NC_025988.3	28
Turquoise killifish	<i>Nothobranchius furzeri</i>	107382917	sgr08	NC_029656.1	27
Mexican tetra	<i>Astyanax mexicanus</i>	103042231	3	NC_035899.1	28
Goldfish	<i>Carassius auratus</i>	113042317	24	NC_039266.1	32
Eastern happy	<i>Astatotilapia calliptera</i>	113028962	9	NC_039310.1	31
Atlantic salmon	<i>Salmo salar</i>	106579725	ssa19	NC_027318.1	26
Guppy	<i>Poecilia reticulata</i>	103456670	LG20	NC_024350.1	29
Tongue sole	<i>Cynoglossus semilaevis</i>	103377016	3	NC_024309.1	31
Australian ghostshark	<i>Callorhynchus milii</i>	103174784	Un	NW_006890060.1	27
Spotted gar	<i>Lepisosteus oculatus</i>	102688457	LG9	NC_023187.1	30
Southern platyfish	<i>Xiphophorus maculatus</i>	102223701	21	NC_036463.1	29
Japanese puffer	<i>Takifugu rubripes</i>	101075823	10	NC_018899.1	28
Nile tilapia	<i>Oreochromis niloticus</i>	100701720	LG9	NC_031974.2	30
Zebrafish	<i>Danio rerio</i>	474351	2	NC_007113.7	28
<i>Insects</i>					
Red flour beetle	<i>Tribolium castaneum</i>	658860	LG3	NC_007418.3	19
Common fruit fly	<i>Drosophila melanogaster</i>	39015	3L(3-27cM)	NT_037436.4	19
<i>Nematode</i>					
Round worm	<i>Caenorhabditis elegans</i>	178118	IV	NC_003282.8	11

GNAv, genomic nucleotide accession version.

tween the cysteine residues. Four disulfide bonds were detected in the TRPA1 structure: C666-C622, C666-C463, C622-C609, and C666-C193. In addition to the conformational changes, the activation mechanism of TRPA1 may also involve disulfide bonding between critical cysteine residues (867). Interestingly, as few as two subunits containing intact cinnamaldehyde binding sites at position C622 are sufficient for channel activation (926).

TRPA1 can be activated by increase in intracellular Ca^{2+} concentration ($[\text{Ca}^{2+}]_i$), through direct interactions of

Ca^{2+} with EF hand motifs present in the NH_2 -terminal domain (213). Point mutations of negatively charged residues (D466 and D477) located between ARD11 and ARD12 (213, 958) abolish Ca^{2+} -dependent activation (958). Deletions in this stretch, however, impair trafficking of the truncated channel to the plasma membrane (596), raising the possibility that these point mutations might actually hinder proper protein expression. Another putative Ca^{2+} -binding domain is formed by residues E1077, D1080, D1081, and D1082 in the distal COOH-terminal region, which have strong effects on the Ca^{2+} - and voltage-depen-

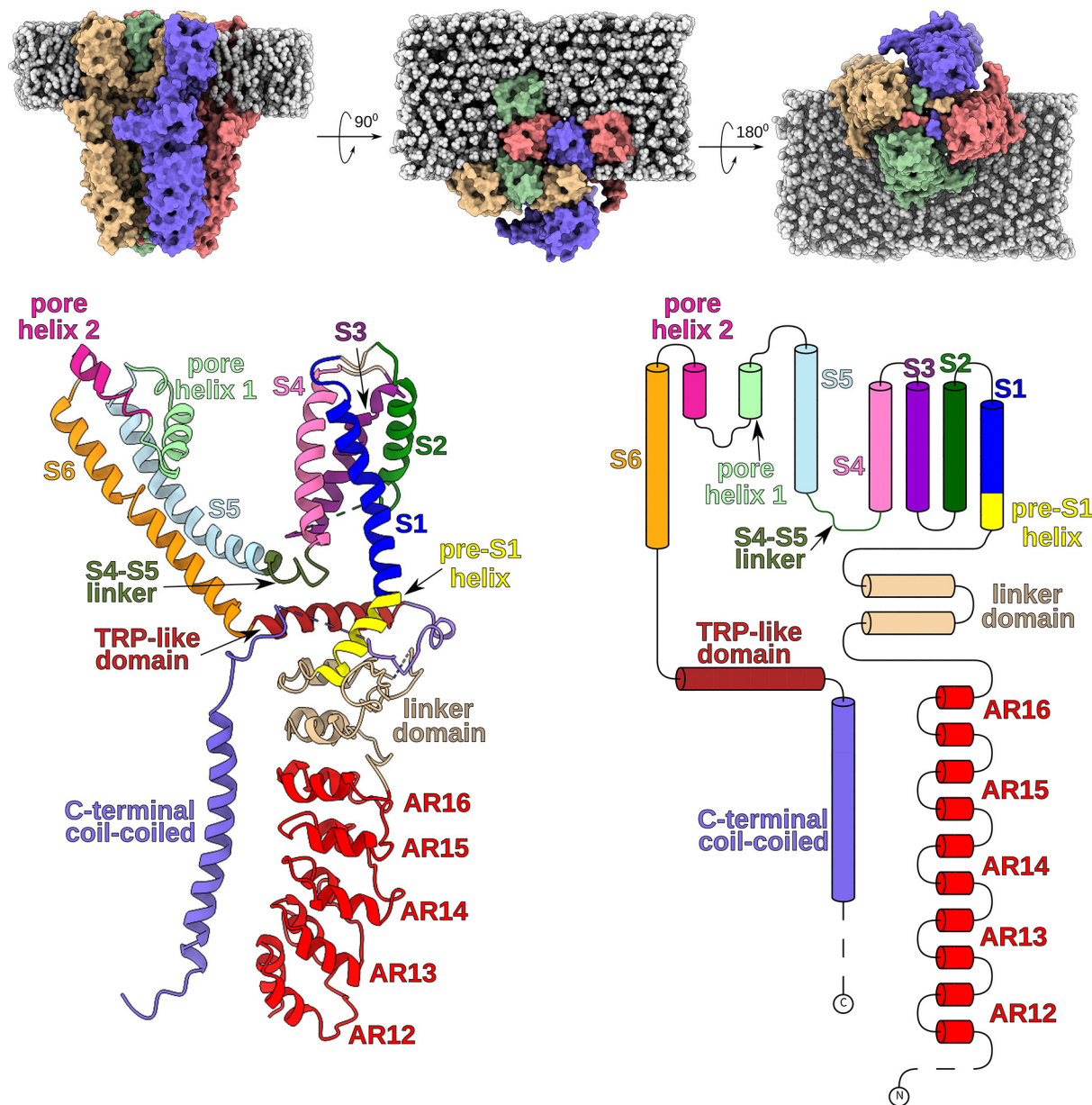


FIGURE 1. Structure of human TRPA1 protein. *Top panels:* side, extracellular, and intracellular views of a three-dimensional density map of TRPA1 treated with allyl isothiocyanate, resolved at 3.5 Å [651]. Each monomer is represented with a different color. *Bottom left:* ribbon diagram of a hTRPA1 monomer. *Bottom right:* linear diagram depicting the major structural domains of TRPA1 (color-coded to match ribbon diagram). [Courtesy of Dr. Ariel Talavera.]

dent potentiation and/or inactivation of agonist-induced responses. Interestingly, this cluster of conserved acidic residues shows partial homology with the Ca^{2+} binding pocket present in large-conductance Ca^{2+} -activated K^+ (BK_{Ca}) channels and may constitute the long-sought Ca^{2+} -sensing domain of TRPA1 (787).

Despite the clear advances made so far, much remains to be done about structural analyses of TRPA1. Future studies should be aimed at the elucidation of the entire protein structure and at the determination of the conformational changes occurring upon stimulation with chemical ago-

nists. This will serve, for instance, to test the hypothesis that nonelectrophilic agonists may act, not by direct binding, but by inducing mechanical alterations in the plasma membrane (see below). Also important will be the identification of the binding sites of channel blockers known so far, which may be instrumental in the rational design of more potent and specific compounds of potential use in TRPA1 therapeutic targeting. Structural analyses may be also useful to unveil the mechanisms underlying several interesting features of this channel that are reviewed below, such as voltage-dependent gating, modulation by thermal stimuli, Ca^{2+} -dependent activation and inactivation, and pore dila-

tion. Importantly, these studies should consider the influence of the lipid environment and the interaction with other proteins on TRPA1 structure and function.

IV. TRAFFICKING AND MEMBRANE EXPRESSION

The tumor suppressor protein CYCL interacts with TRPA1 and deubiquitinates the channel, thereby increasing protein levels in the cell. Oncogenic mutations of CYCL can therefore affect TRPA1 expression. Agonists induce ubiquitination of the channel and, following ligand binding, the interaction between TRPA1 and CYCL renders the channel susceptible to ubiquitination that may terminate its activation (773). Thus association of TRPA1 with the tumor-suppressor protein CYLD may become important in oncogenic mutations in the CYLD gene, which alter cellular levels of TRPA1 (773).

The nocifensive behavior mediated by TRPA1 can be enhanced via protein kinase A/phospholipase C (PKA/PLC) signaling and by activating the channel with the ligand AITC (also known as mustard oil). Both stimuli increased TRPA1 membrane levels *in vitro*. The tetanus toxin reduced the response to the second of two stimuli with AITC in neurons, indicating that vesicle fusion increases the functional expression of TRPA1 in the plasma membrane. Furthermore, capacitance recordings suggest that AITC can induce exocytosis. TRPA1 translocation to the membrane may therefore represent one of the mechanisms controlling TRPA1 functionality upon acute activation or the presence of inflammatory signals (717). Remarkably, the activation of TRPA1 by carvacrol did not increase TRPA1 trafficking. Although the experimental conditions were not identical, this suggests that the electrophilic nature of the agonist determines whether TRPA1 trafficking is induced (515).

The proinflammatory cytokine tumor necrosis factor- α (TNF- α) elevated the plasma membrane content of TRPA1, TRPV1, and the calcitonin gene-related peptide (CGRP). This is mediated by the vesicle-associated membrane protein 1 (VAMP1; but not 2/3) and is inhibited by botulinum neurotoxin (BoNT)/C1 or /A. Thus these neurotoxins may act not only via the known inhibitory effect on the release of pain transmitters, but also by decreasing the exocytotic delivery of TRPA1 and TRPV1, with the concomitant reduction of hypersensitization during inflammation (520). Similar findings were reported using a modified chimeric BoNT, which only required picomolar concentrations to exert an inhibitory effect on TNF- α -mediated TRPA1 and TRPV1 trafficking. This effect is dependent on the cleavage of the synaptosomal nerve-associated protein 25 (SNAP-25), confirming that TRPA1 and TRPV1 trafficking is mediated by driven by vesicle fusion mediated by soluble N-ethylmaleimide-sensitive factor attachment protein receptors (SNARE). Interestingly, this modified BoTN had no effect

on basal levels of TRPA1 and TRPV1, suggesting its therapeutic potential (606). Along the same line, peripherally applied botulinum toxin type A was shown to reduce TRPA1 expression and central antinociceptive activity in rat model of trigeminal neuralgia (901). In addition to the effect of inflammation, changes of temperature have also been reported to induce TRPA1 trafficking. Both cold (4°C) and warm (49°C) stimuli result in higher TRPA1 expression levels (510). Future studies are required to elucidate whether the effects of inflammation and thermal stimuli on TRPA1 trafficking are mediated by the same pathways.

Despite our extensive knowledge about TRPA1 physiology, the molecular players and mechanisms underlying its trafficking remain largely unknown. Recent evidence suggests that TRPA1 trafficking, following activation, depends at least partially on SNARE-mediated vesicle transport. Since the effect of TRPA1 activation on trafficking is dependent on the localized influx of Ca^{2+} (717), it will be imperative to identify the Ca^{2+} -dependent mediators involved in this process and hopefully identify new potential drug targets. This approach requires a combination of pharmacological inhibitors and fluorescent live cell imaging, whereby the effect of inhibiting specific intracellular Ca^{2+} -dependent pathways on TRPA1 trafficking can be studied directly.

With regard to the expression pattern at the level of the plasma membrane, it was shown that activation of 5'-AMP-activated protein kinase (AMPK), an intracellular energy sensor that monitors and modulates energy expenditure, rapidly decreases membrane-associated TRPA1 and its activity within minutes. Given that high glucose decreases AMPK activity and enhances agonist-evoked TRPA1 currents in dorsal root ganglion (DRG) neurons, this regulation was proposed to play a role in painful diabetic neuropathy (872).

Disruption of lipid rafts by cleaving sphingomyelin (SM) with sphingomyelinase (SMase), cholesterol depletion with methyl β -cyclodextrin (MCD), or ganglioside breakdown with myriocin inhibits TRPA1 responses to AITC and formaldehyde in rat trigeminal neurons (701). Furthermore, a carboxamido-steroid that disrupts lipid rafts reduces TRPA1-mediated responses in Chinese hamster ovary (CHO) cells transfected with the human isoform and in rat sensory neurons (699). More recently, total internal reflection fluorescence microscopy and density gradient centrifugation experiments revealed that TRPA1 localizes preferably into cholesterol-rich domains (763). Depletion of cholesterol with an extracellular MCD treatment decreased the maximal response of TRPA1 channels to AITC and reduced the sensitivity of TRPA1 to chemical stimulation with AITC (a 5-fold increased EC_{50}). The former effect may be explained by a reduced channel expression at the plasma membrane that was evidenced by confocal microscopy imaging of cells expressing mCherry-tagged TRPA1 channels.

The latter effect may result from the impairment of direct cholesterol-TRPA1 interactions that enhance the binding affinity of AITC and/or the conformational changes leading to channel opening after AITC binding. Such cholesterol-TRPA1 interactions may be mediated by cholesterol recognition amino acid consensus (CRAC) motifs in the TM2 and TM4 segments that are implicated in the attenuation of chemical activation of TRPA1 by cholesterol-depleting agents. These findings define the membrane context in which TRPA1 is expressed, which may help understanding how the membrane environment may modulate the responses of TRPA1, not only to chemicals, but also to mechanical and thermal stimuli. Future studies should be aimed, for instance, at testing whether cholesterol-TRPA1 interactions are also important for the activation of this channel by other stimuli [e.g., bacterial lipopolysaccharides (LPS), cold, reactive oxygen species, membrane depolarization, etc.], and at determining if these interactions interfere directly with the binding of chemicals or interfere with the gating machinery. In addition, in a more translational direction, it would be interesting to assess whether cholesterol-reducing therapies using statins interfere with the sensory functions of TRPA1.

V. BIOPHYSICAL PROPERTIES OF TRPA1

A. General Properties of TRPA1 Currents

When evoked by a typical voltage ramp stimulus from very negative to very positive potentials, whole-cell TRPA1 currents display a complex but distinctive rectification pattern. There is a slight inward rectification at very negative potentials, which changes to an outward rectification beyond around 0 mV. This behavior does not result from blocking or modulatory effects of Ca^{2+} or Mg^{2+} (864), but from an underlying voltage dependence of channel gating (491). This voltage dependence is manifested as a deactivation tail current resulting from the abrupt hyperpolarization from 0 mV, a voltage at which there is non-zero open probability. At positive potentials, the outward rectification is due to channel activation by membrane depolarization. The activation curve of TRPA1 is shifted to more negative potentials by stimuli such as intracellular Ca^{2+} (958), cold (375), menthol (371), nicotine (801), and LPS (523), rendering the channel active at physiological negative membrane potentials in sensory neurons. These shifts can be very large [e.g., 150 and 400 mV for the effects of intracellular Ca^{2+} (958) and menthol (371), respectively], consistent with the thermodynamic consequence of the very low apparent gating charge of TRPA1 (0.4–0.8 unitary charges) (375, 597, 787, 802, 958). The negative shift of the activation curve is associated at the single-channel level with an increase in the rate of channel activation and a decrease in the rate of channel deactivation (523, 801). Interestingly, human TRPA1 is sensitized upon repetitive application of agonists,

a phenomenon that is related to a progressive negative shift of the voltage dependence of channel activation (515).

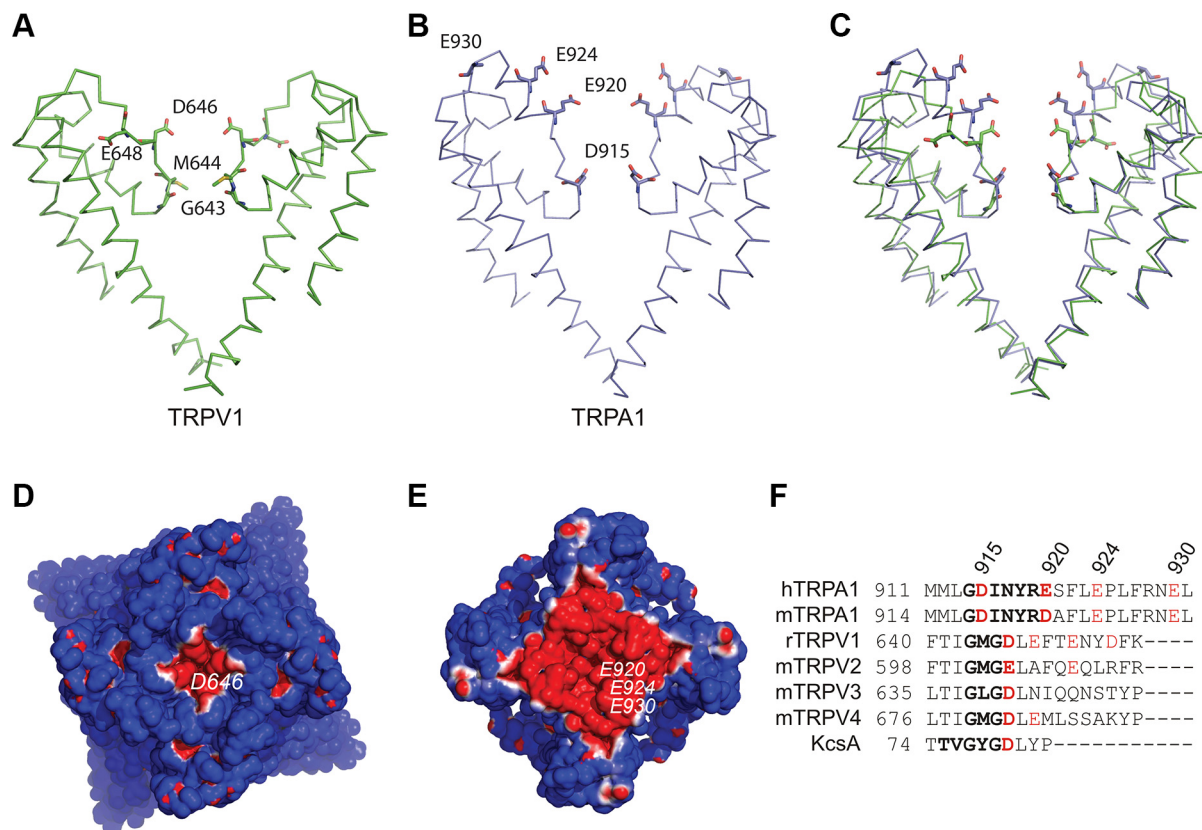
In some conditions, which remain poorly defined, the current decays at very positive potentials, a phenomenon that has been referred to as voltage-dependent inactivation (491, 596, 864). The prominence of this inactivation phase is very variable, and when it is strongly manifested, outward currents are smaller than the inward currents. This phenomenon was reported to be an intrinsic property of the channel and to be mediated by the outer pore helix (864). The voltage dependence was characterized at the single-channel level in cell-attached patches with a voltage for half-maximal inactivation of +34.5 mV and an apparent gating charge of 2.4 unitary charges (slope factor = 10.9) (596).

B. The Pore and Single-Channel Properties

The pore of TRPA1 is formed by the selectivity filter and the S6 transmembrane segments of the four subunits of the channel tetramer. The pore diameter of the nonstimulated channel was estimated at 11 Å, according to the analysis of the relative permeability of cations of different size. Ca^{2+} binding in the pore may hinder monovalent cation permeation and carries ~17% of the mouse TRPA1 inward current. TRPA1 has therefore a relatively high fractional Ca^{2+} current and is the most Ca^{2+} -permeable among TRP channels. The residue D918 determines the Ca^{2+} permeation through the channel (374, 596, 876). The negatively charged residue E920 in human TRPA1 and the corresponding D923 in the mouse isoform are located in the outer side of the pore and may contribute to electrostatic attraction of extracellular cations to the mouth of the pore (FIGURE 2) (161).

Single TRPA1 channel properties depend on the activation mode and experimental conditions (959). TRPA1 has a conductance of ~112 pS at negative and positive potentials in divalent-free solution, but this value is reduced to 55–65 pS at negative potentials in the presence of extracellular Ca^{2+} and Mg^{2+} . Single mouse TRPA1 channels show subconductance states when Ca^{2+} is present in the extracellular solution (564); subconductance states were also reported for human TRPA1 in the absence of Ca^{2+} (315) or in the presence of 0.1 mM Mg^{2+} (161).

Chemical stimulation of TRPA1 induces changes in the selectivity filter that result in progressive but reversible dilation of the channel pore (66). Pore dilatation is characterized by dynamic changes in permeability to *N*-methyl-D-glucamine (NMDG⁺) (152) and an increased divalent cation selectivity and fractional Ca^{2+} current (374). Mutation of the key pore residue D918 prevents AITC-induced increases in Ca^{2+} permeation (374). TRPA1 seems to have therefore at least two open states: restricted and dilated, the latter one allowing the influx of large molecules such as



Yo-Pro (mol wt ~630) and NMDG⁺. Amiloride and its analogue 5-(*N,N*-dimethyl)amiloride (DMA) block more efficiently the dilated state by penetrating deeper into the channel pore (65).

TRPA1 has a relatively high Ca²⁺ selectivity, with a $P_{\text{Ca}}/P_{\text{Na}}$ of ~6 for the constitutive open channel and ~9 for the channel activated by electrophilic agonists. The fractional Ca²⁺ current is ~17% for the constitutively open and 23% for the agonist activated channel (374). Mg²⁺ blocks the open channel but permeates at negative potentials ($P_{\text{Mg}}/P_{\text{Na}} \sim 2$). Ba²⁺ is also able to permeate, with $P_{\text{Ba}}/P_{\text{Na}} \sim 3.5$. The relative inorganic monovalent cation permeabilities follow the sequence Li⁺ > Na⁺ > K⁺ = Rb⁺ > Cs⁺, with values 1.2:1:0.98:0.98:0.95. This suggests a strong field binding cation site in the pore selectivity filter (Eisenman XI). For organic cations the sequence is Na⁺ ~dimethylamine > trimethylamine > tetramethylammonium > NMDG⁺, with values 1:0.99:0.7:0.4:0.1. Electrophilic agonists enhance the permeability to large organic cations due to a pore dilation of 1–3 Å (66, 152, 374).

The single-channel properties of TRPA1 are modulated by coexpression with TRPV1 (767). In CHO cells expressing TRPA1, the ratio of the single-channel conductance values determined at positive and negative potentials was ~1.5, whereas in cells coexpressing TRPA1 and TRPV1, the ratio was ~2.4. Coexpression with TRPV1 also resulted in an increased open probability upon membrane depolarization. These results were recapitulated when comparing single TRPA1 channel properties in trigeminal ganglion (TG) neurons isolated from wild-type and from *Trpv1* knockout (KO) mice. Of note, coexpression of TRPV1 did not change the single-channel conductance of TRPA1 when Ca²⁺ is absent in the extracellular solution. In contrast, intracellular Ca²⁺ does not influence the modulation of TRPA1 properties by coexpression with TRPV1 (767). Although additional research is required to determine the mechanisms underlying these observations, the later indicate that the functional properties of TRPA1 and therefore the pathophysiological roles of this channel are regulated by the coexpression with TRPV1. Key remaining questions regarding this regulation are whether it is mediated by direct in-

teractions of TRPA1 and TRPV1 homotetramers and/or by their heteromerization, and whether it is modulated by the activation state of TRPV1.

C. Structure-Function Relation for Gating

The mechanisms of TRPA1 gating are far from being understood because only very few structure-function data are available. Residues within the TM6 inner pore-forming region are implicated in activation by electrophilic compounds and in voltage-dependent gating of human TRPA1. Substitution of a conserved proline residue (P949) located in the middle of the TM6 by alanine strongly affects the activation and deactivation. Mutation N954A results in a constitutively open channel, suggesting that this residue determines the stability of the closed conformation. Alanine substitutions in a distal bi-glycine motif (GXXXG) reduce the relative Ca^{2+} permeability and affect the activation and deactivation properties. Substitution G958A leads to an increased mean open time, but not to changes in single-channel conductance. On the other hand, the mutant G962A shows short-lived and ill-resolved flickery openings at both positive and negative potentials. These findings highlight the role of the inner pore region in the control of transitions between open and closed states (87).

TRPA1 activity can be modulated by negatively charged ligands such as phosphoinositides and inorganic polyphosphates, most likely through an interaction with as yet unidentified positively charged domain(s). Twenty-seven basic residues all along the COOH-terminal tail of TRPA1 are implicated in activation by electrophilic compounds and voltage. Mutations of proximal COOH-terminal residues K969, R975, K988, and K989 affect channel function. A second significant region was found in a predicted helix, centered on K1048 and K1052. Single alanine mutations in this region completely abolish agonist- and voltage-dependent activation. The charge neutralizations K1092A and R1099A, in the distal portion of the COOH terminus, reduce the sensitivity to electrophilic agonists and increase the voltage-induced steady-state responses. This stretch of basic residues may contain possible interaction sites for negatively charged molecules that are generally considered to modulate TRPA1 (707).

The mutation N855S produces gain-of-function in TRPA1 and is associated with a familial episodic pain syndrome characterized by bouts of severe upper body pain, triggered by physical stress, fasting, or cold (428, 898). It was later found that inverting charge mutation of adjacent residues E854 and K868 results in strong function decrease, whereas charge swapping recovers the channel functionality. It was proposed that these residues form intersubunit salt bridges between adjacent S4-S5 regions that are crucial for stabilizing the conformations associated with chemical- and voltage-dependent activation (954).

Interestingly, mutation of a conserved leucine residue, L906, in the putative pore helix results in a dramatic change in the rectification pattern of mouse TRPA1 (864). Wild-type currents display an outward rectification, whereas the currents carried by the L906C mutant shows inward rectification, independent of divalent cations and irrespective to stimulation by AITC. This phenotype results from the combination of a decrease in the rate of channel deactivation and an enhanced inactivation at positive potentials. The L906C mutant is also resistant to inhibition by HC-030031 and ruthenium red.

Significant understanding of the structure-function relationship of TRPA1 was gained from single-channel recordings of hTRPA1 reconstituted into lipid bilayers (548, 549). Surprisingly, cold, heat, and electrophilic and nonelectrophilic compounds activate this channel with and without its NH_2 -terminal ARD (Delta1–688 hTRPA1). Similar results were found for the inhibitory action of HC-030031. These findings demonstrate that, at least in lipid bilayers, hTRPA1 can function as an intrinsically cold-, heat- and chemosensitive channel (549). The NH_2 -terminal domain is also dispensable for activation of purified *Anopheles gambiae* TRPA1 by heat, AITC, and cinnamaldehyde (788). On the other hand, the integrity of NH_2 -terminal ARD2, ARD6, and ARD11–13 is important for activation by chemical agonists, intracellular Ca^{2+} , and membrane potential (339). Furthermore, other structure-function studies have identified structural elements in the NH_2 terminus that when modified produce severely affected responses to thermal stimuli, including enhanced sensitivity to heat in hTRPA1 (173), distinct thermal sensitivity between splice variants of dTRPA1 (945), abrogation of cold sensitivity in mouse TRPA1 (151), and reversal of the directionality of thermally-induced activation (348). Thus it remains possible that in physiological cellular contexts the thermal and chemical sensitivities of TRPA1 channels are determined by different structural elements and that the NH_2 -terminal ARD is crucial for proper channel functionality.

A recent study proposed the existence of an intracellular water-accessible crevice formed by transmembrane segments 1 to 4. Mutation of polar residues in this region induced complex changes in the sensitivities to voltage, electrophilic agonists, Ca^{2+} , and membrane phosphoinositides (955). These changes are, however, difficult to interpret unambiguously in terms of their relevance to the gating properties of the channel. As was stated at the beginning of this section, the understanding of TRPA1 structure-function relationship is extremely limited, as for instance, the structural bases of voltage-dependent gating and coupling between covalent modification by electrophilic agonists to pore opening remain elusive.

Several other aspects of the structure-function of TRPA1 are discussed in subsequent sections of this review.

VI. MODULATION, REGULATION, AND ANTAGONISM OF TRPA1

TRPA1 is a very attractive therapeutic target for the treatment of pain and inflammation, hence the huge importance of the pharmacology of this channel. However, one has to account often for striking differences between the human isoform and those of rodent species. Several compounds have been identified to have antagonist activity at human TRPA1, but they have large differences in potency as antagonists, no effect or even agonist activity in the rat and mouse isoforms. Thus functional differences have to be taken carefully into account when the modulation of this channel is considered (102). Despite these differences, TRPA1 channels are arguably the most broadly-tuned chemosensory channels known so far (690, 769), being activated by a wide variety of chemical species (TABLE 3).

A. Electrophilic Activators

Due to their reactivity, exogenous and endogenous electrophilic TRPA1 agonists modify the thiol group of cysteine as well as lysine residues in the NH₂ terminus of TRPA1 (317, 491, 793). In the case of human TRPA1, these were reported to be cysteine residues at positions C619, C639, and C663 and to a lesser extent lysine 708, all located in the cytoplasmic NH₂-terminal tail of the channel between the last ankyrin repeat and the first transmembrane segment (317). Mutation of the three cysteines results in weaker responses to electrophilic compounds, but preserves the responsiveness to the nonelectrophilic agonists Δ^9 -tetrahydrocannabinol and 2-APB (317). Mutation of a lysine residue (K708Q) was shown to abrogate the residual response of the triple cysteine human TRPA1 mutant to AITC.

Immunoprecipitated mouse TRPA1 can be covalently modified by electrophilic agonists, and at least 14 other cysteine residues may contribute to the activation of the channel (491). The key cysteine residues of mouse TRPA1 (C415, C422, and C622) are close to or within the ankyrin repeats, illustrating intriguing differences with the human isoform (491). Cysteine residues located in the mouse TRPA1 NH₂ terminus can form disulfide bridges (959) that can be shared (e.g., 666–C622, C666–C463, C666–C193, and C622–C609), suggesting that channel activation may involve different NH₂-terminal conformations (340, 867). Additionally, mass spectrometric analysis of TRPA1 showed that many of these critical cysteines (C193, C415, C463, C622, C634, and C666), along with some other cysteines (C31, C45, C66, C89, C105, C214, C259, C274, C541, C609, and C1087) are modified upon exposure to electrophiles. This suggests that electrophilic binding could lead to disruption of some existing disulfide bonds, as well as to the formation of new such bonds (708). The recent cryo-electron micrograph structure revealed the location of cysteine and lysine residues in the transmembrane core and

facing the lipid environment, which may react with lipophilic electrophiles (651). Currently, it is still unknown what specific cysteine residues, of a total of 28 in human TRPA1, are critical for gating by different electrophiles. Therefore, the mechanism of channel activation after cysteine modification is yet to be fully clarified.

Of note, it was reported that electrophilic compounds can induce hTRPA1 activity in the absence of the NH₂-terminal ANK repeats (549). It was also suggested that the ARD, the pre-S1 helix, the TRP-like domain, and the linker regions of the channel are vital for conformation changes that lead to activation by electrophilic compounds (708).

The class of electrophilic chemical activators of TRPA1 includes, among others, isothiocyanates (the pungent compounds in mustard oil, wasabi, and horseradish) (61, 357) (FIGURE 3), methyl salicylate (in winter green oil) (61), cinnamaldehyde (in cinnamon) (61), allicin and diallyl disulfide (in garlic) (75, 492), acrolein (an irritant in vehicle exhaust fumes and tear gas) (75), chlorobenzylidene malononitrile (44), benzylidenemalononitriles (477), Δ^9 tetrahydrocannabinol (Δ^9 THC, the psychoactive compound in marijuana) (357, 777), chalcones (558), phenylpropanoids (boropinal), 1,4-naphthoquinones (juglone, plumbagin) (314), epoxyeicosatrienoic acids (5,6-EET) (745), and (E)-2-alkenals (109).

Sulfhydryl reacting agents that modify cysteines at an NH₂-terminal site between the last ankyrin repeat and TM1 are necessary for this chemical reactivity (in human C619, C639, C663) (317). These include cinnamaldehyde (CA), super cinnamaldehyde (SC), SC-alkyne (SCA), acrolein, pental, AITC, mustard oil alkyne (MOA), iodoacetamide (IA), IA-alkyne (IAA), and 2-aminoethyl methanethiosulfonate hydrobromide (MTSEA, used for cysteine scanning). The carbon chain length of isothiocyanates is not a main determinant of their ability to activate TRPA1 (815). In mouse TRPA1, out of 31 cysteine residues, the most reactive residues are C415, C422 (between 10th and 11th ARDs), and C622 (between last ARD and TM1). Covalent modification can cause sustained TRPA1 activation (491, 655). Although it is now generally accepted that TRPA1 is activated through covalent modification of specific cysteine residues, the precise mechanism and the chemistry of this covalent modification with unsaturated carbonyl-containing compounds remain unclear. Channel activation occurs with chemicals that could only react with cysteine residues via conjugate addition such as acrylamide, acrylic acid, and cinnamic acid. These compounds react via either conjugate or direct addition, such as acrolein, methyl vinyl ketone, mesityl oxide, acrylic acid *N*-hydroxysuccinimide (NHS) ester, cinnamaldehyde, and cinnamic acid NHS ester. Direct addition occurs via propionic acid NHS ester and hydrocinnamic acid NHS ester. These reaction schemes suggest that TRPA1 is activated preferentially by direct addition of

Table 3. Agonists and bimodal modulators of mammalian TRPA1 channels

Compound Name	EC ₅₀ , μ M	IC ₅₀ , μ M	Isoform, Expression System, Technique	Reference Nos.
Allyl isothiocyanate	64 $\pm$ 3		hTRPA1, oocytes, electrophys. (+80 mV)	317
	11 $\pm$ 1		rTRPA1, oocytes, electrophys. (–60 mV)	357
	22 $\pm$ 3		mTRPA1, CHO, FLIPR	61
		4,100 $\pm$ 800	mTRPA1, CHO, electrophys. (–75 mV)	241
Cinnamaldehyde	61 $\pm$ 9		mTRPA1, CHO, FLIPR	61
	250 $\pm$ 150	3,500 $\pm$ 300	mTRPA1, CHO, Ca ²⁺ imaging	22
	400 $\pm$ 40		hTRPA1, WI-38 fibroblasts, Ca ²⁺ imaging	334
Super cinnamaldehyde	0.8		FLIPR	491
Allicin	1.3		mTRPA1, CHO, FLIPR	492
	51		rTRPA1, CHO, FLIPR	492
	1.9		hTRPA1, CHO, FLIPR	492
	7.5 $\pm$ 0.4		hTRPA1, oocytes, electrophys. (–80 mV)	76
Diallyl disulfide	192 $\pm$ 3		hTRPA1, oocytes, electrophys. (–80 mV)	76
	7.6		hTRPA1, CHO T-Rex, Flex Station II	412
Diallyl sulfide	254		hTRPA1, CHO T-Rex, Flex Station II	412
Diallyl trisulfide	0.49		hTRPA1, CHO T-Rex, Flex Station II	412
Acrolein	5 $\pm$ 1		hTRPA1, oocytes, electrophys. (–60 mV)	75
	85 $\pm$ 9		hTRPA1, WI-38 fibroblasts, Ca ²⁺ imaging	334
	0.8		rTRPA1, HEK293T, Ca ²⁺ imaging	40
2-Chlorobenzylidene malononitrile	0.0009		hTRPA1, HEK293-T-Rex, FDSS	130
	0.214		hTRPA1, HEK293-T-Rex, electrophys. (–30 mV)	130
	0.0007		hTRPA1, HEK293-T-Rex, FLIPR	609
Dibenz[b,f][1,4]oxazepine	0.0003		hTRPA1, HEK293-T-Rex, FDSS	130
	0.063		hTRPA1, HEK293-T-Rex, electrophys. (–30 mV)	130
1-Chloroacetophenone	0.03		hTRPA1, HEK293-T-Rex, FDSS	130
	0.275		hTRPA1, HEK293-T-Rex, electrophys. (–30 mV)	130
Ethyl bromoacetate	0.039		hTRPA1, HEK293-T-Rex, FDSS	130
Bromobenzyl cyanide	0.01		hTRPA1, HEK293-T-Rex, FDSS	130
Camphor		660	rTRPA1, HEK293, electrophys. (–80 mV)	914
	$\leq$ 300		mTRPA1, CHO, Ca ²⁺ imaging	22
Δ 9Tetra-hydrocannabinol	12 $\pm$ 2		rTRPA1, oocytes, electrophys. (–60 mV)	357
	0.23 $\pm$ 0.03		rTRPA1, HEK293T, Ca ²⁺ imaging	192
PF-4840184	0.097 $\pm$ 0.005		rTRPA1, FLIPR	695
	0.023 $\pm$ 0.0006		hTRPA1, FLIPR	695
Plumbagin	0.46 $\pm$ 0.05		hTRPA1, HEK293, Ca ²⁺ imaging	314
Boropinal A	10 $\pm$ 3		hTRPA1, HEK293, Ca ²⁺ imaging	314
Juglone	1.7 $\pm$ 0.5		hTRPA1, HEK293, Ca ²⁺ imaging	314
Nicotine	17	4,000	mTRPA1, HEK293, electrophys. (–75 mV)	801
4-Hydroxyhexenal	40 $\pm$ 12		mTRPA1, CHO, Ca ²⁺ imaging	35
4-Hydroxy-2-nonenal	1.9 $\pm$ 0.7		mTRPA1, CHO, Ca ²⁺ imaging	35
4-Hydroxynonenal	20 $\pm$ 3		mTRPA1, CHO, Ca ²⁺ imaging	35
	13		mTRPA1, CHO, Ca ²⁺ imaging and FLIPR	491
	27		mTRPA1, HEK293T, Ca ²⁺ imaging	829
	9.9 $\pm$ 1.2		hTRPA1, HEK293T, FLIPR	103
	6.6 $\pm$ 1.5		mTRPA1, HEK293T, FLIPR	103
	6.0 $\pm$ 0.8		rTRPA1, HEK293T, FLIPR	103

Continued

Table 3.—Continued

Compound Name	EC ₅₀ , μ M	IC ₅₀ , μ M	Isoform, Expression System, Technique	Reference Nos.
15-Deoxy- Δ 12,14-prostaglandin J2 (15d-PGJ ₂)	5.6 $\pm$ 1.1		mTRPA1, CHO, Ca ²⁺ imaging	35
	40 $\pm$ 16		hTRPA1, HEK293T, FLIPR	103
	60 $\pm$ 20		mTRPA1, HEK293T, FLIPR	103
	5.4 $\pm$ 1.1		rTRPA1, HEK293T, FLIPR	103
Hydrogen peroxide	1200 $\pm$ 400 (at 90 s)		mTRPA1, CHO, Ca ²⁺ imaging	35
	230 (at 600 s)		mTRPA1, CHO, Ca ²⁺ imaging	35
	290 $\pm$ 90		hTRPA1, HEK293T, Ca ²⁺ imaging	97
	297 $\pm$ 9		mTRPA1, HEK293T, Ca ²⁺ imaging	714
Chloramine-T (<i>N</i> -chloro-sodium- <i>p</i> -toluenesulphenamide)	11 $\pm$ 1		hTRPA1, HEK293T, Ca ²⁺ imaging	97
Formaldehyde	357		mTRPA1, CHO, Ca ²⁺ imaging and FLIPR	491
	0.016 $\pm$ 0.001‰		hTRPA1, HEK293T, Ca ²⁺ imaging	513
	0.015 $\pm$ 0.001‰		rTRPA1, HEK293T, Ca ²⁺ imaging	513
Hypochlorite	11 $\pm$ 1 ppm		hTRPA1	97
	7 $\pm$ 1 ppm		mTRPA1, HEK293T, Ca ²⁺ imaging	97
Icilin	>25			211,777
Ozone	3		hTRPA1, HEK293T, Ca ²⁺ imaging	809
Toluene diisocyanate	10,000		hTRPA1, HEK293T, Ca ²⁺ imaging	807
2-Chloro- <i>N</i> -(4-(4-methoxyphenyl)thiazol-2-yl)- <i>N</i> -(3-methoxypropyl)-acetamide (JTO10)	0.00065		mTRPA1, HEK293T, Ca ²⁺ imaging	797
	0.047		hTRPA1, HEK293T, Ca ²⁺ imaging	308
<i>p</i> -Benzoquinone	0.36 $\pm$ 0.02		mTRPA1	32
	0.44 $\pm$ 0.02		hTRPA1, CHO, Ca ²⁺ imaging	32
	3.2 $\pm$ 0.6		mTRPA1, CHO, electrophys. (–60 mV)	32
<i>N</i> -Acetyl- <i>p</i> -benzoquinoneimine	0.9 $\pm$ 0.3		mTRPA1	32
	1.33 $\pm$ 0.04		hTRPA1, CHO, Ca ²⁺ imaging hTRPA1, CHO, Ca ²⁺ imaging	32
Crotalphine	0.046		hTRPA1, mTRPA1, rTRPA1, COS1, Ca ²⁺ imaging	125
Methyl <i>p</i> -hydroxybenzoate	4,400		mTRPA1, HEK293T, electrophys. (–60 mV)	258
3'-Carbamoylbiphenyl-3-yl cyclohexylcarbamate (URB597)	24 $\pm$ 3		hTRPA1	591
	70 $\pm$ 8		rTRPA1, HEK293T, Ca ²⁺ imaging	591
Flufenamic acid	24 $\pm$ 3		hTRPA1, WI-38 fibroblasts, Ca ²⁺ imaging	334
	57 $\pm$ 5		hTRPA1, HEK293T, Ca ²⁺ imaging	334
	44 $\pm$ 11		hTRPA1, WI-38 fibroblasts, electrophys. (+100 mV)	334
	55 $\pm$ 4		hTRPA1, WI-38 fibroblasts, electrophys. (+100 mV)	334
Niflumic acid	28 $\pm$ 3		hTRPA1, WI-38 fibroblasts, Ca ²⁺ imaging	334
Mefenamic acid	61 $\pm$ 5		hTRPA1, WI-38 fibroblasts, Ca ²⁺ imaging	334
Diclofenac	210 $\pm$ 20		hTRPA1, WI-38 fibroblasts, Ca ²⁺ imaging	334
Flurbiprofen	342 $\pm$ 6		hTRPA1, WI-38 fibroblasts, Ca ²⁺ imaging	334
	310 $\pm$ 70		hTRPA1, HEK293T, WI-38 fibroblasts, electrophys. (+100 mV), Ca ²⁺ imaging	334
Indomethacin	470 $\pm$ 50		hTRPA1, WI-38 fibroblasts, Ca ²⁺ imaging	334
Ketoprofen	>500		hTRPA1, WI-38 fibroblasts, Ca ²⁺ imaging	334
Nifedipine	157 $\pm$ 8		hTRPA1, WI-38 fibroblasts, Ca ²⁺ imaging	334
	140 $\pm$ 20		hTRPA1, HEK293T, Ca ²⁺ imaging	334
	0.40 $\pm$ 0.02		mTRPA1, CHO, Ca ²⁺ imaging	245
Nimodipine	0.8 $\pm$ 1.3		mTRPA1, CHO, Ca ²⁺ imaging	245
Nicardipine	0.5 $\pm$ 0.07		mTRPA1, CHO, Ca ²⁺ imaging	245

Continued

Table 3.—Continued

Compound Name	EC ₅₀ , μ M	IC ₅₀ , μ M	Isoform, Expression System, Technique	Reference Nos.
Nitrendipine	3.8 $\pm$ 0.3		mTRPA1, CHO, Ca ²⁺ imaging	245
($\pm$) BAY K 8644	32.7 $\pm$ 0.2		mTRPA1, CHO, Ca ²⁺ imaging	245
Lidocaine	5,700 $\pm$ 200		rTRPA1, HEK293T, electrophys. (–60 mV)	460
	24,000 $\pm$ 600		hTRPA1, HEK293T, electrophys. (–60 mV)	460
5-Nitro-2-(3-phenylpropylamino) benzoic acid (NPPB)	0.32		hTRPA1, HEK293T, FLIPR	481
Propofol	65.4		hTRPA1, HEK293, electrophys. (–60 mV)	599
	2.4	19.5	mTRPA1, CHO, electrophys.	372
	17		mTRPA1, Sf21, Flexstation III	897
Thymol	64		hTRPA1, HEK293T, Ca ²⁺ imaging	458
	127		hTRPA1, HEK293T, electrophys. (–80 mV)	458
	20		hTRPA1, HEK293T, FLIPR	458
	<100	>100	mTRPA1, CHO, electrophys.	371
Menthol	95 $\pm$ 15	56 $\pm$ 8	mTRPA1, CHO, electrophys.	371
		68	mTRPA1, CHO, electrophys. (–60 mV)	493
	278 $\pm$ 30	>1,000	hTRPA1, HEK293T, FLIPR	103
	5.2 $\pm$ 0.7	950 $\pm$ 80	mTRPA1, HEK293T, FLIPR	103
	7.1 $\pm$ 1.1	511 $\pm$ 25	rTRPA1, HEK293T, FLIPR	103
1-Hexanol	7,900 $\pm$ 900		hTRPA1, HEK293T, Ca ²⁺ imaging	416
1-Heptanol	2,700 $\pm$ 400		hTRPA1, HEK293T, Ca ²⁺ imaging	416
1-Octanol	810 $\pm$ 20		hTRPA1, HEK293T, Ca ²⁺ imaging	416
Apomorphine	7.1		hTRPA1, HEK293T, Ca ²⁺ imaging	719
6-(Methylsulfinyl)hexyl isothiocyanate (6-MSITC)	150 $\pm$ 30		mTRPA1, HEK293T, electrophys. (–60 mV)	837
	39 $\pm$ 4		hTRPA1, HEK293T, electrophys. (–60 mV)	837
6-(Methylthio)hexyl isothiocyanate (6-MTITC)	30 $\pm$ 3		mTRPA1, HEK293T, electrophys. (–60 mV)	837
	34 $\pm$ 3		hTRPA1, HEK293T, electrophys. (–60 mV)	837
Cannabinol	0.18 $\pm$ 0.02		rTRPA1, HEK293T, Ca ²⁺ imaging	190
Cannabichromene	0.06 $\pm$ 0.02		rTRPA1, HEK293T, Ca ²⁺ imaging	192
Cannabidiol	0.096 $\pm$ 0.012		rTRPA1, HEK293T, Ca ²⁺ imaging	192
	0.11 $\pm$ 0.05		rTRPA1, HEK293T, Ca ²⁺ imaging	190
Δ 9-Tetrahydrocannabinol acid	0.24 $\pm$ 0.03		rTRPA1, HEK293T, Ca ²⁺ imaging	192
	2.7 $\pm$ 0.9		rTRPA1, HEK293T, Ca ²⁺ imaging	190
Cannabidiol acid	12 $\pm$ 9		rTRPA1, HEK293T, Ca ²⁺ imaging	192
	5.3 $\pm$ 1.5		rTRPA1, HEK293T, Ca ²⁺ imaging	190
Cannabigerol	3.4 $\pm$ 1.0		rTRPA1, HEK293T, Ca ²⁺ imaging	192
	0.7 $\pm$ 0.03		rTRPA1, HEK293T, Ca ²⁺ imaging	190
Cannabigerol acid	8 $\pm$ 4		rTRPA1, HEK293T, Ca ²⁺ imaging	190
Cannabigvarin	1.60 $\pm$ 0.01		rTRPA1, HEK293T, Ca ²⁺ imaging	190
Tetrahydrocannabivarin	1.5 $\pm$ 0.6		rTRPA1, HEK293T, Ca ²⁺ imaging	190
Tetrahydrocannabivarin acid	16 $\pm$ 2		rTRPA1, HEK293T, Ca ²⁺ imaging	190
Anandamide	10 $\pm$ 2		rTRPA1, HEK293T, Ca ²⁺ imaging	191
9-Hydroxyoctadecadienoic	32 $\pm$ 4		rTRPA1, HEK293T, Ca ²⁺ imaging	191
13-Hydroxyoctadecadienoic	13 $\pm$ 2		rTRPA1, HEK293T, Ca ²⁺ imaging	191
Arachidonic acid	13 $\pm$ 4		hTRPA1, HEK293T, Ca ²⁺ imaging	679

Continued

Table 3.—Continued

Compound Name	EC ₅₀ , μ M	IC ₅₀ , μ M	Isoform, Expression System, Technique	Reference Nos.
<i>R</i> -(+)-(2,3-dihydro-5-methyl-3-[(4-morpholinyl)methyl]pyrolo[1,2,3-de]-1,4-benzoxazin-6-yl)-(1-naphthalenyl) methanone mesylate (WIN)	18 20 $\pm$ 6		mTRPA1, CHO, electrophys. (–60 mV) rTRPA1, HEK293T, Ca ²⁺ imaging	11 751
(<i>R,S</i>)-3-(2-iodo-5-nitrobenzoyl)-1-(1-methyl-2-piperidinylmethyl)-1H-indole (AM1241)	48		mTRPA1, CHO, electrophys. (–60 mV)	11
<i>N</i> -(2-chloroethyl)-5Z,8Z,11Z,14Z-eicosatetraenamide (ACEA)	12		mTRPA1, CHO, electrophys. (–60 mV)	11
AM251	0.86 $\pm$ 0.06		rTRPA1, HEK293T, Ca ²⁺ imaging	751
AM630	1.9 $\pm$ 0.2		rTRPA1, HEK293T, Ca ²⁺ imaging	751
Deacylasadisulfide propionate	11.0 $\pm$ 1.4		rTRPA1, HEK293T, Ca ²⁺ imaging	739
Deacylasadisulfide arachidate	11.0 $\pm$ 1.4		rTRPA1, HEK293T, Ca ²⁺ imaging	739
Asadisulfide alcohol	10.9 $\pm$ 0.8		rTRPA1, HEK293T, Ca ²⁺ imaging	739
Fetisulfide A	11 $\pm$ 4		rTRPA1, HEK293T, Ca ²⁺ imaging	739
Isovelleral	0.50 $\pm$ 0.13 2.6 $\pm$ 1.1		hTRPA1 mTRPA1, HEK293T, Ca ²⁺ imaging	240 240
Polygodial	0.40 $\pm$ 0.07 0.059 0.67		hTRPA1, HEK293T, Ca ²⁺ imaging hTRPA1, CHO, Ca ²⁺ imaging mTRPA1, HEK293T, electrophys. (–60 mV)	240 240 240
Miogatral	0.13 0.63		hTRPA1, CHO, Ca ²⁺ imaging mTRPA1, HEK293T, electrophys. (–60 mV)	240 240
Miogadial	0.2 0.4		hTRPA1, CHO, Ca ²⁺ imaging mTRPA1, HEK293T, electrophys. (–60 mV)	240 240
Crotonaldehyde	23		rTRPA1, HEK293T, Ca ²⁺ imaging	40
Hydroxy- α -sanshool	69		hTRPA1, HEK293T, Ca ²⁺ imaging	681
6-Shogaol	11.2 16 $\pm$ 2	16.7 $\pm$ 0.4	hTRPA1, HEK293T, Ca ²⁺ imaging rTRPA1, HEK293T, Ca ²⁺ imaging	681 552
6-Paradol	71		hTRPA1, HEK293T, Ca ²⁺ imaging	681
Linalool	117		hTRPA1, HEK293T, Ca ²⁺ imaging	681
Carvacrol	750 $\pm$ 110 7		WC frog TRPA1, oocytes, electrophys. (–60 mV) hTRPA1, HEK293T, FLIPR	703 458
Eugenol	260		hTRPA1, HEK293T, electrophys. (–60 mV)	164
1'S-1'-acetoxychavicol acetate	0.16		hTRPA1, HEK293T, Ca ²⁺ imaging	576
2-Tert-butyl-5-methylphenol	3		hTRPA1, HEK293T, FLIPR	458
2,6-Dimethylphenol	31		hTRPA1, HEK293T, FLIPR	458
2,5-Dimethylphenol	57		hTRPA1, HEK293T, FLIPR	458
3,4-Dimethylphenol	67		hTRPA1, HEK293T, FLIPR	458
2,6-Diisopropylphenol	4		hTRPA1, HEK293T, FLIPR	458
Caffeine	96 $\pm$ 11 62 $\pm$ 3 1,000–2,500	990 $\pm$ 120	rTRPA1 mTRPA1 hTRPA1, HEK293T, FLIPR mTRPA1, HEK293T, Ca ²⁺ imaging	103 103 566 566
Trinitrophenol	107 $\pm$ 6 30 $\pm$ 5		hTRPA1, HEK293T, FLIPR mTRPA1, HEK293T, FLIPR	103 103

Continued

Table 3.—Continued

Compound Name	EC ₅₀ , μ M	IC ₅₀ , μ M	Isoform, Expression System, Technique	Reference Nos.
Farnesyl thiosalicylic acid	4.9 $\pm$ 0.9		hTRPA1, HEK293T, FLIPR	103
	86 $\pm$ 13		mTRPA1, HEK293T, FLIPR	103
	100 $\pm$ 10		rTRPA1, HEK293T, FLIPR	103
3'-Carbamoylbiphenyl-3-yl cyclohexylcarbamate (URB597)	8 $\pm$ 2		hTRPA1, HEK293T, FLIPR	103
	74 $\pm$ 20		mTRPA1, HEK293T, FLIPR	103
	129 $\pm$ 23		rTRPA1, HEK293T, FLIPR	103
4-Methyl-N[2,2,2-trichloro-1-(4-nitro-phenylsulfa-nyl)-ethyl]-benzamide (CMP1)	0.93 $\pm$ 0.05		mTRPA1, HEK293T, FLIPR	103
	0.88 $\pm$ 0.03		rTRPA1, HEK293T, FLIPR	103
		1.0 $\pm$ 0.1	hTRPA1, HEK293T, FLIPR	103
		2.7 $\pm$ 0.3	rhTRPA1, HEK293T, FLIPR	103
6-Gingerol	10.4 $\pm$ 0.03	>100	rTRPA1, HEK293T, Ca ²⁺ imaging	552
L-Carveol	190 $\pm$ 30		hTRPA1, Flp-In 293, Ca ²⁺ imaging	546
trans-p-Methoxycinnamaldehyde	30 $\pm$ 15		hTRPA1, Flp-In 293, Ca ²⁺ imaging	546
Methyl eugenol	160 $\pm$ 20		hTRPA1, Flp-In 293, Ca ²⁺ imaging	546
4-Allylanisole	1,500 $\pm$ 300		hTRPA1, Flp-In 293, Ca ²⁺ imaging	546
p-Anisaldehyde	550 $\pm$ 70		hTRPA1, Flp-In 293, Ca ²⁺ imaging	546
Piperine	30		hTRPA1, HEK293T, Ca ²⁺ imaging	628
Isopiperine	33		hTRPA1, HEK293T, Ca ²⁺ imaging	628
Isochavicine	71		hTRPA1, HEK293T, Ca ²⁺ imaging	628
Piperanine	150		hTRPA1, HEK293T, Ca ²⁺ imaging	628
Piperolein A	7.8		hTRPA1, HEK293T, Ca ²⁺ imaging	628
Piperolein B	11		hTRPA1, HEK293T, Ca ²⁺ imaging	628
N-isobutyl-(2E,4E)-tetradeca-2,4-diamide (Ntetra)	19		hTRPA1, HEK293T, Ca ²⁺ imaging	628
Curcumin	3.3		rTRPA1, HEK293T, Ca ²⁺ imaging	573
Oleocanthal	2.8		hTRPA1, HEK293T, electrophys. (-60 mV)	657
Umbellulone	19 $\pm$ 4		rTRPA1, HEK293T, Ca ²⁺ imaging	583
	28 $\pm$ 7		hTRPA1, HEK293T, Ca ²⁺ imaging	583
		410 $\pm$ 50	mTRPA1, CHO, electrophys.	943
Dihydroumbellulone	22	340 $\pm$ 80	mRPA1, CHO, electrophys.	943
Tetrahydroumbellulone	ND	380 $\pm$ 30	mTRPA1, CHO, electrophys.	943
β -Umbellulol	ND	420 $\pm$ 40	mTRPA1, CHO, electrophys.	943
Acetyl tetrahydroumbellulone	ND	490 $\pm$ 60	mTRPA1, CHO, electrophys.	943
Acetyl β -umbellulol	ND	>1,000	mTRPA1, CHO, electrophys.	943
Ligustilide	44	1,500	mTRPA1, CHO, electrophys.	944
Dehydroligustilide	540	23	mTRPA1, CHO, electrophys.	944
Capsiate	2.76 $\pm$ 0.08		hTRPA1, HEK293T, electrophys. (-60 mV)	738
Dihydrocapsiate	2.9 $\pm$ 0.2		hTRPA1, HEK293T, electrophys. (-60 mV)	738
Nordihydrocapsiate	2.82 $\pm$ 0.16		hTRPA1, HEK293T, electrophys. (-60 mV)	738
Artepillin C	1.8		hTRPA1, HEK293T, Ca ²⁺ imaging	305
Baccharin	16		hTRPA1, HEK293T, Ca ²⁺ imaging	305
Drupanin	>250		hTRPA1, HEK293T, Ca ²⁺ imaging	305
Methyl syringate	510		hTRPA1, Flp-In 293, Ca ²⁺ imaging	757
Perillaldehyde	41 $\pm$ 8		rTRPA1, HEK293T, Ca ²⁺ imaging	73
	42 $\pm$ 8		rTRPA1, HEK293T, Ca ²⁺ imaging	72
	22 $\pm$ 2		rTRPA1, HEK293T, Ca ²⁺ imaging	73
Perillaketone	20 $\pm$ 2		rTRPA1, HEK293T, Ca ²⁺ imaging	72

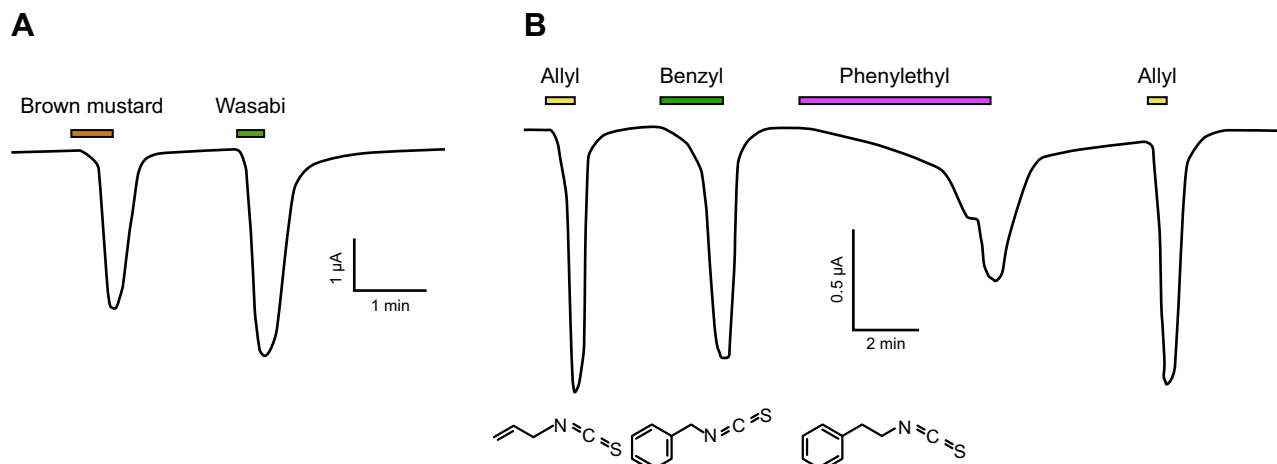


FIGURE 3. Activation of TRPA1 by electrophilic compounds. Activation of human TRPA1 currents in oocytes by natural extracts of brown mustard and wasabi (A) as well as allyl-, benzyl-, or phenylethyl-isothiocyanate (B). Agonist structures are indicated below each current trace. [Modified from Jordt et al. (357). Reprinted by permission from Springer Nature.]

the thiol group of TRPA1 cysteines to the agonist carbonyl carbon of unsaturated carbonyl-containing compounds (696). Unraveling of the chemistry behind activation and deactivation of TRPA1 via covalent modifications remains an exciting challenge.

TRPA1 activity is modulated by O_2 in a complex manner. A stimulatory action of hyperoxia is mediated by glutathione-sensitive oxidation of cysteine residues C633 and C856 located in the NH_2 terminus of the channel (792).

Notably, human TRPA1 is also activated by hypoxia via a decrease in the activity of prolyl hydroxylases (PHDs) and the resulting reduction of the tonic inhibition of the channel induced by hydroxylation of the NH_2 -terminal residue P394. Hypoxia also increases the insertion of nonhydroxylated channels in the plasma membrane. TRPA1 is therefore activated by both hypoxia and hyperoxia, displaying minimal basal activity at O_2 concentrations close to the physiological values. Interestingly, this type of modulation may impact the activation of the channel by several factors. These include cold (because O_2 levels in solution increase at lower temperatures), Zn^{2+} (because this ion interacts with cysteine residues), and the function of mitochondria (because these organelles consume the majority of cellular O_2 and hence regulate the concentration of this gas in the cytosol). Thus O_2 seems to be yet another physiological tuning factor of TRPA1 activity.

H_2S is one of the most important TRPA1 activators (617). NaHS, a donor of H_2S , activates TRPA1, but not TRPV1. H_2S may be involved via this pathway in several physiological processes including nociception (535), as this gas functions as an endogenous transmitter in humans. TRPA1 activation by H_2S seems to underlie the vasoactive effects of this compound, as TRPA1 mediates the release of CGRP from sensory nerves of rat tracheae and to cutaneous vaso-

dilatation in the mouse ear (670). TRPA1 receptor activation should be considered as a potential mechanism of H_2S . Furthermore, polysulfides (H_2S_n) produced by the interaction between H_2S and nitric oxide (NO) also activate TRPA1 (534). Interestingly, dimethyl trisulfide induces analgesia via activation of TRPA1 and the release of somatostatin and its subsequent action on sst4 receptors (74, 671).

H_2S increases cAMP levels in neuronal and glial cell lines and primary neuron cultures with hyperpolarization. In addition to its action on TRPA1, H_2S may be involved in multiple signaling pathways and produce various effects on ion channels [e.g., T-type Ca^{2+} channels, ATP-sensitive K^+ (K_{ATP}) channels], which may inhibit or promote nociception. It is also conceivable that H_2S may affect the *N*-methyl-D-aspartate (NMDA) receptor complex (748). Activation of $Ca_v3.2$ contributes to the H_2S -induced mechanical hyperalgesia and allodynia in mice. Thus T-type Ca^{2+} channel and TRPA1 channel blockers may be an efficient way to attenuate NaHS/ H_2S -induced mechanical hyperalgesia and allodynia (627). The functional link between these channels is also evidenced in the context of H_2S -induced colonic pain and referred hyperalgesia in mice (833). Dual effects of H_2S are also found in rat pancreatic islet-derived cells (RIN14B), which express K_{ATP} and TRPA1 channels. It was proposed that activation of K_{ATP} channels by the H_2S donor NaHS reduces spontaneous oscillation of $[Ca^{2+}]_i$, whereas activation of TRPA1 results in a delayed sustained increase in $[Ca^{2+}]_i$ (839).

In general, most reactive oxygen species (ROS, such as peroxide causing cysteine oxidation), reactive nitrogen species (RNS, such as NO mediating *S*-nitrosylation), and reactive carbonyl species (RCS, such as prostaglandin J_2 and unsaturated aldehydes mediating cysteine carbonylation) behave as TRPA1 activators (425, 737, 794). The typical ROS,

H₂O₂, is contained in industrial products and also generated within cells, and induces pain and activates TRPA1. The effects of H₂O₂ on TRPA1 are mimicked by other ROS and by RNS. Cysteine-reducing agents suppress H₂O₂-induced TRPA1 activation, whereas cysteine-oxidizing agents activate TRPA1 (714). Activation by H₂O₂ is also inhibited by scavengers of the hydroxyl radical such as the typhostin AG-related compounds AG555 and AG556 (818). Oxidative stress leads to the production of oxidized phospholipids (OxPAPC) that are able to activate TRPA1 via cysteine modification, leading to acute pain, hyperalgesia, and induced pronociceptive peptide release (479, 614).

Among RNS, NO is a potent TRPA1 activator. NO can induce acute pain in humans and plays an important role in pain sensitization caused by inflammation and injury in animal models. NO acts both in the central nervous system via a cGMP pathway and in the periphery on sensory neurons through direct activation of TRPA1 (and also TRPV1). Tetrahydrobiopterin (or BH₄) is an essential cofactor for NO production that stimulates a subset of DRG neurons by TRPA1 activation (536). Nitro-oleic acid (9-OA-NO₂), an electrophilic fatty acid byproduct of NO and nitrite reactions, is a potent TRPA1 activator (79). Excessive NO during inflammation (nitrative stress) leads to the nitration of phospholipids resulting in the formation of this highly reactive cysteine modifying agents. 9-OA-NO₂ fails to activate TRPA1 in which the cysteines at position C619, C639, C663, and the lysine at 708 are mutated (806). The effects of OA-NO₂ are blocked by dithiothreitol, but cannot be prevented or reversed by the NO scavenger carboxy-PTIO (2-(4-carboxyphenyl)-4,4,5,5-tetramethylimidazoline-1-oxyl-3-oxide) (723). *N*-nitroso-2-exo,3-exo-ditrifluoromethyl-7-azabenzobicyclo[2.2.1]heptane (NNO-ABBH1) *S*-nitrosylates TRPA1 and activates this channel, with this effect sensitive to specific cysteine mutations but not to scavenging of NO (424, 426).

The list of TRPA1 agonists comprises also electrophilic prostaglandins (PG). The PGD₂ metabolite 15-deoxy- δ (12, 14)-prostaglandin J₂ (15dPGJ₂) activates human TRPA1 expressed in human embryonic kidney (HEK) cells and in mouse trigeminal neurons (176). This effect is blocked with ruthenium red and the selective TRPA1 inhibitor HC-030031. This effect is not mimicked by their non-electrophilic precursors, PGE₂ and PGD₂, or PGB₂, which differs from PGA₂ only in that its electrophilic carbon is rendered unreactive through steric hindrance. Nonetheless, TRPA1 is required for the nociceptive effects of PGE₂ in vivo (181).

The analgesic action of nonsteroidal anti-inflammatory drugs (NSAIDs) results from inhibition of cyclooxygenases and blockade of PG biosynthesis. Cyclopentenone PGs, 15dPGJ₂, PGA₂, and PGA₁, formed by dehydration of their respective parent PGs, PGD₂, PGE₂, and PGE₁, possess a

highly reactive unsaturated carbonyl group required to activate TRPA1. Cyclopentenone PGs produce pain by direct stimulation of nociceptors via TRPA1 activation. Some PGs are proalgesic. Thus TRPA1 antagonism may contribute to suppress pain evoked by PG metabolites without the adverse effects of inhibiting cyclooxygenases (508).

TRPA1 is also activated by alkylating compounds such as the chemical warfare agent sulfur mustard (SM) and 2-chloroethyl-ethylsulfide (CEES) (772). *N*-acetyl-L-cysteine, but not glutathione, prevents activation of human TRPA1 by SM, suggesting that the former interacts directly with the channel (771). CEES-induced cytotoxicity and Ca²⁺ responses in human A549 lung epithelial cells are reduced by AP18. It was therefore proposed that TRPA1 inhibition may serve to reduce SM-induced cell damage. In this line, a later study showed that mouse skin lesions induced by CEES are reduced by the TRPA1 inhibitors HC-030031 and A-967079 and a by the CGRP inhibitor MK-8825 (2).

TRPA1 activators can be generated during tissue damage. 4-Hydroxynonenal (or *trans*-4-hydroxy-2-nonenal or 4-HNE or HNE) is an α,β -unsaturated hydroxyalkenal produced by lipid peroxidation in cells. HNE is found in higher quantities during oxidative stress in inflamed tissues due to the increase in the lipid peroxidation chain reaction. HNE acts as an endogenous agonist for TRPA1 and promotes acute pain, release of substance P and CGRP from nerve endings, and neurogenic inflammation. HNE acts via covalent modification of the cysteine and lysine residues in the TRPA1 NH₂ terminus because quadruple mutations TRPA1-3C/K-Q mutants are insensitive to HNE (829). Endogenous HNE effects are abolished with dithiothreitol (DTT), indicating that they act by formation of disulfide bonds. In contrast, the actions of alkenyl aldehydes and 15d-PGJ₂ are not reversed by DDT, suggesting that these agents form Michael adducts (35). Oxidative stress, a pathological feature of many respiratory diseases, causes the endogenous formation of a number of reactive electrophilic alkenals via lipid peroxidation, such as alkenal, HNE, and another alkenal, 4-oxononenal, which is far more electrophilic than HNE (808).

TRPA1 is activated by several lipid compounds (farnesyl thiosalicylic acid, farnesyl thioacetic acid, and 5,8,11,14-eicosatetraenoic acid) and two marketed compounds: disulfiram (Antabuse; a compound used in the treatment of alcohol abuse) and the anti-fungal agent chlordantoin (496). Also constituents of tear gases induce TRPA1 activation. The release of methyl isocyanate in Bhopal, India, caused the worst industrial accident in history (99). Other components of tear gas such as 1H-dibenz[b,e]azepines (morphanthridines) and dibenz[b,f][1,4]oxazepines are highly potent TRPA1 activators (4, 275).

It is widely known that inhalation of ozone is a major health risk in industrialized countries and impairs lung function through sensory neural-mediated pathways (vagal nociceptive C-type bronchopulmonary nerves). Ozone stimulates TRPA1 but not TRPV1 (809). It must be noted, however, that no evidence of TRPA1 involvement in ozone-induced cough hyperresponsiveness was found in rabbit or guinea pig (167).

Formalin causes pain and is therefore widely used in animal models for testing the effects of analgesics (250). Although formalin directly activates TRPA1 (513), this compound also induces Ca^{2+} release from intracellular stores in sensory neurons, primary keratinocytes, and non-neuronal cell lines in a TRPA1-independent manner. This mechanism may underlie formaldehyde-induced neuronal excitation and subsequent inflammation (252).

Toluene diisocyanate is used as a chemical intermediate in the production of polyurethane products such as foams, coatings, and elastomers. It is a reactive hazardous irritant that causes respiratory symptoms such as cough, rhinitis, dyspnea, and chest tightness in exposed workers. TRPA1 was found to be activated by this compound (206, 807), and its inhibition to reduce airway hyperreactivity, neutrophilia and eosinophilia, and Th2-mediated responses (924).

A recent screening study identified three electrophilic molecules sharing a 2-chloro-*N*-(thiazol-2-yl)acetamide structure as very potent agonists of TRPA1, with effective concentrations below 1 nM. One of these compounds, 2-chloro-*N*-(4-(4-ethoxyphenyl)thiazol-2-yl)-*N*-(3-methoxypropyl)acetamide, named JT010, is ineffective on TRPV1, TRPV3, TRPV4, TRPM2, TRPM8, and TRPC5 up to 1 μM (797) and has been proposed as a model to trigger TRPA1-mediated pain in humans (308). In a series of experiments using a biotinylated analogue of JT010, this compound activated TRPA1 by binding covalently to the residue C621 (797). This particular cysteine residue was shown to have an exceptionally high reactivity that largely surpasses that of antioxidant enzymes (58).

Capsazepine, the renowned TRPV1 inhibitor, activates TRPA1 via cysteine modification and exerts a strong desensitizing effect on the channel. Notably, TRPA1 is required for the anti-inflammatory and antinociceptive effects of systemic administration of capsazepine, suggesting for a novel therapeutic strategy against inflammation and pain (394).

Interestingly, also animal-produced electrophiles are able to activate TRPA1. The electrophilic arthropod defensive compound, *para*-benzoquinone (pBQN) potently activates human TRPA1 (from 10 nM), with the critical implication of three cysteine residues (C621S, C641S, C665S) that are distinct from reported to be crucial for the action of other electrophiles. At concentrations higher than 300 nM, the

activator effect is reduced and is followed by a very fast desensitization phase, which is also dependent on the above-mentioned cysteine residues (340). Another animal-derived TRPA1 agonist is crotalphine, a structural analogue of an analgesic peptide found in the venom from the South American rattlesnake *Crotalus durissus terrificus* (125). Crotalphine activates the mouse, rat, and human TRPA1 isoforms with low efficacy but high affinity, requiring specific cysteine residues in the NH_2 terminus of the channel. Importantly, crotalphine produces a strong desensitization to subsequent application of AITC or carvacrol in a Ca^{2+} -dependent manner. Furthermore, this peptide inhibits AITC-induced release of CGRP from mouse trachea and reduces hypersensitivity to cold produced by ciguatoxin, and the mechanical hypersensitivity induced by bradykinin or by the yeast cell wall polysaccharide zymosan (125). Along a similar line, a peptidergic scorpion toxin (WaTx) activates TRPA1 by acting on same intracellular site modified by reactive electrophiles (286, 475).

B. Nonelectrophilic Modulators

In addition to the huge number electrophilic activators, an ever-increasing list of compounds that are unlikely to induce covalent modifications of the channel protein can activate TRPA1 (119, 959).

Parabens, alkyl esters of *p*-hydroxybenzoate, are added to pharmaceuticals, cosmetics, and food products as antibacterial agents. Methyl-paraben (methyl *p*-hydroxybenzoate) is an activator of TRPA1 and produces pain in mice that is blocked with ruthenium red (258).

3'-Carbamoylbiphenyl-3-yl cyclohexylcarbamate (URB597) is a potent inhibitor of the anandamide-degrading enzyme fatty acid amide hydrolase (FAAH). URB597, but not two other FAAH inhibitors URB532 and Compound 7, activates human and rat TRPA1. URB597 activates rat TRPA1 in inside-out patch-clamp experiments and stimulates TRPA1-expressing rat DRG neurons that also respond to AITC. In contrast, URB597 inhibits TRPM8 and is ineffective on TRPV1 and TRPV4 (591).

The super-cooling chemical agent icilin (AG-3-5) activates not only TRPM8 but also TRPA1. This compound triggers a rapid, long-lasting, and dose-related hyperthermia in rats, which is attenuated by pretreatment with the nitric oxide synthase (NOS) inhibitor N^G -nitro-L-arginine methyl ester hydrochloride (L-NAME) or by a selective NMDA receptor antagonist. Thus TRPM8/TRPA1-induced hyperthermia requires both NO production and NMDA receptor activation (211).

Surprisingly, also some frequently used therapeutic compounds such as Ca^{2+} antagonists have been identified as TRPA1 activators. Four 1,4-dihydropyridines (nifedipine, nimodipine, nicardipine, and nitrendipine), and the struc-

turally related L-type Ca^{2+} channel agonist BAY K 8644, activate TRPA1 (245). TRPA1 activation in perivascular nerves of resistance arteries produces vasodilation via the release of CGRP (76). However, it remains to be investigated whether the stimulatory action of 1,4-dihydropyridines on this channel has any relevance in the action of these compounds as antihypertensive drugs. A relatively simple experiment to be performed along this line is to test whether the vasodilation response of resistance arteries to 1,4-dihydropyridines is weaker in *Trpa1*-deficient than in wild-type animals. Of note, it would be also interesting to determine the role of TRPM3, another sensory TRP channel that is activated by nifedipine (218) and whose activation in perivascular nerves induces dilation of resistance arteries (19).

Another surprise came with the identification of local anesthetics (LAs), which suppress cellular excitability by inhibiting voltage-gated Na^+ channels, as activators of TRPA1. Lidocaine activates TRPA1 in a concentration-dependent manner, and its action is blocked by HC-030031. However, lidocaine can also act as an inhibitor of TRPA1, an effect that is more obvious in rodent than in human TRPA1. This species-specific difference is probably related to the transmembrane domains 5 and 6 forming the pore region (460). At high concentrations (30 mM), lidocaine induces Ca^{2+} release from intracellular stores, an effect that may mediate TRPA1- and TRPV1-independent CGRP release and cell death mouse DRG neurons (229). Lidocaine, procaine, and tetracaine induce glutamatergic spontaneous excitatory transmission in substantia gelatinosa (SG) neurons (660, 661). Lidocaine dose-dependently and reversibly increases the frequency, but not the amplitude of spontaneous excitatory postsynaptic current (sEPSC) in SG neurons. This presynaptic enhancement is due to activation of TRPA1 in nerve terminals presynaptic to SG neurons that increases the spontaneous release of L-glutamate (660). Of note, TRPA1 is activated and permeated by the lidocaine derivative QX-314, resulting in a concentration- and use-dependent inhibition of cotransfected $\text{Na}_v1.7$ channels. These findings further support the development of selective inhibition of action potential firing in nociceptive neurons (779).

Importantly, nonelectrophilic compounds may have electrophilic metabolites. For instance, when acetaminophen (*N*-acetyl-*p*-aminophenol, APAP), the most common antipyretic/analgesic medicine worldwide is overdosed, its metabolite *N*-acetyl-*p*-benzo-quinoneimine (NAPQI) causes liver damage. NAPQI is an electrophilic molecule and stimulates TRPA1 causing airway neurogenic inflammation. These inflammatory responses evoked by NAPQI and APAP can be abolished by TRPA1 antagonists (580).

NPPB [5-nitro-2-(3-phenylpropylamino)-benzoate], a classic Cl^- channel blocker, potently activates human TRPA1.

The action of NPPB suggests a possible close interaction between S5 and NH_2 -terminal domains of the channel. As indicated by the analysis of NPPB derivatives, NPPB activates TRPA1 through a structure-specific mechanism (481).

Anesthetic agents generally induce a paradoxical activation and sensitization of TRPA1. Propofol (2,6-diisopropylphenol) and etomidate are pungent and elicit intense pain upon injection. The former compound is commonly used as intravenous anesthetic (251). This compound was identified as TRPA1 agonist in a study that also described the agonist effect of other alkyl phenols, such as thymol, 2-*tert*-butyl-5-methylphenol, and carvacrol (458) (see below). The effects of propofol on sensory neurons may contribute to peripheral sensitization to nociceptive stimuli in traumatized tissue and may be therefore clinically relevant (936). TRPA1 activation mediates the decrease in mean arterial pressure and dilation of murine coronary microvessels induced by propofol, via a mechanism involving activation of NOS and BK_{Ca} channels (742, 743) and restores the sensitivity of TRPV1 via NOS-dependent activation of protein kinase $\text{C}\epsilon$ ($\text{PKC}\epsilon$) (744). Interestingly, the related compound fospropofol does not produce pain upon injection and is unable to evoke depolarizing currents in sensory neurons (650). Propofol-induced desensitization of TRPA1 was proposed to mediate the inhibition of responses of wide dynamic range (WDR) neurons to noxious heat (analgesia) and the reduction of AITC sensitization of WDR neurons (antihyperalgesia), which agrees with clinically observed reduction in postoperative pain in surgical patients anesthetized with propofol (799). It must be noted, however, that TRPA1 is not the sole target of propofol in nociceptive neurons, as it was reported that this compound may produce pain via TRPV1 activation and voltage-gated Ca^{2+} channels downstream of GABA_A receptor activation (251, 599). Recent structure-function, molecular modeling, and photoaffinity labeling studies strongly suggest that propofol binds TRPA1 at a site located within the S5-S6 pocket by forming H-bond and halogen-bond interactions with the amino acid residues S876, M915, and M956 (820, 897).

General anesthetics (GAs) can activate peripheral nociceptive neurons in addition to their known depressing action on the central nervous system. There are, however, differences between the effects of these molecules, as for instance isoflurane and desflurane are pungent and able to activate TRPA1, whereas sevoflurane and halothane are not pungent and ineffective on this channel. Intravenous administration and inhalation of pungent GAs at clinical concentrations excite sensory neurons and induce pain-related responses and activation of tracheobronchial sensory nerves, triggering CGRP release and neurogenic inflammation (233, 264, 386, 509, 563). Furthermore, TRPA1-dependent neurogenic inflammation is greater in mice anesthetized with pungent compared with nonpungent anesthetics. The pronociceptive effects of GAs combined with surgical

tissue damage can lead to a paradoxical increase in postoperative pain and inflammation (386, 509).

A similar paradoxical effect is known for fenamate NSAIDs. Several NSAIDs activate TRPA1, including flufenamic, niflumic, and mefenamic acid, as well as flurbiprofen, ketoprofen, diclofenac, and indomethacin. The response to fenamate agonists was blocked by TRPA1 antagonists. Fenamate NSAIDs also potentiate the activation of TRPA1 by electrophilic compounds (334). It should be noted that the NSAIDs diclofenac, ketorolac, and xefocam suppress thermal and mechanical hyperalgesia following TRPA1 activation, seemingly due to channel inactivation or desensitization (603, 830, 831). Along the same line, etodolac activates via covalent modification of cysteine residues, and subsequently desensitizes the channel (871).

TRPA1 activation was proposed to mediate the irritation caused by primary alcohols in skin, eye, and nasal mucosa. Higher alcohols also activate TRPA1 with potency proportionally increasing with the carbon chain length. Interestingly, although these alcohols are not electrophilic, the residues C665 and H983 in the NH₂ terminus are required for their stimulatory action on TRPA1. Straight-chain secondary alcohols and primary and secondary alcohols activate hTRPA1 with a potency correlated with their octanol/water partition coefficients (416).

Human TRPA1 is activated in an irreversible manner by low micromolar concentrations of apomorphine, a non-narcotic derivative of morphine. At higher concentrations, this compound produces an irreversible activation of the channel, resulting from a destabilization of the open state. TRPA1 activation may therefore be implicated in the ulceration and pain reactions at the injection site during treatment with apomorphine (719).

TRPA1 is activated by taurolidine, a compound clinically used as an antimicrobial or to exert newly recognized antineoplastic actions. This action results in release of CGRP from nociceptive nerve endings in isolated mouse trachea. The metabolite taurultam and its oxathiazine derivative had comparatively weaker effects (387).

Glibenclamide is an anti-diabetic drug that stimulates insulin release through the inhibition of K_{ATP} channels in pancreatic beta-cells. However, this compound may produce abdominal pain, gastrointestinal disturbances, and nocturia. Glibenclamide activates human TRPA1 at concentrations similar to those acting on K_{ATP} channels in vitro, and pharmacological and genetic evidence indicates that TRPA1 is implicated in the stimulatory effects of glibenclamide in mouse primary sensory neurons (55).

TRPA1 activation may account for the frequent adverse effects observed in patients treated with phenytoin. This

compound activates a nonselective cationic current in human gingival fibroblasts with typical features of TRPA1 channels, including inhibition by HC-030031 and reduction by shRNAs against hTRPA1. Phenytoin effects on these cells are strongly decreased by the antioxidant vitamins ascorbic acid, folic acid, and alpha-tocopherol. Phenytoin induces accumulation of collagen in the extracellular matrix, but did not enhance cell proliferation (486).

C. Activation by Natural Compounds

A significant number of sensory TRP channel activators appeared during evolution as chemical weapons produced by plants against herbivores. As recently reviewed (766), this fascinating subject has as notable examples the activation of TRPA1 by compounds produced by *Brassica* (mustard) and *Allium* (onions and garlic) plants, such as pungent isothiocyanates and allicin, respectively. Of note, other plants such as those of the genus *Cinnamomum* (cinnamon) produce TRPA1 agonists, but these compounds have more of an antibacterial function than a repellent one. What these plants do have in common is that they have found a very special place in our culinary cultures and in the history of Mankind through their importance in commerce and therefore in the dominance over trade routes (592, 593, 878).

As already mentioned above, the spice wasabi (from *Wasabia japonica*) acquires pungency through its active compound AITC. The related compounds 6-(methylsulfinyl)hexyl isothiocyanate (6-MSITC) and 6-(methylthio)hexyl isothiocyanate (6-MTITC) have lower pungency, are responsible for the fresh flavor of wasabi, and act as TRPA1 electrophilic activators (837). The other most prominent TRPA1 agonist is cinnamaldehyde, the main constituent of cinnamon oil from *Cinnamomum verum* and other cinnamon species (61). This spice is widely used as a condiment and flavoring component in chocolate; many dessert recipes, such as apple pie and donuts; spicy candies, tea, hot cocoa, liqueurs, and even in toothpastes. Given orally, it causes burning and tingling due to activation of TRPA1. When applied topically on the skin, cinnamaldehyde produces spontaneous pain, heat and mechanical hyperalgesia, cold hypoalgesia, neurogenic axon reflex erythema, intensified warm sensations, reduced heat pain threshold, moderate itch, flare, hyperkinesis, allokinesis, and increased skin blood flow and temperature (30, 53, 135, 322, 575, 630, 740). In the human tongue, cinnamaldehyde enhances both heat- and cold-induced pain (14, 685), increases the local temperature (353), and produces burning sensation (301). When applied to human airways, this compound produces smart (stinging pain) (16), cough (365), and weak burning sensation (134). From the first report on its action on TRPA1, cinnamaldehyde was described as more specific than AITC (61). Hence, the responses to cinnamaldehyde in vitro and in vivo had been widely taken as proof for functional expression of TRPA1 (889). However, this com-

pound may have off-target effects, such as the described inhibitory effect on voltage-gated Ca^{2+} channels in resistance arteries and concomitant vasodilation, which could make an impact on nociception and local circulatory responses and thermoregulation (26).

As TRPV2, TRPA1 is one of the newly described ionotropic cannabinoid receptors (10). It is activated by the nonelectrophilic Δ^9 -tetrahydrocannabinol (Δ^9 -THC or THC), the psychoactive compound in marijuana (*Cannabis sativa*) (357). This THC effect does not require the presence of polyphosphates in the intracellular solution, in contrast to that of AITC. This suggests distinct mechanisms underlying the action of these compounds (145). The cannabinoid receptor agonists WIN 55,212-2 and AM1241 activate TRPA1 and thereby decrease the responses to capsaicin and AITC. Moreover, TRPA1 is required for the inhibitory action of these compounds on the nocifensive behavior elicited by capsaicin in mice (11). The activation of TRPA1 by fitocannabinoids was further confirmed in a study testing the action of cannabidiol, THC, cannabidiol acid, THC acid, cannabichromene, and cannabigerol, which produce intracellular Ca^{2+} responses comparable in amplitude to that evoked by AITC (192). Synthetic analogues of phytocannabinoids were also reported to activate the channel (129, 485, 666). TRPA1 mediates part of the inhibition of the ongoing activity of ON and OFF neurons of the rostral ventromedial medulla in anesthetized rats induced by cannabidiol and cannabichromene, two nonpsychoactive cannabinoids. This effect is emulated by AITC. Cannabidiol and cannabichromene also elevate endocannabinoid levels in the ventrolateral periaqueductal grey. These findings open the possibility of using these compounds to target TRPA1 to produce antinociceptive effects at the supraspinal level (498).

TRPA1 also mediates THC-induced CGRP release and vasorelaxant responses to sensory nerve stimulation. Notably, morphine and its derivatives are commonly in pain control but have proalgesic effects at high concentrations, as for instance the induction of short-lasting painful sensations upon dermal application and CGRP release via TRPA1 activation. Naloxone has analogous effects (254). Spinal TRPA1 also contributes to morphine antinociceptive tolerance via a positive feedback mechanism that upregulates the H_2O_2 -producing astroglial enzyme D-amino acid oxidase (884). TRPA1 is a target of the farnesyl prenylogue of cannabigerol found in the waxy fraction from the variety Carma of fiber hemp (*Cannabis sativa*). This plant may contain additional prenylogous versions of medicinally relevant cannabinoids, for which their biological profiles could be of potential therapeutic interest (667). Of note, also cannabinoid receptor antagonists such as AM251 and AM630 activate TRPA1, illustrating the importance of their off-target effects (647).

Allium sativum (garlic) produces the organo-sulfur pungent electrophile allicin, from which other active derivatives are produced: diallyl sulfide, diallyl disulfide, and diallyl trisulfide. Allicin is a protective compound acting against endothelial cell dysfunction in a rat model of right ventricle hypertrophy (786). TRPA1 activation by allicin (76, 492) could mediate this effect, if considering that TRPA1 is expressed in heart (773). The allicin derivative ajoene, despite containing reactive electrophilic groups, is unable to activate TRPA1, but enhances the activation by other electrophiles such as AITC and allicin (925). It must be noted that allicin and its derivatives also activate TRPV1, although with lower potency than for TRPA1 and with a lower efficacy compared with capsaicin (412, 492). Nevertheless, this unspecificity should be taken into account when extracting conclusions over the role of TRPA1 in visceral sensitivity and motility from experiments based on the use of garlic (and allicin) as agonist of this channel (257). Interestingly, allicin-induced activation of TRPV1 is mediated by a single cysteine residue in the NH_2 terminus of the channel, suggesting for a similarity with the mechanisms of action operating in TRPA1 (706). TRPA1 activation was proposed to mediate the hyperthermic effects of the durian fruit (from *Durios zibethinus* Murr.), which contains the allicin-related compound diethyl disulfide (814). S-alkyl-S-alkenyl disulfides are a rare class of natural products contained in the foul-smelling gum-resin of *Ferula assa-fetida* L. These compounds showed a transthiolation capacity and potentially activate TRPA1, suggesting this channel as the culprit of potential beneficial effects associated with the use of asafetida as a spice and medicine (739).

Over 80 terpenes are produced by plants, fungi, and animals, serving as deterrents of foragers and predators. These compounds feature reactive unsaturated dialdehyde moieties. The noxious fungal sesquiterpene isovelleral activates TRPA1, thereby exciting sensory neurons and eliciting nocifensive responses in mice. Another drimane sesquiterpene, polygodial (from *Polygonum hydropiper* leaves), also activates TRPA1 (240). Miogadial, miogatrial, and polygodial, all with an unsaturated 1,4-dialdehyde moiety, are stronger TRPA1 agonists than AITC (346). Interestingly, although isovelleral contains a β -unsaturated dicarbonyl moiety potentially capable of forming Michael adducts, it activates TRPA1 with mutated reactive cysteines, suggesting that dialdehyde sesquiterpenes may act through a mechanism distinct to that operating in other electrophiles (68). Interestingly, nonelectrophilic sesquiterpene such as α -, β -eudesmol, and γ -eudesmol also activate TRPA1 (622). These compounds are found in hop essential oil and are thought to confer a spicy note to beer. The activation of hTRPA1 by β -eudesmol requires the residues T813, Y840, and S873.

Another prominent TRPA1 agonist is menthol (from *Mentha piperita*). This compound was first described to have no effect on TRPA1 (777) or to be an inhibitor of TRPA1

responses triggered by cold or cinnamaldehyde (493). However, it was later shown to stimulate basal currents in the low micromolar range and to have an inhibitory action beyond a few hundred micromolar (371) (FIGURE 4). Wash-out of inhibitory concentrations of menthol leads to a significant rebound of TRPA1 activity, which is consistent with a very fast unblocking event. The agonist effect of menthol is due to a shift of the voltage dependence of channel activation to negative potentials. The inhibitory effect relates to an increased probability of null traces and the appearance of fast flickering of single-channel currents, raising the possibility that menthol acts as an open pore blocker.

Notably, several modulators of TRPA1 act in such a bimodal fashion. Importantly, hTRPA1 is activated by menthol but not inhibited by it, whereas TRPA1 from non-

mammalian species are reported to be insensitive to this compound. Structure-function studies on mouse, human, and *Drosophila* TRPA1 isoforms led to the identification of the pore region TM5 and TM6 as the critical domain determining whether menthol can act as an inhibitor and of specific residues in TM5 (S873 and T874) required for menthol-induced activation. It remains, however, unclear whether these structures are actually involved in binding or channel gating (908). Nevertheless, human TRPA1 emerged as a highly sensitive menthol receptor that very likely contributes to the diverse psychophysical sensations after topical application such as warmth, burning, irritation, and pain (168, 575). These findings were crucial to re-interpret conflicting data on the sensory roles of TRPM8 and TRPA1 that raised from the wrong assumption that menthol was specific for the former channel (860). They help explaining, for instance, the existence of men-

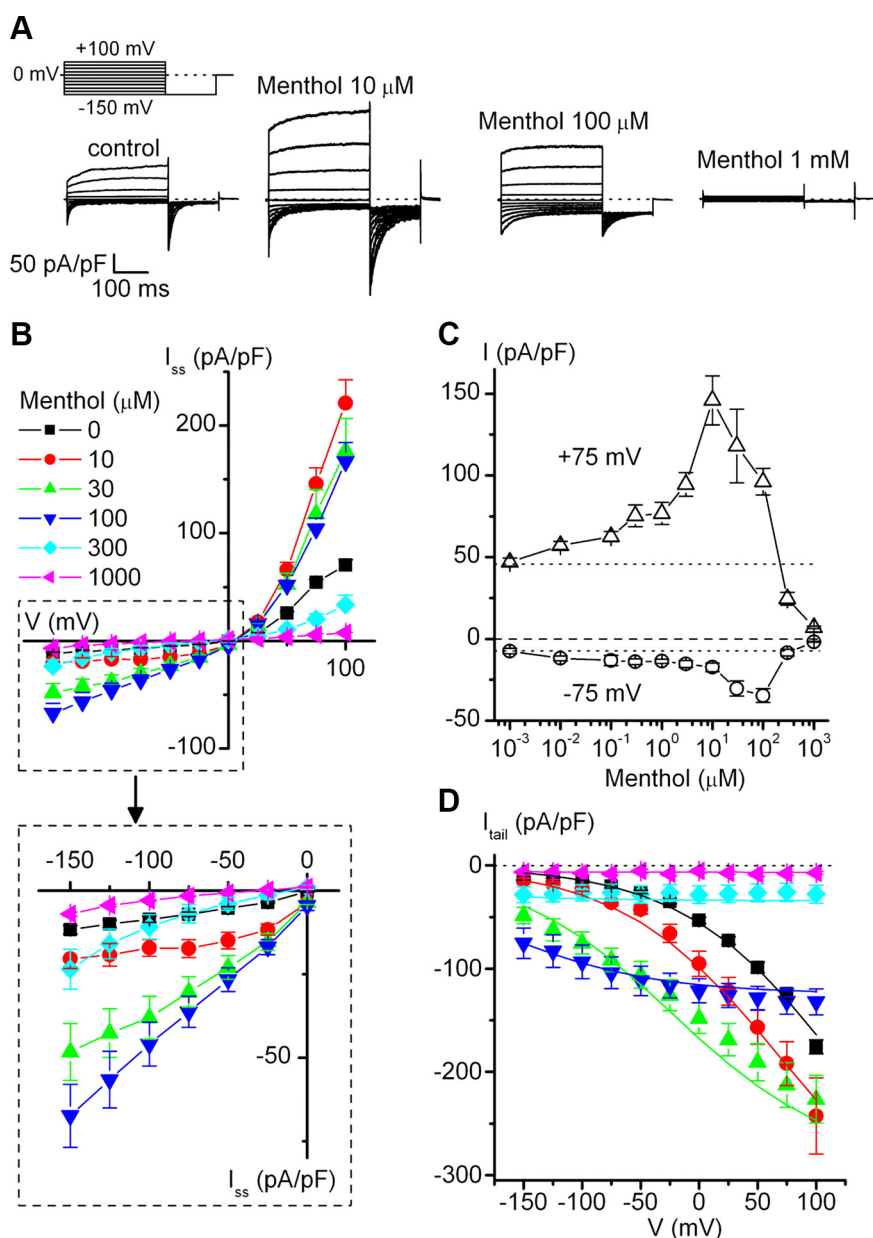


FIGURE 4. Bimodal action of menthol on the gating properties of TRPA1. **A:** whole-cell TRPA1 currents in response to the indicated voltage protocol under control conditions and during application of 10 and 100 μM and 1 mM menthol. **B:** current (I)-voltage (V) relationships obtained at the end of the 300-ms voltage steps shown in A. **C:** dose-response relationships of menthol on TRPA1 at the indicated voltages. Dotted lines indicate constitutive currents before menthol application. **D:** tail currents at -150 mV in control and during application of the different concentrations of menthol. [Modified from Karashima et al. (371).]

thol-responsive neurons in *Trpm8*-deficient animals (170, 207) and the role of TRPA1 as cold sensor in visceral nociceptive neurons (244).

TRPV1, and to a lower extent TRPA1, are molecular targets of hydroxy- α -sanshool, from Szechuan peppers (*Zanthoxylum piperitum*), an effect that was proposed to underlie some of the irritant, cooling, tingling, and paresthetic sensations these spices produce in the tongue (420). This view was, however, challenged by a subsequent study reporting that hydroxy- α -sanshool has minor effects of these channels and only at high concentrations, but on the other hand inhibit the 2-pore potassium channels KCNK3, KCNK9, and KCNK18 (77). The later effect was then proposed to mediate Szechuan peppercorns sensory properties referred as “tingling and numbing,” “mild electric shock,” or “pins and needles” through the combination of the targeting of capsaicin-sensitive nociceptors and mechanosensitive neurons. This view was further supported by psychophysical experiments showing that application of isobutylalkenyl amide induce multifaceted and dynamic sensations in human tongue (15), as well as complex aversive responses in the rat (398). Nevertheless, these initial findings served as starting point for the development of new hydroxy- α -sanshool-derived alkylamides manifesting higher selectivity for TRPA1. The *cis*-C6 double bond in the polyenic chain of a-SOH is critical for TRPA1 activation and no structural specificity is required for TRPV1 activation (521). Alkylamides and 6-shogaol act on TRPA1 by covalent bonding, whereas none of these compounds activate TRPV1 through such interactions (391, 681).

Thymol (2-isopropyl-5-methylphenol) is a natural monoterpene phenol derivative of cymene, isomeric with carvacrol. This compound is enriched in the oil of the thyme plant (*Thymus vulgaris*) and has a pleasant aromatic sharp odor as well as a pungent flavor. Thymol potently activates TRPA1 and desensitizes it to further exposure to thymol or AITC. Other phenols such as 2-tert-butyl-5-methylphenol, 2,6-diisopropylphenol (propofol) and carvacrol also activate hTRPA1. Phenols with more bulky carbon substitutions and hydrophobicity are more potent activators (458). TRPA1 is also activated by the related monoterpenoids *p*-cymene-3-carboxylic acid and 3-amino-*p*-cymene (632) and limonene (361). Thymol was later found to have in fact a bimodal action on TRPA1 (371) and to share the same required structural domains for channel activation with menthol (908). This compound increases the spontaneous release of L-glutamate onto spinal substantia gelatinosa neurons, an effect that is antagonized by HC-030031 (917). TRPA1 was proposed to mediate the pungency of carvacrol, a compound found in the spice oregano, derived from *Origanum vulgare*, which is used in the kitchen for the aromatic, warm, and slightly bitter taste of its leaves (458, 915). Intraepidermal injection of carvacrol in humans produces concentration-dependent pain sensations that are re-

duced by the TRPA1 antagonist A-967079 (722). TRPA1 is activated by eugenol (164) and gingerol-related compounds (552). The former is a phenylpropene found in cloves (*Syzygium aromaticum*) and it is used in perfumes, flavorings, and essential oils and in medicine as a local antiseptic and anesthetic. Gingerol is a phenol-derived compound extracted from the essential oils of clove and fresh ginger (*Zingiber officinale*). The related compounds L-carveol, *trans*-*p*-methoxycinnamaldehyde, methyl eugenol, 4-allylanisole, and *p*-anisaldehyde, contained in the stem and leaves of *Agastache rugosa* (of the *Labiatae* family), selectively activate hTRPA1 over hTRPV1 (546). Of note, thymol, carvacrol, and eugenol are well-known agonists and sensitizers of TRPV3 in skin keratinocytes (915). The ability of these chemicals to increase the avoidance of warmer temperatures supported a role for TRPV3 in warmth detection (399). However, the recent reports on heat-induced activation of mammalian TRPA1 and on its implication in heat sensing (see below) indicate that this channel may also underlie the modulation of warmth sensing by such compounds.

Another example for a species-dependent activation of TRPA1 is caffeine (from *Coffea arabica*). This compound activates mouse TRPA1 but suppresses human TRPA1. The region between the amino acid residues 231 and 287, in the distal NH₂-terminal region of mTRPA1, is critical for this effect. The point mutation M268P changed the effect of caffeine from activation to suppression in mTRPA1 (565, 566).

Also (–)-nicotine (from *Nicotiana tabacum*) acts as a bimodal TRPA1 modulator. TRPA1 activation may therefore mediate part of the irritation induced by topical application of nicotine as used in nicotine replacement therapies on the mucosa and skin (801) or by the use of electronic cigarettes (165). It is thus thought that TRPA1 block may facilitate the development of smoking cessation therapies with less adverse effects. The nicotine congenitor anabasine (from *Nicotiana glauca*) acts in the same way as nicotine (801). (+)-Nicotine was later reported to activate hTRPA1 (718).

1'-Acetoxychavicol acetate is the major pungent component in galangal (galanga, blue ginger), a rhizome of plants of the genus *Alpinia officinarum* originated from Indonesia that has medicinal and culinary uses. This compound activates TRPA1 more potently than AITC, but not TRPV1 (576). It remains unclear, however, whether any of the properties of galangal, which include antioxidative, anti-gastric ulcer and antitumor effects, and potentially enhanced thermogenesis, are related to TRPA1 activation.

The spiciness of black pepper (*Piper nigrum* from the family *Piperaceae*) is due to the chemical piperine. This and the related compounds isopiperine, isochavicine, piperanine, piperolein A, piperolein B, and *N*-isobutyl-(2*E*,4*E*)-tetra-

deca-2,4-diamide strongly activate TRPA1 (628). This effect, and the complementary action on TRPV1, was proposed to mediate the beneficial effects of piperine through stimulation of oropharyngeal sensory neurons that may serve as novel therapeutic strategy against dysphagia (25, 688) and piperine-induced contraction in guinea pig ileum, urinary bladder, and trachea (86).

Curcumin is an active principle of the turmeric root (*Curcuma longa*) of the ginger family (*Zingiberaceae*). This compound is used in many cuisines as “yellow curry” and can form Michael adducts. Curcumin activates human TRPA1 but has no effect on TRPM8 or TRPV1 (453). Within a series of 33 synthetic curcumin analogues, 20 acted as TRPA1 modulators, 6 of the 1,3-dicarbonyl and acyclic series behaved as TRPM8 antagonists, and only a few were able to either inhibit or activate TRPV1 (573).

Oleocanthal is a natural phenolic compound known to mediate the pungency of extra-virgin olive oil (166). Human TRPA1 is selectively activated by oleocanthal and is required for the stimulation of rodent trigeminal neurons. Results from immunohistochemical localization of TRPA1 and psychophysical experiments in humans point to a restricted functional expression of this channel in the pharynx (657). Similar findings obtained with the over-the-counter analgesic ibuprofen agree with shared chemesthetic qualities with olive oil (tickle) (93), and with the fact that the former activates human TRPA1, though not directly, via its electrophilic metabolite ibuprofen-acyl glucuronide. Interestingly, ibuprofen-acyl glucuronide impairs subsequent channel activation by AITC, suggesting that it may mediate part of the analgesic and anti-inflammatory activities of ibuprofen (186).

The “headache tree,” California bay laurel (*Umbellularia californica*) is known for causing severe headache crises upon inhalation of its vapors. The monoterpene ketone umbellulone is the major volatile constituent of the leaves of this plant and has irritating properties, probably due to its ability to rapidly bind to thiol groups. Umbellulone has a bimodal action on TRPA1, stimulating at low concentrations and inhibiting at high concentrations. The former effect is reduced, but not completely abolished, in the TRPA1 mutant C622S. Analogues of umbellulone with removed Michael acceptor properties have reduced stimulatory effects but increased inhibitory potency. Several umbellulone derivatives induce weak TRPM8 activation (943). Multiple sensory responses to umbellulone, such as stimulation of nociceptive neurons, release of CGRP, and nocifensive behavioral responses, are abrogated in *Trpa1*-deficient mice. Systemic administrations of TRPA1 and CGRP receptor antagonists inhibit the increased rat meningeal blood flow induced by intranasal application or intravenous injection of umbellulone. These data suggest that other noxious agents such chlorine, cigarette smoke, and formaldehyde

produce headaches via TRPA1 activation in trigeminal nerves (583), and illustrate how the study of the mechanisms underlying the action of herbal compounds may help in the understanding of pain (278).

Ligustilide is the major contributor of the aroma of celery (from *Apium graveolens*, *Levisticum officinale*, from popular medicinal plants used in traditional Chinese medicine such as *Angelica sinensis*, *Ligusticum chuanxiong*, and North American traditional medicine from *Ligusticum portieri*). This electrophilic volatile dihydrophthalide is a potent activator and modest inhibitor of TRPA1. Similar to what was found for umbellulone, the aromatization of ligustilide to dehydroligustilide reduces the stimulatory effect and enhances the inhibitory effect on TRPA1. It was proposed that TRPA1 contributes to the flavor and pharmacological actions of celery and other important plants from the Chinese and native American traditional medicines (944). Later on, a series of ligustilide analogues (3-ylidenephthalides) were synthesized and found to be strong modulators of TRPA1 and TRPM8 (633).

Capsiate, produced by “sweet” *Capsicum annuum* L., is a nonpungent cultivar of red pepper, and, like capsaicin, it is considered to boost metabolism by activating the sympathetic nervous system and suppressing inflammation. TRPA1 is activated by the capsinoids capsiate, dihydrocapsiate, and nordihydrocapsiate, although with potency lower than that for TRPV1 activation (738). Interestingly, benzylamide-to-tetrazole substitutions in the vanilloids olvanil, rinvanil, and phenylacetylrinvanil resulted in nonelectrophilic compounds with the ability to activate TRPA1 (199).

Artepillin C, a prenylated derivative of cinnamic acid, is the main pungent ingredient in Brazilian green propolis, a popular health supplement used for its various biological properties. Artepillin C activates TRPA1 more potently than AITC. Other cinnamic acid derivatives such as baccharin and drupanin also activate TRPA1, whereas *p*-coumaric acid has no effect (305).

The hydroalcoholic extracts of the first leaves of *Kalopanax pictus* Nakai (*Araliaceae*) activate human TRPA1 and TRPV1 channels (758). Out of six commercially available compounds (methyl syringate, coniferyl alcohol, protocatechuic acid, hederacoside C, alpha-hederin, and eleutheroside B) found in *K. pictus*, methyl syringate was shown to be ineffective on hTRPV1 and to activate hTRPA1, although with lower potency than AITC (757). It was later shown that methyl syringate inhibits the hypoxic induction of cyclooxygenase-2 (COX-2) expression and invasion through TRPA1 activation in adenocarcinomic human alveolar basal epithelial A549 cells, suggesting that this compound may be of value to abrogate hypoxia-induced inflammation (641).

Parthenolide is a major constituent of feverfew (*Tanacetum parthenium*), a traditional medicinal herb that has been used for centuries to treat pain and headaches. This compound stimulates recombinant and native rat and mouse TRPA1 channels, behaving as a partial agonist. Interestingly, parthenolide desensitizes TRPA1 causing it to be unresponsive to peptidergic TRPA1-expressing nerve terminals, an effect that may contribute to the antimigraine effect of parthenolide (506).

Furylketones derivatives are another class of chemical structures active with high potency on TRPA1 (73). These compounds relate to the naturally-occurring perillaldehyde and perillaketone extracted from *Perilla frutescens*, a plant widely used in Asian cuisine. Both are activators of TRPA1, which may explain the chemesthetic properties of this plant. *Perilla* leaf (zisuye) cause warm sensations and are widely used in traditional Chinese medicine and against stomach dysfunction (72).

TRPA1 and TRPV1 are activated by theasinensins A and D, two auto-oxidation products of epigallocatechin gallate, a polyphenol contained in green tea. This effect might contribute to the astringent sensation produced by this compound in the tongue (440, 441).

Despite their beneficial effects, such as in the treatment of skin disorders and cancer, retinoids produce irritation side effects. The retinoid receptor RAR β antagonist LE135 stimulates DRG neurons and produces pain-related behaviors via activation of TRPA1 and TRPV1. A point mutation K170R abrogates the stimulatory effect on TRPA1 (927).

Anethole, present in anise, fennel, and licorice, activates native mouse TRPA1 in DRG and TG neurons and recombinant human TRPA1 in HEK293 cells. The mechanism of activation by this compound seems to require the same residues involved in stimulation by menthol, S873 and T874 in TM5. Anethole desensitizes and for unknown reasons fails to produce nocifensive behaviors in mice (518).

Other reported TRPA1 agonists of natural origin are leucettamols, bifunctionalized sphingoid-like compounds from a marine sponge *Leucetta* sp (157), C₁₄ polyacetylenes from the plant *Echinophora platyloba* (*Apiaceae*) (158), and compounds from the Himalayan ritual medicinal plant *Waldheimia glabra* (277).

D. Ca²⁺-Dependent Modulation

The initial studies on functional properties of TRPA1 were already consistent with a modulation of channel activation by intracellular Ca²⁺ (357). It was shown that human TRPA1 mediated a ruthenium red-sensitive large and rapid increase in intracellular Ca²⁺ concentration in response to carbachol in HEK293 cells coexpressing this channel and

M1 muscarinic acetylcholine receptors (357) and in response to thapsigargin in TRPA1-expressing HEK293 (357), Chinese hamster ovary cells, and sensory neurons (649). Furthermore, although Ca²⁺ was shown not to be critical for channel activation, its presence in the extracellular solution enhanced the responses to AITC and THC (357). The later results were further supported and extended to establish that Ca²⁺ also induces mouse TRPA1 current inactivation that is relieved by membrane depolarization (564). Taken together, these findings led to a model whereby extracellular Ca²⁺ permeating through the channel induce potentiation and inactivation by binding to a site within or very close to the pore (564).

This model was confirmed by the observation of TRPA1 activation by increase in intracellular Ca²⁺ concentration through the patch pipette (213, 958) or via ultraviolet (UV)-triggered uncaging (876) in whole-cell recordings and by applying Ca²⁺ directly on the cytoplasmic side of the membrane in the inside-out configuration (213, 876, 958). Furthermore, the characterization of the rat TRPA1 pore mutant D918A in which Ca²⁺ permeability is greatly reduced resulted in no potentiation or inactivation by extracellular Ca²⁺ (876). The activation and inactivation processes display distinct Ca²⁺ sensitivities (787, 876), and rat TRPA1 currents are inactivated by Mg²⁺, Ba²⁺, and Ca²⁺, but potentiated only by Ba²⁺ and Ca²⁺ (876), all of which suggests that both processes are independent (596, 876).

The molecular mechanisms underlying Ca²⁺-dependent modulation of TRPA1 have been a matter of debate and require further study, especially in native conditions (596, 876). It was initially proposed that Ca²⁺ activates TRPA1 by binding directly to a putative NH₂-terminal EF-hand (213, 958), but later work reported that mutations in this region do not interfere with neither Ca²⁺-dependent processes (596, 876). Later structure-function analyses of human TRPA1 revealed a functional similarity with the rat isoform and the implication of a distal COOH-terminal region of the human TRPA1 in Ca²⁺- and voltage-dependent gating (787). Interestingly, the Ca²⁺-dependent inactivation of both channels is faster at negative potentials (787, 876), suggesting that the inactivation is accompanied by a shift of the voltage dependence of channel activation to more positive potentials. The deletion of the last 26 amino acid residues of the COOH terminus of human TRPA1 results in a nonfunctional protein, whereas deletion of the last 20 has no significant effects on Ca²⁺-induced potentiation, but decreases the rate of Ca²⁺-dependent inactivation, again another argument for potentiation-inactivation uncoupling (787). Single mutations in a negatively charged amino acid stretch located in a preceding COOH-terminal region result in channels with various and intriguing properties, including shift of the voltage dependence of channel activation to less positive potentials (E1077A and E1077K) and delayed Ca²⁺-induced potentiation in the mutants

E1073A, D1080A, D1081A, and D1082A. The possibility of this region to bind Ca^{2+} was supported by molecular dynamics simulations performed on a structural model of this stretch built from a Ca^{2+} -binding motif found in the human Ca^{2+} - and voltage-activated BK channel (hSlo1). Taken together, these findings indicate that the conserved acidic motif in the COOH terminus of TRPA1 is directly implicated in the regulation of this channel by Ca^{2+} (787). However, it seems clear that much further work is necessary to determine the precise role of this region in the interplay of the effects of chemical agonists, membrane potential and permeating Ca^{2+} in the gating mechanisms of TRPA1. Intriguingly, another study reported direct interaction of TRPA1 and calmodulin that were only observed in the presence of Ca^{2+} (or Ba^{2+} , but not Mg^{2+}) (304). This interaction was shown to be mediated by the C-lobe of calmodulin and a noncanonical calmodulin-binding domain formed by 17 amino acid residues in the COOH terminus of TRPA1 and to be required for Ca^{2+} -dependent potentiation and inactivation of the channel.

Ca^{2+} is thus one of the most important endogenous modulators of TRPA1, producing activation secondary to signaling events leading to increase in cytosolic Ca^{2+} levels, amplification of initial activation by other stimuli of the channel, rapid termination of channel activation via the inactivation process, and refractoriness to subsequent channel stimulation. Furthermore, TRPA1 modulation by Ca^{2+} is a key element of the interaction of this channel with TRPV1 (332, 649) (see sect. VII). It is also worth noting that the Painless channel from *Drosophila* and the TRPA1 channels from the green anole lizard, chicken, and rat snake are also modulated by Ca^{2+} (435, 436, 753).

E. Modulation by Metals

The gating promiscuity of TRPA1 is further illustrated by the stimulatory effects of several metals, such as Zn^{2+} , Cd^{2+} , and Cu^{2+} . Zinc, an essential biological trace element known to modulate the function of multiple proteins (902), activates mouse and human TRPA1. A series of permeability and structure-function experiments yielded as most plausible mechanism of Zn^{2+} -induced activation one in which Zn^{2+} permeating through the pore interacts with cysteine residues in the NH_2 and COOH termini and a histidine residue the COOH terminus (67, 333). The EC_{50} for activation of hTRPA1 by intracellular Zn^{2+} in inside-out patches is ~ 50 nM (333). This value was shown to be even lower for mouse TRPA1 heterologously expressed in CHO cells (7.5 nM) in a study that also reported that this channel and its human isoform can be activated by extracellular application of the $\text{Cu}^{2+}/\text{Zn}^{2+}$ ionophores clioquinol (an antifungal and amoebicidal drug) and zinc pyritione (34). The in vivo relevance for these findings is supported by the findings that *Trpa1* KO mice display reduced nocifensive responses to hindpaw injections of zinc acetate

(333) and do not show the clioquinol-induced reduction in the threshold to paw pressure or the shortened withdrawal latency to cold (10°C) (34), which were otherwise clearly observed in wild-type animals. Furthermore, intratracheal instillation of ZnCl_2 reduces the respiratory rate in mice, an effect that is absent in *Trpa1* KO animals (289). It was suggested that activation of TRPA1 by Zn^{2+} induces secretion of cholecystokinin and gastric emptying in rats (568). On the other hand, TRPA1 is not involved in TRPV1 inhibition by intracellular Zn^{2+} (489), nor in Zn^{2+} -induced toxicity in HEK cells, or A549 (adenocarcinomic human alveolar basal epithelial) cells endogenously expressing this channel (770).

Zn^{2+} , Cd^{2+} , and Cu^{2+} stimulate mouse pulmonary sensory neurons via a TRPA1-dependent mechanism (289). Cd^{2+} activates human (333, 531) and mouse (531) TRPA1 channels. As for Zn^{2+} , two cysteine residues and one histidine residue are implicated in Cd^{2+} -induced activation; this ion permeates the channel and TRPA1 is required for pain behaviors induced by intraplantar Cd^{2+} injection in mice (531).

F. Modulation by pH

Intracellular alkalosis activates mouse (259) and human (208) TRPA1 channels, and these effects were proposed to mediate the channel stimulation by extracellular NH_4Cl . The mouse isoform is activated from a concentration of extracellular NH_4Cl of 1 mM, with an EC_{50} of 9.2 mM. Increasing the intracellular pH to 8.0 enhances channel opening in inside-out patches of HEK293 cells expressing mouse TRPA1 (259). The human and rat channel isoforms are less sensitive to alkalosis than the mouse one: hTRPA1 is activated only at pH 9.5 (208), whereas rTRPA1 is not activated even at such high pH values (875). These differences may be also due to the use of distinct experimental conditions. Human TRPA1 is less sensitive than rat TRPV1, which was activated at pH 7.8 (208).

An implication of NH_2 -terminal cysteine residues was consistently proposed to underlie the mechanism of alkalosis-induced activation. Intracellular alkalization does not activate the double mutant C422S/C622 of mouse TRPA1 (259), whereas two human TRPA1 mutants show either complete (C621S) or partial (C665S) ablation of activation by NH_4Cl (208). Although it was proposed that a high intracellular pH enhances the action of endogenous electrophilic agonists that are retained in cell-free membrane patches (259), the mechanisms whereby the proposed modifications of cysteine residues induce channel activation remain unknown.

The assessment of the effects of NH_4Cl on sensory neurons has yielded inconsistent results. On one hand, the increase in currents observed in a fraction of wild-type mouse DRG

neurons stimulated by 10 mM NH_4Cl was otherwise not observed in cells isolated from *Trpa1* KO animals (259). On the other hand, another study reported that the fraction of mouse DRG neurons responding in intracellular Ca^{2+} imaging experiments to 50 mM NH_4Cl was not different for cells isolated from wild-type, *Trpv1* KO, and *Trpa1* KO animals, but there was more than 50% reduction of the fraction of responsive neurons in *Trpa1* KO cells treated with the TRPV1 inhibitor BCTC (208). Altogether, and not considering the obvious differences in experimental design, it seems that TRPA1 may be required for the responses of mouse DRG neurons to NH_4Cl in the lower range of concentration.

The results of testing the implication of TRPA1 in nociceptive responses triggered by intraplantar injection of NH_4Cl in mice also gave apparently contrasting results. It was on one side shown that injection of 20 μl of 100 mM NH_4Cl induces clear nocifensive behaviors in wild-type mice, with this response virtually absent in *Trpa1* KO animals (259). On the other hand, injection of 10 μl of 375 mM NH_4Cl was reported to be the lowest concentration inducing robust nocifensive responses in wild-type mice. Furthermore, these responses were not significantly different in *Trpv1* KO, *Trpa1* KO, and *Trpa1* KO mice treated with BCTC or another TRPV1 antagonist AMG 9810 (208). Thus it was proposed that other receptors may determine the pain induced by NH_4Cl . Taken together, it is clear that much more detailed evaluation of these responses needs to be made to determine what the precise contribution of TRPA1 to NH_4Cl -induced neuronal and behavioral responses really is.

Later in vitro studies concluded that rat TRPA1 is activated by intracellular acidification and that is implicated in the responses of trigeminal sensory neurons to CO_2 (875) and weak acids such as acetic, propionic, formic, and lactic acids (874). Such findings support the idea that noxious TRPA1 agonists activate the channel via specific chemical properties, such as electrophilicity, oxidative power, and in this case, the ability to acidify the intracellular milieu, rather than through specific chemical structures (268).

The effects of extracellular protons represent another paradigm of interspecies differences regarding the modulation of TRPA1. Indeed, extracellular acidosis activates and sensitizes human TRPA1, whereas this stimulus is ineffective on the rhesus monkey isoform (184) and inhibits rat TRPA1 (875). Species differences in activation of TRPA1 by extracellular protons seem to result from differences in the amino acid sequence in the S5 and S6 transmembrane segments. Protons seem to interact with an extracellular site of human TRPA1 and not through modifications of NH_2 -terminal cysteine residues (184). These findings led to the proposal of TRPA1 as a sensor for extracellular acidosis in human sensory neurons, but a subsequent study revealed that the ex-

perimental TRPA1 antagonist A-967079 does not reduce the reported pain to a continuous intraepidermal injection of a pH 4.3 solution (722). Although it was concluded the TRPA1 can be excluded as a critical player in human acidosis-induced pain, the possibility that A-967079 does not inhibit the activation of human TRPA1 by extracellular protons has not been discarded.

G. Activation by Cold and Heat

TRPA1 was initially described as a cold-activated channel (61, 777) (FIGURE 5), but this view was immediately controversial (75), and already discussed in multiple occasions (141, 446, 450–452, 512, 775, 776, 855, 861, 891, 959).

One of the key points in the discussion about the cold sensitivity of TRPA1 is whether TRPA1 channels have intrinsic sensitivity to cold. When trying to find an answer for this question, it is even difficult to define what “to have intrinsic sensitivity” really is, because to understand the function of an ion channel in a biologically relevant context, this protein should not be separated from its native membrane environment, its endogenous regulators, or its interaction partners. A recent study showed that human TRPA1 inserted in an artificial lipid bilayer is cold-sensitive, with and without the NH_2 -terminal ankyrin repeat domains (549). It was concluded that this channel is therefore “intrinsically sensitive” to cold. However, one may suggest that perhaps human TRPA1 is cold-activated when it is expressed in artificial membranes, but actually not so in its native environment. Thus this may be an endless discussion. Nevertheless, there is fairly convincing evidence for activation of certain TRPA1 isoforms by cold in relatively intact cellular conditions. For instance, cold-induced activation of mouse TRPA1 was shown to occur in a Ca^{2+} -independent manner and at the single-channel level in cell-attached and in inside-out patches (375, 713). Thus it could be argued that this phenomenon is at least membrane-delimited and does not require diffusible mediators. Nevertheless, the effects of positive and negative feedback loops due to Ca^{2+} -dependent activation and desensitization on the response to cold cannot be neglected in intact cells (876, 958).

Recent studies indicate that the thermal modulation of TRPA1 is even more complex. Purified human TRPA1 expressed in artificial membranes is activated by both cold and heat, with minimal open probability around 22°C. Interestingly, the activation by cold and heat is modulated by redox modification. Reducing agents decrease the responses to heat and cold, whereas H_2O_2 produces a response at 22°C, decreases the response to heating to 30°C, and increases that to cooling to 15°C. Furthermore, the redox status and channel agonists modulate the responses of human TRPA1 in HEK293 cells, and warming to 36 °C under oxidative stress triggers a robust TRPA1-dependent CGRP release from mouse trachea (548). These results contrast with the

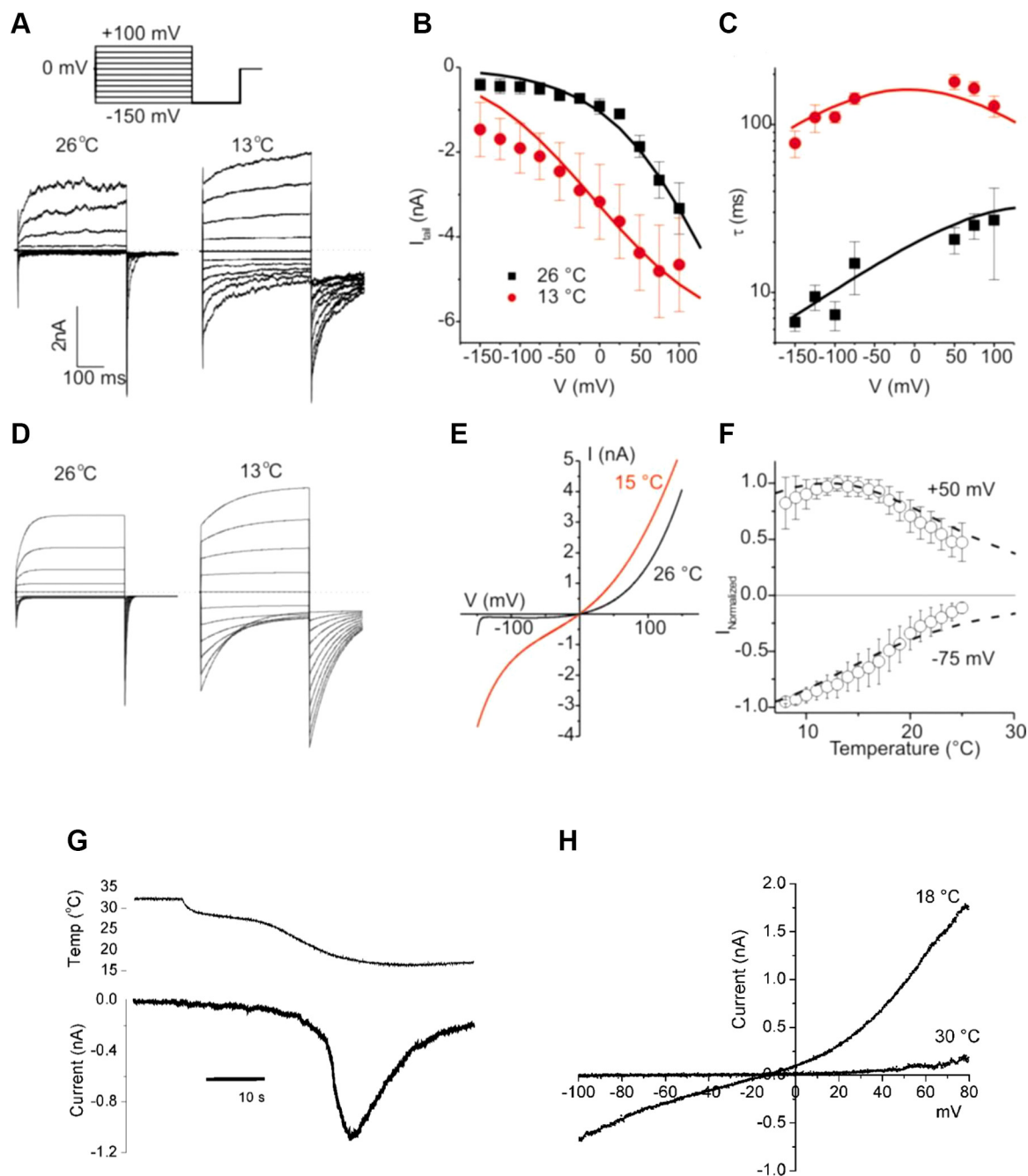


FIGURE 5. Activation of TRPA1 by cold. *A*: effects of cooling on the voltage-dependent gating and kinetics of TRPA1. Whole-cell currents in Ca^{2+} -free intracellular and extracellular solutions in response to the indicated voltage step protocol applied at 26 and 13°C. *B*: average peak inward tail currents at -150 mV at 26 and 13°C. *C*: average time constants obtained from monoexponential fits to the time course of current relaxation at different voltages and temperatures. *D*: model predictions of TRPA1 currents at 26 and 13°C in response to the voltage step protocol in *A*. *E*: model predictions of TRPA1 currents during 400-ms voltage ramps. *F*: average TRPA1 currents at different temperatures and at -75 and +50 mV, normalized to the maximal current in the tested temperature range. *G*: representative traces of inward current evoked by cooling in TRPA1-expressing CHO cells. *H*: current (I)-voltage (V) relationship of TRPA1 at 30 and 18°C. [Modified with permission from Karashima et al. (375) and Story et al. (777), with permission from Elsevier.]

conclusion of a previous report that warm temperatures suppress rat and human TRPA1 activity (873). Intriguingly, warming accelerates the activation of the channel induced by AITC, suggesting that heating favors the chemically-induced activation as previously suggested (375), but a con-

comitant more dramatic increase in the rate of channel desensitization may result in lower maximal responses to the agonist. Taken together, these findings highlight the complexity of the thermal modulation of TRPA1, demonstrate that environmental factors strongly shape the responses of

this channel to thermal stimuli, and more importantly, clearly illustrate that much further research is required to fully understand the basic properties of TRPA1. Interestingly, TRPA1 may play a role in heat nociception (320, 847), possibly by contributing to the definition of the threshold for heat-induced responses in nociceptors (320). The mechanism underlying this role may be related to the ability of the channel to sense oxidative stress produced by heat (847).

We find several factors that may help to reconcile the different views on the role of TRPA1 in cold sensing. First, it seems clear that there are interspecies differences regarding the mammalian TRPA1 sensitivity to cold (151, 451). Second, the experimental conditions may vary significantly across different studies. For instance, HEK293 cells have reduced tolerance to transient transfection with mouse TRPA1, displaying intracellular Ca^{2+} overload, probably due to constitutive channel activation, and consequently less ability to further respond to cold and any other stimulus (375). There is also no homogeneity in the age of animals used to isolate sensory neurons (75, 319, 357, 375), and different laboratories may produce primary cultures of these cells in relatively different ways, using or not growth factors and complements, which could make a strong impact on the function of this heavily modulated channel (57, 548). Third, the experimental protocols may not always be optimal for detecting cold-induced activation of TRPA1. For instance, if cold is applied during stimulation with a strong agonist such as AITC, the open probability of the channel is not expected to increase much further, and only the negative effect of cold on the open channel conductance will be apparent (375). Moreover, cooling protocols vary significantly from one study to the other, in terms of starting temperature and speed of temperature change. It is expected that these factors strongly affect the responses of a channel whose activation receives both positive and negative feedback from Ca^{2+} as permeating ion (876). In addition, to explore the sensitivity of an ion channel in a physiologically meaningful way, the responses to thermal stimuli should be determined starting from the physiological temperature. Fourth, the mixed pharmacology of TRPA1 with TRPM8, exemplified by the lack of specificity of menthol for the later channel, has probably resulted in the underestimation of TRPA1-mediated responses to cold (371). Fifth, some negative conclusions about TRPA1 activation by cold have been made from quite a long range, by looking at very complex in vivo scenarios, sometimes wrongly assuming full specificities of TRPA1 agonists such as cinnamaldehyde and AITC (20, 26, 124, 241, 269, 616, 677, 890). Sixth, because of its diverse regulation, TRPA1 is not expected to contribute equally to cold sensitivity in all expression systems, pathophysiological settings, and animal models. As examples, different animal strains may behave distinctly to intense cold [the jumping phenotype observed in C57BL/6J mice seems not to be present in other strains (375)]; the

implication of TRPA1 (and TRPM8) in cold allodynia was inferred to differ between cold injury and neuropathic pain patients (574); cold hypersensitivity induced by menthol, but not by icilin or lysophospholipids, was dependent on expression of TRPA1 (270); and the relative contribution of this channel compared with that of TRPM8 varies across different sensory neurons (244). Doubtlessly, there are multiple examples of TRPM8 as the main/sole determinant in the responses to cold (189, 385, 401, 402, 512, 635, 644, 825, 843, 896, 948), but this does not mean that the cold sensitivity of TRPA1 is not relevant in other scenarios.

From a mechanistic point of view, activation of mouse TRPA1 by cold (375) and *Drosophila* TRPA1 by heat (866) is associated with a negative shift of the activation curve to more negative potentials (FIGURE 5), an effect reminiscent of that documented for other thermosensitive TRP channels, such as TRPM8, TRPV1, TRPM4, and TRPM5 (803, 859).

The role of TRPA1 as a cold sensor has been discussed in detail (858). There is strong genetic and pharmacological evidence that activation of TRPA1 by cold translates to cellular, tissue, and in vivo behavioral responses, especially in pathological conditions (49, 70, 104, 163, 179, 198, 219, 247, 318, 325, 355, 375, 378, 397, 427, 428, 445, 532, 533, 579, 610, 646, 680, 825, 831, 853, 919, 941, 956). However, many of these responses may not be necessarily fully mediated by direct activation of TRPA1 by cold, but indirectly through the production of a stimulatory/sensitizing factor (70, 325, 517, 532, 533, 853, 896). Evidence for functional TRPA1 expression and responsiveness to cold has been also shown in human dental pulp fibroblasts (235), human odontoblasts (236), and adenocarcinomic human alveolar basal epithelial A549 cells (784, 785). TRPA1 also contributes to cold-induced contractions in the isolated rat colon preparations (217) and is the major mediator of cold-evoked responses in mouse and rat visceral sensory neurons (244, 326, 577).

TRPA1 isoforms from fly (298, 367, 857), mosquito (298), frog (703), lizard (5, 435, 437, 703), chicken (436, 702), snake (281, 436, 639), and the fishes medaka (Japanese rice fish) (613) and takifugu (pufferfish) (611) are activated by heat.

H. Mechano-activation

The study of the role of TRPA1 in mechanosensation was sparked by the proposal and immediate discussion of its implication in mechanosensitivity of inner-ear hair cells (174, 175, 276, 564), by the potential molecular spring properties of the NH_2 -terminal ankyrin repeats (46, 330, 759), and by the similarities between this channel and the hair cell transducer (265, 564). TRPA1 was, however, shown not to be essential for mechanosensing in hair cells

(75, 445, 776, 905). To the best of our knowledge, it has not been reported that recombinant mammalian TRPA1 channels can be activated by classical mechanical stimulation, i.e., membrane indentation, shear stress, or membrane stretch (see also Ref. 959). On the other hand, there are multiple reports on the role of TRPA1 in the detection of mechanical stimuli in native cells, including sensory neurons (380), human peridontal ligament cells (836), Merkel cells (760), and odontoblasts (734). This suggests that TRPA1 is mechanosensitive only in its native environment, which is after all not surprising.

The TRPA1 inhibitor HC-030031 strongly reduces the mechanically-evoked action potential firing in rat and wild-type mouse C fibers, particularly at high-intensity forces. TRPA1 inhibition does not influence the mechanical responsiveness of A δ fiber nociceptors (380). Interestingly, TRPA1 is expressed in thin-caliber axons and intraepidermal nerve endings, as well as in large-caliber axons, lanceolate and Meissner endings, and epidermal and hair follicle keratinocytes. The firing rates of action potential generated in C-fiber nociceptors in response to a wide range of forces are much lower in *Trpa1*-deficient mice than in wild-type animals. A δ mechanonociceptors of *Trpa1*-deficient mice also display reduced firing, but only to large forces. Thus TRPA1 seems to mediate mechanotransduction via a cell-autonomous mechanism in nociceptor terminals. It is also possible that TRPA1 function in keratinocytes contributes to skin mechanosensation (447). Other studies showed that slowly-adapting mechanically activated currents recorded in mouse DRG neurons are inhibited by HC-030031 (856), that intermediately-adapting mechanically activated currents in neurites of mouse DRG neurons are reduced by genetic or pharmacological ablation of TRPA1, whereas rapidly- and slowly-adapting currents are unaltered (127). There is evidence for mechanosensory function of TRPA1 in *C. elegans*, where this channel is activated by application of negative, but not positive, pressure through the recording patch pipette in the whole-cell configuration (393).

The activation of TRPA1 by hypertonic solution (HTS), but not hypotonic solutions, could be also taken as argument for mechanosensitivity of this channel (939). Interestingly, other reports are consistent with TRPA1 being activated also by hypotonic solutions in odontoblasts (234, 835) and in the heterologous expression system HEK293T cells (260).

There is evidence for activation of TRPA1 by another type of mechanical stimulation, that arising from the alteration on membrane structure by insertion of exogenous molecules. For instance, TRPA1, but not TRPV1, TRPV4, or TRPM8, is activated by trinitrophenol, a negatively charged amphiphilic compound that produces crenation of the plasma membrane upon its preferential insertion in the outer leaflet of the bilayer (315). On the other hand, chlor-

promazine, which is positively charged and inserts preferentially in the inner bilayer leaflet producing membrane cupping, has no effect on TRPA1 channels with low basal activity. However, this compound induces a large increase in inward currents previously stimulated with AITC. This effect is accompanied by a complex modulation of the outward currents that seem to be consistent with an increase in the rates of channel activation and inactivation at positive potentials. In addition, GsMTx-4, a tarantula toxin that is thought to inhibit mechanosensitive channels by inserting in the outer membrane leaflet, also activates TRPA1 (315). Furthermore, primary and secondary alcohols activate human TRPA1 in a carbon chain length-dependent manner, and with a strength that highly correlated with the molecule lipophilicity (416). This suggests that nonelectrophilic TRPA1 agonists may act via a mechanism that involves the detection of physical alterations in the plasma membrane. This idea is further supported by the close relation between the abilities of bacterial lipopolysaccharides to activate TRPA1 and to produce mechanical perturbations in the plasma membrane (523, 764, 765).

Taken together, the hypothesis of intrinsic mechanosensitivity of mammalian TRPA1 remains to be further investigated, but it seems clear that this channel is implicated in mechanosensation in physiological and pathological conditions (see sect. VII).

I. Activation by Light

Human TRPA1 is activated by near UV light in a wavelength-dependent and membrane-delimited manner, through a mechanism that implicates the production of ROS. The sensitivity of TRPA1 to light is enhanced by the photosensitizing agents acridine orange and hypericin. These findings suggest that TRPA1 activation underlies the pain and burning sensations triggered by photodynamic therapy (316).

UV light evoked a current in human epidermal melanocytes that was inhibited by ruthenium red, camphor, and HC-030031 and that was ~90% reduced by TRPA1-targeted miRNA that decreased channel expression in 85%. TRPA1 inhibition decreased melanin synthesis as well. The mechanism underlying TRPA1 activation is not mediated by ROS production, but downstream of an opsin-mediated G protein signaling cascade involving PLC activation (82). Further studies revealed that UV-induced activation of TRPA1 results in melanocyte membrane depolarization, which was proposed to reduce the photoactivated current inactivation, thereby prolonging the Ca²⁺ entry required for melanin production (84). Whether the cell depolarization is fully mediated by TRPA1 or also by other channels is still unresolved (142). Also obscure remains the actual contribution of the complex interplay between Ca²⁺- and voltage-dependent gating of TRPA1 to the regulation of Ca²⁺ signaling

and membrane potential in these cells. A more detailed study of the transduction signaling pathway indicates that UV light activates $G\alpha_{q11}/PLC\beta$ signaling, leading to the hydrolysis of phosphatidylinositol 4,5-bisphosphate (PIP_2) and the consequent generation of diacylglycerol (DAG) and inositol 1,4,5-trisphosphate (IP_3). In turn, on one hand PIP_2 depletion seems to enhance TRPA1-mediated photocurrents via a relief of a tonic inhibition of channel activation, and on the other, IP_3 stimulates Ca^{2+} release from intracellular stores. Hence, melanocytes were proposed to feature a UV transduction mechanism resembling the phototransduction cascades of the eye (83).

A behavioral assay in zebrafish embryos used to screen molecules that could modify the startle reflex to light led to the identification of optovin as a reversible photoactivated agonist of TRPA1. Further experiments suggested that optovin activates human TRPA1, not via the generation of single oxygen, but partly through direct covalent modification of cysteine residues. Interestingly, treatment with optovin allowed light-mediated control of the motor activity of paralyzed extremities in spinalized zebrafishes and elicited nocifensive behaviors in mice (415). These findings have research applications in the identification of functional expression of this channel in cardiomyocytes (488) and cortex (382) and may have potential for clinical applications as a strategy for precise spatiotemporal control of endogenous TRPA1 activation (243, 415).

The clinical relevance of the implication of TRPA1 in light-induced responses was further supported by the elucidation of the molecular transduction mechanism underlying cutaneous porphyria, a condition in which patients suffer from burning pain upon exposure to sunlight or from the effects of photodynamic therapy. Ultraviolet and blue light generates singlet oxygen that in turn acts as TRPA1 and TRPV1 agonist (56). Furthermore, human TRPA1 is activated and photosensitized by 7-dehydrocholesterol (7-DHC), a precursor of cholesterol and vitamin D_3 that is found in very high plasma levels of patients suffering from the autosomal Smith-Lemli-Opitz syndrome. TRPA1 and TRPV1 mediate responses of mouse DRG to acute application of 7-DHC in the dark- and light-induced responses in the presence of 7-DHC. Illumination with 405-nm light induces release of CGRP from isolated mouse trachea, an effect that is enhanced by preapplication of 7-DHC. The latter effect is absent in preparations from double *Trpa1/Trpv1* KO animals. Finally, application of 405-nm light causes an increase in firing of mouse cutaneous C-fibers and habituates a response to 7-DHC, which is on its own ineffective. These findings led to the proposal of a mechanism whereby TRPA1 and TRPV1 are activated by ROS, RNS, and/or RCS generated from Fe^{3+} -catalyzed conversion of oxysterols that are produced by peroxidation of 7-DHC by singlet oxygen resulting from photochemical reactions involving endogenous chromophores (54). In contrast, TRPV1,

but not TRPA1, appears to be implicated in the action potential firing of mouse DRG neurons triggered by protoporphyrin IX phototoxicity produced by 630-nm light (900).

The role of TRPA1 in the responses to light has received support from studies in *Drosophila*, which have shown that this channel is required for cell-autonomous light transduction in class IV dendritic arborization neurons and avoidance to light in larvae (907), for the sensitivity of neuroendocrine cells to UV light (267, 293), for rapid light-dependent feeding deterrence (220), and for blind females to avoid laying eggs under UV light (292). Furthermore, a TRPA1 homologue is required for the extraocular photophobic response of planaria to near UV light (105).

Interestingly, activation of TRPA1 and TRPV1 seems to mediate X-ray-induced mechanical and heat allodynia in mice (177).

J. Modulation by PIP_2 and Polyphosphates

PIP_2 is an acidic phospholipid of the inner leaflet of the plasma membrane comprising ~1% of plasma membrane phospholipids. Most if not all TRP channels are capable to interact or are indirectly modulated by PIP_2 . Although PIP_2 effects on TRPA1 remain debated, it seems clear that it is involved in channel activation and desensitization (373, 389). A typical feature of TRPA1 is its rapid desensitization following activation by agonists such as AITC, CA, and high intracellular Ca^{2+} concentrations. TRPA1 desensitization is delayed when PIP_2 is supplemented via the patch pipette, whereas the PIP_2 scavenger neomycin accelerates desensitization. Preincubation with the phosphatidylinositol 4-kinase inhibitor wortmannin reduces both constitutive TRPA1 channel activity and the response to AITC. These data indicate that PIP_2 modulates TRPA1, although to a lower extent than other TRP channels, such as TRPM4 (373).

Other reports showed that in inside-out patches, PIP_2 does not activate TRPA1 (389). When TRPA1 was electrophilic-activated, addition of PIP_2 produces a concentration-dependent inhibition of TRPA1. It seems that PIP_2 may act as an inhibitor of TRPA1, reducing the sensitivity of TRPA1 to its activators (389). Another study showed that capsaicin induces cross-desensitization of TRPA1 upon dialysis of PIP_2 through the patch pipette (9). On the other hand, other studies reported no effect of PIP_2 in excised patches (388) or inhibition in the presence of inorganic polyphosphate (389). Furthermore, rapamycin-induced PIP_2 depletion does not inhibit TRPA1, while it inhibits TRPM8 (876). Of note, one of the tools used to study PIP_2 signaling is the PLC inhibitor U73122, but this compound was shown to activate directly TRPA1 via covalent modification, and to induce release of CGRP from mouse skin in a TRPA1-dependent manner (373, 587).

This view is further complicated by the possibility of TRPV1-TRPA1 interactions as well as by distinct experimental conditions. For instance, channel modulation by PIP₂ under chronic inflammation, usually lasting many days, could be different from previously described after acute alterations within minutes. Inflammation results in a long-lasting (chronic) PIP₂ depletion. Chronic PIP₂ production can be stimulated by overexpression of phosphatidylinositol-4-phosphate-5-kinase, and the PIP₂-specific phospholipid 5'-phosphatase can reduce plasma membrane levels of PIP₂. It was proposed that the responses of TRPA1 to agonists are not significantly influenced by chronic changes in PIP₂ (648). However, if TRPA1 and TRPV1 are present, chronic PIP₂ reduction leads to pronounced tachyphylaxis of both channels. Thus the chronic effect of PIP₂ on TRPA1 activity depends on presence of the TRPV1 (648).

One of the problems to study in detail effects of phosphoinositides, such as PIP₂, is the difficulty to work for long time on inside-out patches. TRPA1 seems to require an unidentified cytosolic factor whose action can be mimicked by inorganic polyphosphates. Multiple intracellular molecules are able to rescue activation of TRPA1 by covalent modification in inside-out patches. These compounds include polyphosphates [e.g., pyrophosphate (PPi)] and polytriphosphates (PPPi). Application of polyphosphates to inside-out patches not only stabilizes TRPA1 activity but also increases the single-channel conductance (596). Structure-function analysis of polyphosphates indicates that at least four phosphate groups are needed to render their activity (388). Finally, IP₃ and inositol-hexaphosphate partially rescue TRPA1 activation. This indicates that intracellular factors are required for TRPA1 to adopt a functional channel conformation (388). It is suggested that TRPA1 can exist in different functional states: a native state (cell-attached patch) and a non-native state (excised patch). As mentioned above, THC can activate TRPA1 even in the absence of polyphosphates, whereas electrophilic pungent chemicals and Ca²⁺ require it for activation (145).

K. Modulation by Phosphorylation

Although TRPA1 is likely to be modulated by phosphorylation, this process is not well understood, and only little information is available. TRPA1 can be activated by bradykinin, and *Trpa1* KO animals show impaired responses to bradykinin as well as to noxious mechanical stimuli (75). Bradykinin activates and sensitizes sensory nerves through mechanism involving G protein-coupled receptor-mediated PLC, PLA, cyclooxygenase, and 12-lipoxygenase signaling pathways. These signaling pathways subsequently interact with downstream ion channels, including TRPA1, thereby changing the excitability of sensory nerves (12, 39, 870).

Recently, possible PKA phosphorylation sites were identified in the sequence of hTRPA1 (516). The channel mutant

S1101A displays increased current density, whereas the mutant S804A shows a reduction of the time to peak in response to agonist. Another mutation, S227A, results in increased PKA-mediated sensitization. Furthermore, four mutations at positions S86, S317, S428, and S972 were shown to be possible phosphorylation sites for PKA, contributing to significant suppression of TRPA1 sensitization. These residues are highly conserved between different species, including *Drosophila*, zebrafish, mouse, rat, rhesus macaque, and human (516).

The mouse TRPA1 amino acid residue S87 is a target of PKA-mediated phosphorylation. Furthermore, phosphorylation of the TRPA1 residues S119, T281, and T529 by PKC is essential for the normal sensitization of the channel. During both PKA- and PKC-mediated phosphorylation of TRPA1, the scaffolding protein A-kinase anchoring protein 79/150 (AKAP) plays an essential role in anchoring these kinases to TRPA1 (123).

TRPA1 was also recently shown to have putative phosphorylation cyclin-dependent kinase 5 (Cdk5) sites within its ankyrin repeats, which could potentially modify channel opening or conductance (782). Cdk5 phosphorylation is blocked by TFP5, a Cdk5 inhibitory peptide. These sites are highly conserved in mammals, mosquitos, and fruit flies.

L. Modulation by Protein-Protein Interactions

The interactions of TRP channels with diverse modulatory proteins represent an increasingly important research topic (see <http://trpchannel.org/>). So far, the best studied examples of protein-protein interactions in this field are those between TRPV1 and the ubiquitin hydrolase CYCL (see above), the PKA anchor protein AKAP5, and secretogranin 3, a member of the chromogranin/secretogranin family of neuroendocrine secretory proteins (see also <http://trpchannel.org/>).

Some features of neuronal TRPA1 are not present in heterologous expression systems, but can be restored when TRPA1 and TRPV1 channels are coexpressed (8, 705). Furthermore, coexpression of these channels results in unique activation profiles that can be distinct from those of cells expressing only TRPA1 or TRPV1 (697). TRPV1 and TRPA1 function together, and resiniferatoxin-mediated "neurosurgery" removes both sensor molecules. In adult mice, resiniferatoxin causes desensitization to heat and sensitization to cold (653), and experiments on facial skin capsaicin injection to rats suggest that TRPV1 signaling in TG neurons sensitizes TRPA1 and thereby induces cold hyperalgesia (325). Approximately 30–50% of TRPV1-expressing small- to medium-sized peripheral sensory neurons coexpress TRPA1, and almost all TRPA1-positive neurons coexpress TRPV1 (404, 777). Ca²⁺-triggered activation of

TRPA1 is attenuated by TRPV1 in the presence of extracellular Ca^{2+} , but not in Ca^{2+} -free conditions. TRPV1 mutations at residue Y671 affect probably TRPA1 permeation properties, but the mutations in TRPV1 did not affect association of the TRPA1 and TRPV1 channels. The TRPV1 mutation Y671K alters the magnitude of currents through TRPA1, the sensitivity to extracellular $[\text{Ca}^{2+}]$, and the voltage dependency (649). Some TRPV1-selective cannabinoids such as WIN 55,212 are effective inhibitors of TRPA1. The synthetic cannabinoid arachidonol-2 chloroethanolamine (ACEA) activates TRPV1, but inhibits TRPA1 probably via a TRPV1-dependent mechanism. Some cannabinoids mediate their peripheral analgesic properties, at least in part, via the TRPV1 and TRPA1 channels (693). Conversely, knock-down of TRPA1 in sensory neurons by specific small interfering RNA abolishes the WIN effect on TRPV1 dephosphorylation, suggesting that WIN acts through TRPA1 (354). TRPA1 stimulation enhances TRPV1 phosphorylation via the putative PKA phosphorylation site S116, and this cross-sensitization contributes to higher pain sensitivity in inflamed tissues (761).

Transmembrane protein 100 (Tmem100) is a potential modulator of TRP channels and is often coexpressed and forms complexes with TRPA1 and TRPV1 in DRG neurons. Tmem100-deficient mice display a reduction in the inflammatory mechanical hyperalgesia and TRPA1- but not TRPV1-mediated pain (888, 892). Tmem100 can selectively potentiate TRPA1 activity by weakening the TRPA1-TRPV1 association (888). In contrast, the Tmem100-3Q mutant enhances the association of TRPA1 and TRPV1 and therefore inhibits TRPA1. Notably, a cell-permeable peptide (CPP) containing the COOH-terminal sequence of Tmem100-3Q inhibits persistent pain, suggesting the possibility of targeting TRPA1-TRPV1 complexes for pain management (888, 892). Interestingly, TRPV1-TRPA1 interactions may explain intriguing clinical findings such as the association between a functional knockdown of TRPV1 with hypersensitivity to garlic (642).

Analysis of the effects of icilin on TRPA1 and sensitivity of mice to cold stimuli, which is inhibited by blockers of iPLA₂ (BEL, bromoenol lactone), suggests for a possible interaction of TRPA1 with this lipase (270).

Prokineticins PK1 and PK2, and their G protein-coupled receptors prokineticin receptor 1 (PKR1) and prokineticin receptor 2 (PKR2), play important roles in several biological processes such as gut motility, neurogenesis, angiogenesis, circadian rhythms, hematopoiesis, and nociception. *Pkr* KO animals display impaired PK1-, PK2-mediated hyperalgesia that partially depend on TRPA1 activity (588). Another link between pain and TRPA1 comes from studies on the protease activated receptor 2 (PAR2) (180, 345, 586, 931).

Annexin A2 (AnxA2) is highly expressed in the sensory neurons expressing TRPA1. Membrane TRPA1 levels and TRPA1-mediated Ca^{2+} responses are enhanced in a AnxA2-deficient mouse. In vivo experiments showing reduced TRPA1-dependent acute and inflammatory pain in AnxA2-deficient mice led to the conclusion that coexpression with AnxA2 in a subpopulation of sensory neurons may impair TRPA1-dependent nociceptive signaling in vertebrates (52).

Although direct interactions have not been demonstrated, pharmacological evidence suggests functional interaction between T-type Ca^{2+} channels and TRPA1 in sensory neurons. Activation of the former results in Ca^{2+} entry that may activate and desensitize TRPA1, thereby modulating nociceptive signaling (600).

Notably, a direct binding of TRPA1 NH₂-terminal ankyrin repeats to the COOH-terminal proline-rich motif of the fibroblast growth factor receptor 2 (FGFR2) activates this receptor, resulting in lung adenocarcinoma progression and metastasis. In turn, TRPA1 is depleted by the transfer of TRPA1-targeting exosomal microRNA (miRNA-142-3p) from brain astrocytes to cancer cells (94).

M. TRPA1 Antagonism

The huge importance of TRPA1 in pain, inflammation, and many other potential indications in acquired diseases triggered an increasing demand for antagonists (410, 673, 674). Examples of these antagonists are listed in **TABLE 4**.

TRPA1 is inhibited by the natural compounds camphor (derived from *Cinnamomum camphora*) (22, 39, 63, 678, 713, 914) and the related molecule borneol (731). The former compound, however, induces a rebound effect upon washout that is typical for bimodal TRPA1 modulators (22). A series of 19 analogues (1b-5) of racemic [6]-gingerol (1a) in which three pharmacophoric regions were altered resulted in the identification of some full TRPA1 antagonists (552). Lutein is a natural tetraterpene xanthophyll, and 1 of 600 known naturally occurring carotenoids that incorporates into membranes. The methylated- β -cyclodextrin (RAMEB) complex of this compound reduces TRPA1 activation by electrophilic agonists (329).

Human TRPA1 currents stimulated by AITC, menthol, flufenamic acid, or octanol are inhibited by 1,8-cineole (eucalyptol), a cyclic ether monoterpenoid found in eucalyptus oil. This action, together with a stimulatory effect on TRPM8, may mediate the analgesic and anti-inflammatory effects on this compound in humans. Interestingly, 1,4-cineole activates, rather than inhibits, hTRPA1 (795). A subsequent study on monoterpene analogues showed that borneol, 2-methylisoborneol, and fenchyl alcohol, but not norcamphor, inhibit hTRPA1 activation by menthol (796). The

Table 4. Synthetic TRPA1 inhibitors

Name	Company	IC ₅₀	References	Comments
HC-030031	Hydra Biosciences (USA)	≤1.8–20 μM	W02007073505 (2007) (232, 491, 513)	Nonelectrophilic; effective in human, rat, mouse, guinea pig; ineffective on frog and zebrafish channels (294)
Hydra 7	Hydra Biosciences (USA)	≤10 μM	W02009002933 (2008)	
Chembridge-5861528	Hydra Biosciences (USA)	14–18 μM	(750)	
CB-625	Cubist Pharmaceuticals/ Hydra Biosciences	ND		Phase I clinical trial completed. Discontinued due to solubility concerns (673)
Glenmark 10	Glenmark Pharmaceuticals (Switzerland)	50–100 nM	US2009325987 (2009)	
Glenmark 15	Glenmark Pharmaceuticals (Switzerland)	<50 nM	US2009325987 (2009)	
Glenmark 37	Glenmark Pharmaceuticals (Switzerland)	<50 nM	US2009325987 (2009)	
Glenmark 17	Glenmark Pharmaceuticals (Switzerland)	<250 nM	W02009118596 (2009)	
Glenmark 23	Glenmark Pharmaceuticals (Switzerland)	0.5–1 μM	W02009118596 (2009)	
Glenmark 8	Glenmark Pharmaceuticals (Switzerland)	<500 nM	W02009144548 (2009)	
Glenmark 39	Glenmark Pharmaceuticals (Switzerland)	<500 nM	W02009144548 (2009)	
GRC-17536	Glenmark Pharmaceuticals (Switzerland)	<10 nM	(562)	Phase IIa clinical trial (NCT01726413); diabetic peripheral neuropathy/respiratory disorders
2-{1,3-Dimethyl-2,4-dioxo-1,2,3,4-tetrahydro-pyrrolo[3,2- <i>d</i>]pyrimidin-5-yl}- <i>N</i> -(4-trifluoromethyl-phenyl)-acetamide (3h)	University of Ferrara	400 nM	(69)	
<i>N,N</i> -bis-(2-hydroxybenzyl)-2,5-diamino-2,5-dimethylhexane	IRM LLC, A Delaware Limited Liability Company (Bermuda)	ND	W02007098252 (2007)	
Tramadol	Grünenthal	0.1–10 μM	(537)	Formerly known as Tramal; first launched and marketed by Grünenthal in 1977
AMD_09	University of Florence, Italy	10.3–13.2 μM	(290)	
AMD_12	University of Florence, Italy	7.3–8.2 μM	(290)	
AP-18	IRM LLC, A Delaware Limited Liability Company (Bermuda)	3.1 μM	W02007098252 (2007) (656)	
A-967079	Abbott Laboratories	67–290 nM	W02009089082 (2009) (150)	
Renovis 11	Renovis (a wholly owned subsidiary of Evotec)	2.7 μM	(197)	
AZ456	AstraZeneca	30–305 nM	W02012050512 (2012) (609, 844)	
AMG7160	Amgen	51 nM	(400)	
AMG2504	Amgen	35 nM	(400)	
AMG9090	Amgen	21 nM	(400)	

Continued

Table 4.—Continued

Name	Company	IC ₅₀	References	Comments
AMG5445	Amgen	91 nM	(400)	
AMG0902	Amgen	IC ₉₀ = 300 nM	(461)	
CMP1	Abbott Laboratories	2 μ M	(153)	
CMP2	Abbott Laboratories	1.4 μ M	(153)	
CMP3	Abbott Laboratories	1.1 μ M	(153)	
2B10	Amgen	90–260 nM	(456)	(Monoclonal antibody)
SZV-1287	University of Pécs, Hungary	2.4 μ M	(652)	
JNJ-41477670	Janssen Pharmaceuticals	7.2 nM	(101)	
Thiadiazole derivatives	Pfizer	0.05–0.93 μ M	(832)	

ND, not determined.

menthol analogue 4-isopropylcyclohexanol inhibits TRPA1, but also the Ca²⁺-activated Cl[−] channel ANO1, TRPV1, TRPV4, and TRPM8 (798).

Resolvins are anti-inflammatory and pro-resolving lipid molecules that can be endogenously produced from omega-3 polyunsaturated fatty acids such as eicosapentaenoic acid and docosahexaenoic acid. Resolvin D1 (RvD1) potently inhibits cinnamaldehyde-induced TRPA1 responses at submicromolar levels. RvD1 alleviates TRPA1-mediated acute nociception in mice triggered by intradermal hindpaw injection of cinnamaldehyde or formalin. RvD1 also strongly inhibits TRPV3 and TRPV4 (64). A subsequent study revealed that RvD2 is a potent inhibitor of TRPV1 and TRPA1 in primary sensory neurons. On the other hand, RvE1 selectively inhibits TRPV1, and RvD1 is selective for TRPA1. The very interesting roles of resolvins and related lipid mediators in inflammation and pain were recently reviewed (473). Of note, hindpaw inflammation experiments in male Wistar rats showed that the antinociceptive potency of RvD1 and TRPA1 blockers is weakened by either perturbation of the balance of endogenous pro- and anti-nociceptive mechanisms in inflammation or via TRPA1-opioid receptors interactions (615).

The natural stilbenoids resveratrol and pinosylvin inhibit TRPA1 activation by AITC and decrease AITC-induced paw inflammation and production of the proinflammatory cytokine interleukin-6 (IL-6) in mice (542, 570, 930). A more recent study showed that from a series of 20 stilbenoids, none modulates TRPV1 and most of them have stronger action than resveratrol, with maximal potency observed with ortho monooxygenated stilbenes 6 and 17 (572). mTRPA1 is inhibited by tannic acid (612) and by the extract of the medicinal plant *Pterodon pubescens Benth*, containing a mixture of nine sesquiterpenes and seven diterpenes (605).

Several potent TRPA1 inhibitors are already available from Hydra Biosciences, Abbot, AMGEN, Glenmark, The

Scripps Research Institute, and Renovis. The first synthetic antagonists appeared in 2007 and were all xanthine alkaloid based, e.g., the Hydra compound HC-030031, which is the most frequently used TRPA1 antagonist. The similarity of this compound to caffeine is intriguing. Other structurally similar compounds were released by Hydra (Hydra 7, Chembridge 5861528), Glenmark (nanomolar blockers such as Glenmark 10, 15, 37). Phthalimide derivatives (high nanomolar, Glenmark 17, 34) and imidazo-ourine derivatives (Glenmark 8, 39) were released by Glenmark. All these inhibitors are nonelectrophilic. Other nonelectrophilic antagonists have been developed from diamino-dimethylhexanes. Other reported TRPA1 antagonists are the analgesics tramadol and its metabolite M1 (537), a novel series of α -aryl pyrrolidine sulfonamides (850), and the paracetamol (acetaminophen) analog 6a/b (256). The antipyretic effect of paracetamol, however, is not related to its action of TRPA1 that lead to hypothermia, but to prostaglandin inhibition in the brain (529). ADM_12, a novel compound formed by lipoic and homotaurine residues covalently linked, also antagonizes TRPA1 (585). The later reverts oxaliplatin-induced neuropathy in rats, without displaying toxicity on astrocytes or cardiotoxicity (585). The treatment of rats with the Japanese herbal medicine Gosha-jinkigan improved oxaliplatin-induced hypersensitivity to menthol, AITC, and cold, associated with suppression of mRNA TRPM8 and TRPA1 overexpression in DRG neurons (377, 540). (+)-Borneol, a bicyclic monoterpene found in the essential oil of plants used for analgesia and anesthesia in traditional Chinese medicine, induces antihyperalgesic effects in a mouse model of oxaliplatin-induced neuropathic pain, associated with the block of TRPA1 (946). A high-throughput screening of quinazolinone-based compounds that optimized antagonistic potency and increasing polarity yielded a purinone (AM-0902). This compound has pharmacokinetic properties allowing for >30-fold coverage of the rat TRPA1 IC₅₀ in vivo and produces dose-dependent inhibition of AITC-induced flinching in rats (716).

Among electrophilic compounds, the oxime AP18 was disclosed as an antagonist of TRPA1. AP18-related oximes have both agonist and antagonist activity (197). Another oxime, A967079, is a potent nanomolar blocker of human TRPA1 that attenuates cold allodynia produced by nerve injury, but does not alter noxious cold sensation or body temperature (150). Interestingly, this compound does not inhibit frog TRPA1, which led to the identification of two amino acid residues located within the putative fifth transmembrane domain as determinants of the inhibitory action of A967079 on mammalian TRPA1 (571). More recently, 3-(4,5-diphenyl-1,3-oxazol-2-yl)propanol oxime (SZV-1287) was shown to inhibit both TRPA1 and TRPV1 and TRPA1-dependent CGRP release from the peripheral sensory nerve endings (652). Other electrophilic antagonist, such as Abbott A; Renovis 11; Amgen AMG7160, 2504, 9090, 5445; and the Abbott CMP1, 2, 3, operate in the nanomolar range (678).

Pyrazolone and its derivatives dipyrone, propyphenazone, and antipyrine inhibit Ca^{2+} responses and currents in TRPA1-expressing cells and mouse acute nocifensive responses induced by TRPA1 agonists. Dipyrone and propyphenazone decrease TRPA1-mediated nociception and mechanical allodynia in models of inflammatory and neuropathic pain (91, 578). Of note, not dipyrone itself, but its metabolites 4-*N*-methylaminoantipyrine and 4-aminoantipyrine, activate and sensitize TRPA1 and TRPV1 in a redox-dependent manner (715).

Pha1 β , a peptide from the venom of the armed spider *Phonutria nigriventer*, and its recombinant analogue CTK 01512-2, inhibit activation of TRPA1 by AITC. This action is likely to explain how these peptides attenuate acute pain and mechanical and cold hyperalgesia elicited by AITC and the TRPA1-dependent neuropathic pain induced by the chemotherapeutic agent bortezomib (821, 869).

The quest for TRPA1 blockers is steadily increasing (see sect. VII, D–H).

VII. PATHOPHYSIOLOGY OF TRPA1

A. TRPA1-Related Channelopathies

So far, the only described TRPA1 channelopathy is a rare autosomal dominant Familial Episodic Pain Syndrome (FEPS) (428). FEPS appears in infancy and is characterized by episodes of incapacitating upper body pain that can occasionally radiate to the abdomen and legs. These episodes are usually triggered by fasting or physical stresses such as cold and can last up to 90 min. FEPS patients show hypersensitivity to AITC and present enhanced cutaneous flare responses and secondary hyperalgesia to punctate stimuli (428). Sequencing of candidate genes served to identify a gain-of-function missense mutation (N855S) in the S4 of TRPA1 as probable cause of the disease. This mutant channel displays a more than fivefold larger inward current at normal resting potentials compared with the wild-type channel (428) (FIGURE 6).

The N855 residue is involved in the Ca^{2+} -dependent activation, and the TM4-TM5 linker contributes to agonist- and voltage-dependent activation and regulates the gating of the channel in a state-dependent manner via a Ca^{2+} -sensitive mechanism (428, 954). Structural modeling and patch-clamp analysis of the N855S mutant revealed intersubunit salt bridges between residues E854 in TM4 and K868 in TM5 that stabilize the open state of the channel (954). Specific TRPA1 antagonists inhibit the abnormal in vitro response of the mutant channel, suggesting that they could be used as therapy for this syndrome (428, 898). Of note, mutation of N855 to an arginine residue (N855R) was reported to enhance the response to cinnamaldehyde and the rate of inactivation, without affecting the voltage dependence of channel activation (122).

Single nucleotide polymorphisms (SNPs) that may be important for somatosensory abnormalities in neuropathic pain patients have been discovered in a search for the con-

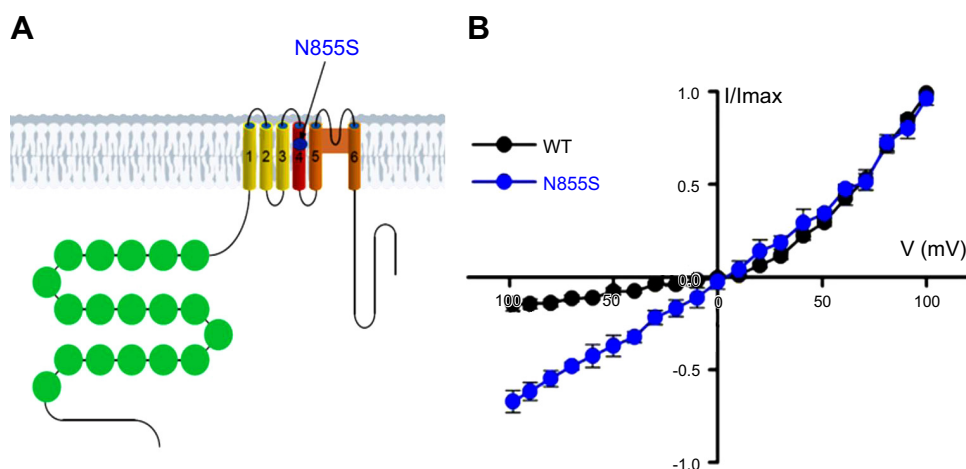


FIGURE 6. Structure and activity of human TRPA1 and N855S mutant underlying Familial Episodic Pain Syndrome (FEPS). *A*: schematic representation of the TRPA1 channel displaying mutation defined in the FEPS that is located in putative transmembrane segment S4. *B*: average current (I/I_{max}) versus voltage (V) relationship of hTRPA1-WT and hTRPA1-N855S in the presence of 100 μM cinnamaldehyde. Currents are normalized to +100 mV. [Modified from Kremer et al. (428).]

tribution of genetic variants. A mutation, E179K, in the TRPA1 NH₂ terminus was associated with the presence of paradoxical heat sensation (510).

A study using intracellular Ca²⁺ imaging analyzed the functional properties of four SNP variants of human TRPA1 (Y69C, A366D, E477K, and D573A) in comparison with gain-of-function channels bearing the SNPs R797T and N855S (553). Only Y69C displayed enhanced sensitivity to agonists, similar to R797T, but to a lower extent than N855S. The other nonconservative substitutions showed poor responses, which could be rescued by pretreatment with the Src family inhibitor PP2 or Zn²⁺. The TRPA1 variants and several experimental mutants (TRPA1 Y97F, Y226F, and YY654–655FF) expressed poorly in neuroblastoma SH-SY5Y cells compared with HEK293 cells.

Another study aimed to determine the influence of human SNPs on the responses of TRPA1 to known chemical agonists and components of diesel exhaust particles and insoluble coal fly ash (CFA) particles. The variants R3C and R58T, present in the NH₂ terminus displayed gain-of-function, with enhanced responses to these stimuli (196). Several variants such as E179K and K186N located in the ankyrin repeat domain-4 and at the predicted N-linked glycosylation site residues N747A and N753A showed reduced response to CFA (196). These findings suggest roles for ankyrin domain-4 and cell surface N-linked glycans in the mechanism underlying the activation of TRPA1 by insoluble particles. Importantly, the polymorphisms R3C and R58T correlated with reduced asthma control in children, suggesting a link between TRPA1 and the severity of asthma in the context of high air pollution (196).

The TRPA1 agonist menthol is often used in commercial nicotine-containing products to attenuate the sensation of burn associated with nicotine (384, 801). Preference for smoking menthol cigarettes is variable between individuals and populations, suggesting that differences in the *Trpa1* gene could contribute to the preference of mentholated products. A common *Trpa1* haplotype defined by 1 missense and 10 intronic SNPs is associated with preference for mentholated cigarettes in heavy smokers (838).

Future studies should be directed at understanding the structural bases of the perturbed functional properties of TRPA1 variants associated with human disease. Likewise, a more comprehensive characterization of the effects of channel modulators should be performed for these variants. Altogether, this may serve to design therapeutic strategies aimed at correcting TRPA1 channelopathies.

B. TRPA1 in Pain and Inflammation

The sensation of pain results from somatosensory stimuli generating a cascade of adaptive responses in the body.

Feeling pain is essential for our survival as it provides a warning signal. This sensation involves a complex interaction between specialized nerves, the spinal cord, and the brain. The detection of painful mechanical, thermal, or chemical stimuli is attributed to the activation of nociceptors present in the primary afferent nerve fibers of the somatosensory system (358, 854). Experiments in the skin indicate that chemical activation of TRPA1 produces pain, heat sensation, and mechanical hyperalgesia, cold hyperalgesia, and a neurogenic axon reflex erythema (258, 541, 575). A sustained activation of TRPA1 by endogenous agonists (408) implicates this channel in persistent and chronic pain in a wide variety of conditions, such as neuropathy, inflammation, osteoarthritis, migraine, diabetes, fibromyalgia, bronchitis, and emphysema (581, 855, 959) and therefore may be an excellent target for novel analgesic and anti-inflammatory molecules (39, 203, 266).

1. TRPA1 in neuropathic pain

Peripheral sensory neuropathy is a neurological deficit that may result in decreased sensation of the peripheral nervous system (338). Patients with this condition present symptoms of pain as well as decreased or loss of touch, vibration, and thermal sensation (337). A loss of protective sensation puts patients at risk of undetected injury. The management of neuropathy-related pain symptoms is challenging, as it often involves strong opioid-containing drugs that could lead to dependence and addiction.

TRPA1 mediates pain evoked by mechanical stimuli in peripheral neuropathy, and block of this channel reduces mechanical hypersensitivity induced by peripheral diabetic neuropathy (409, 879, 880) or spinal nerve injury (232, 881). In the diabetic and spared nerve injury (SNI) models of neuropathy a selective TRPA1 inhibitor, Chembridge-5861528 (CHEM) fails to induce conditioned place-preference (CPP) (882). In another study, systemically administered morphine and pregabalin reduced mechanical hyperalgesia and the spontaneous discharge rate of the presumed pain-relay neurons of diabetic animals, without inducing CPP (694). This indicates that ongoing pain, as revealed by CPP, is less sensitive to treatment by the TRPA1 channel antagonist than mechanical hypersensitivity in peripheral neuropathy (882). It was latter shown that prolonged peri-injury block of spinal TRPA1 with CHEM attenuates maintenance, but not development, of mechanical allodynia following nerve injury (883).

More evidence of TRPA1 function in the development of neuropathic pain was obtained from a rat model of a spinal nerve ligation in which administration of diluted bee venom to the spinal nerves reduced the expression of a few TRP channels, including TRPV1, TRPA1, TRPM8, and c-Fos in the ipsilateral DRG. This treatment also decreased the development of the mechanical, thermal, and cold allodynia (406). Interestingly, the expression levels of TRPA1 mRNA

in intact nerves adjacent to the injured nerves are upregulated, indicating their possible contribution to development of neuropathic pain (682). Artemin, a neurotrophic factor derived from glial cells, inhibits the TRPA1 activity and pain behaviors upon short-term application (929), but in long-term application increases the expression of TRPV1 and TRPA1 in cultured DRG neurons (341). Peripheral nerve injuries induce pain and hypersensitivity that can be attenuated by blocking TRPA1 channels (232, 610). Recently, it was described that microinjections of antioxidants or selective TRPA1 antagonists into the amygdala decrease pain and mechanical hypersensitivity in nerve-injured animals (698). Along the same line, the painlike behaviors elicited by a constriction of the infraorbital nerve mouse model are fully mediated by TRPA1, via oxidative stress by-products released from monocytes and macrophages recruited at the injury site (826). Notably, a subsequent study on mice with partial sciatic nerve ligation reported that TRPA1 channels expressed in Schwann cells, and not those in nociceptors, are the ones activated by the NADPH oxidase 2 (NOX2)-dependent oxidative burst produced by macrophages recruited to the perineural space. In turn, activation of TRPA1 in Schwann cells stimulates the NOX1 pathway resulting in the release of H_2O_2 that ultimately activates TRPA1 in nociceptors (187). Notably, TRPA1 activation in these cells by the ethanol metabolite acetaldehyde initiates a NOX1-dependent production of H_2O_2 and 4-hydroxynonenal, which in turn may induce allodynia via activation of this channel in nociceptors. These findings suggest that TRPA1 in Schwann cells mediates some symptoms of ethanol-related pain (185).

The sensitivity of TRPA1 to ROS was proposed to mediate also the key role of this channel in a mouse model of neuropathic pain induced by sciatic nerve chronic constriction (662). This model features a supporting involvement of the sympathetic nervous system via α -adrenoceptors that leads to hypersensitivity to mechanical stimulation, cold, and local injection of AITC or norepinephrine. Similarly, H_2O_2 -induced activation of TRPA1 was proposed to mediate the inflammatory response in an acute gout attack rodent model (827).

The vasoactive peptide endothelin-1 has been suggested as a target for pain treatment, as it is implicated in several pain conditions related to inflammation, cancer, diabetic neuropathy, and sickle cell disease (749). TRPA1 inhibition reduces the painlike behaviors induced by intraplantar injections of endothelin-1 in mice, possibly via a PLC-dependent and PKC-independent pathway. Furthermore, endothelin-1 enhances the pain induced by cinnamaldehyde (469), which could result from a sensitization of TRPA1 via the endothelin A receptor and protein kinase A pathway (942), and/or from the its stimulatory effect on tetrodotoxin-resistant voltage-gated Na^+ channels in DRG neurons

(950). TRPA1 seems to be also implicated in the spontaneous pain behavior elicited by acid-induced oral ulcer in the rat, a model in which endothelin-1 signaling plays a key role (601).

Anticancer treatments with bortezomib, oxaliplatin, cisplatin, and paclitaxel elicit severe cold and mechanical allodynia via oxidative stress-dependent TRPA1 activation (526, 579, 828). Blocking TRPA1 attenuates cold and mechanical allodynia induced by anti-cancer drugs, such as paclitaxel or oxaliplatin (155, 579, 585). Oxaliplatin and cisplatin increase TRPA1 mRNA expression in rodent TG and DRG neurons (163, 526, 790, 817, 919). Recent studies postulate that the acute hypersensitivity to cold induced by oxaliplatin is mediated by human TRPA1 sensitization to ROS via mechanisms that, depending on the dose, are mediated by inhibition of propyl hydroxylases (533, 567) or by oxidation of cysteine residues (532). Pharmacological evidence indicates that TRPA1 is implicated in abnormal local cold-induced vascular responses observed during oxaliplatin-induced peripheral neuropathy (636). It must be noted, however, that these compounds act on several other ion channels controlling neuronal excitation and cold sensing: TREK1, TRAAK and HCN (204), TRPM8 (402, 538), and $Na_v1.6$ (205). Treatment with oxaliplatin enhances the responses of adult rat DRG neurons to icilin, but not to the TRPM8 agonist WS-12, suggesting sensitization of TRPA1 (28). Recently, oxaliplatin was shown to induce an intracellular acidification in DRG neurons (683). As described above, changes in the cytosolic pH are linked to a sensitization of TRPA1 channels, suggesting another mechanism of oxaliplatin induced neurotoxicity.

Paclitaxel-evoked mechanical allodynia is partially reduced by the TRPA1 antagonist HC-030031 and the TRPV4 antagonist HC-067047 and fully attenuated by the combination of both inhibitors (507). Paclitaxel treatment induces release of CGRP and acts via oxygen radical formation, that in turn target TRPA1 and TRPV4. Whereas TRPA1 and TRPV4 are needed for the delayed development of mechanical allodynia, the cold allodynia is exclusively dependent on TRPA1 activity (507). Treatment of cultured DRG neurons with paclitaxel at low concentration of paclitaxel (10 nM) increases CGRP release, but at a 30-fold higher concentration (300 nM) it induces the opposite effect (665).

TRPA1 is also a key mediator of the proinflammatory/proalgesic effects of exemestane, letrozole, and anastrozole, three aromatase inhibitors used in breast cancer therapy (262), and of the aromatase substrate androstenedione (188). TRPA1 is activated also by two other compounds with chemotherapeutic potential: the lipoxygenase inhibitor and antioxidant nordihydroguaiaretic acid (NDGA) and its derivative terameprocol. When injected in mice, none of these compounds induces spontaneous pain behaviors, but enhance the responses to evap-

orative cooling induced by topical application of acetone (679). Vinca alkaloids, anti-mitotic and anti-microtubule compounds derived from the periwinkle plant *Catharanthus roseus*, also induce TRPA1 activation, thereby inducing an immediate pain syndrome in fruitflies and mice (114).

The evidence for the roles of TRP channels in chemotherapy-induced peripheral pain has been recently reviewed (586) (**FIGURE 7**).

2. TRPA1 and inflammation

TRPA1 is a main player in the fast onset and maintenance of inflammation. Many endogenous TRPA1 agonists are produced during inflammation, e.g., as the result of lipid oxidation, peroxide formation, oxidative stress (e.g., oxygen ions, free radicals, and peroxides), unsaturated aldehydes (e.g., HNE), electrophilic prostaglandins, and RNS (279, 829) (**FIGURE 8**).

TRPA1 is linked to occurrence of the inflammatory symptoms of rosacea (48). Along the same line, a recent study revealed the implication of TRPA1 in the immunomodula-

tory effects of cinnamaldehyde in an in vivo mouse model of LPS-induced systemic inflammatory response syndrome (519).

Furthermore, TRPA1 activation in epidermal keratinocytes enhances the expression of pro-inflammatory cytokines known to be key contributors to skin inflammation (47). By triggering the release of CGRP, TRPA1 activation causes vasodilation, a key symptom in inflamed tissue (432). Inflammation is also mediated by TRPA1 via increase in the levels of tumor necrosis factor (TNF)- α , a key player in joint inflammation. Blockade of TRPA1 may be therefore beneficial in reducing chronic arthritic pain (248).

In the context of tissue wound healing in a mouse cornea alkali burn injury model, it was found that lack of TRPA1 in cultured ocular fibroblasts attenuates the expression of transforming growth factor β 1, IL-6, and α -smooth muscle actin. TRPA1 ablation results in reduced inflammation and fibrosis/scarring in the corneal stroma during wound healing (626). At the level of the skin, it was recently shown that HC-030031 has antinociceptive and anti-inflammatory effects after thermal injury (183, 287).

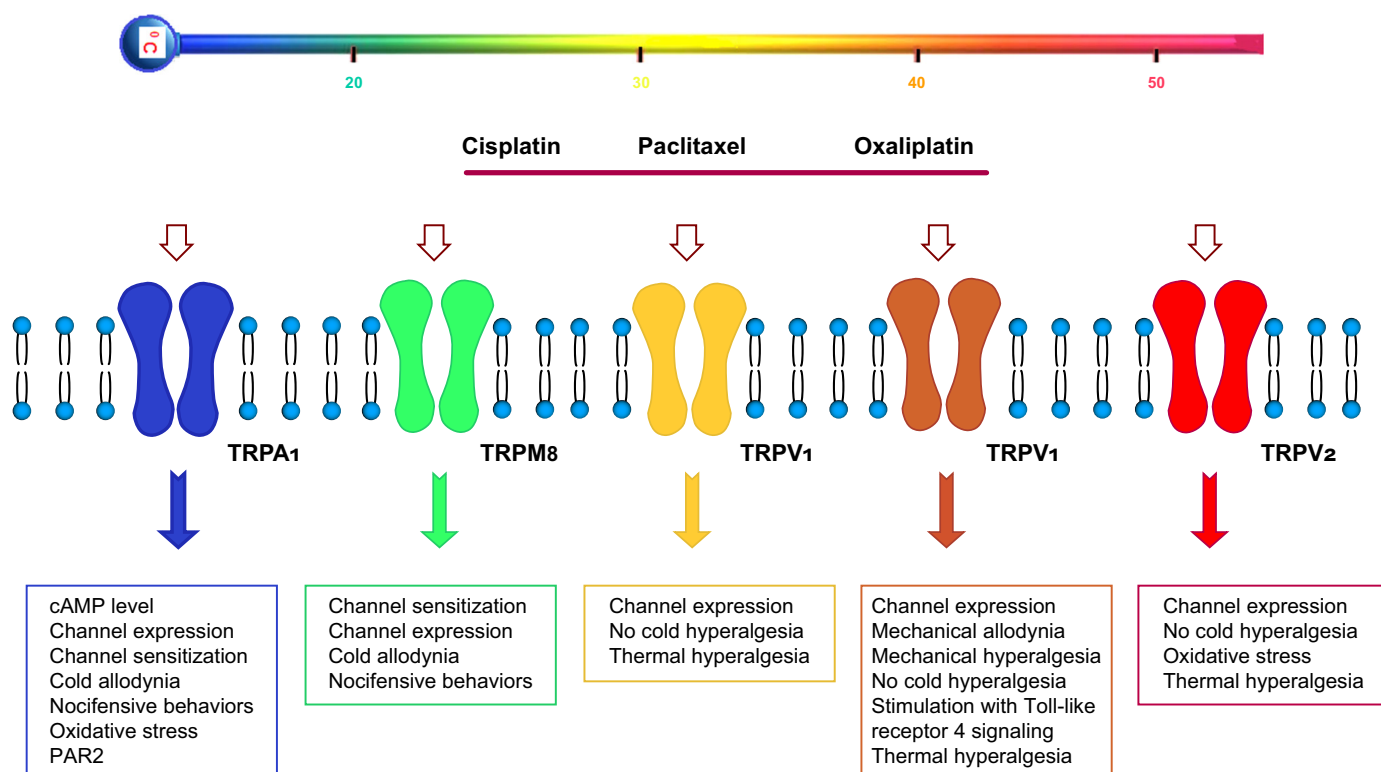


FIGURE 7. Summary of chemotherapeutic agents effects such as cisplatin, oxaliplatin, and paclitaxel on TRP channels (TRPA1, TRPM8, TRPV1, TRPV2, and TRPV4) in the dorsal root ganglion (DRG) neurons. Activation of the TRP channels by chemotherapeutics can induce changes in the channel expression levels, nociceptive behaviors, oxidative stress, mechanical, heat and cold hypersensitivity. Moreover, channel levels are influenced by activation of some secondary molecular mechanisms such as glutathione [GSH] or proteinase-activated receptor 2 [PAR2]. The proinflammatory agents trypsin and mast cell tryptase cleave PAR2, inducing its activation. PAR2 is expressed in sensory nerves and functionally coupled with TRPA1 channel. PAR2 activation was shown to increase the TRPA1 currents in native DRG neurons. [Modified from Naziroğlu and Braidy (586).]

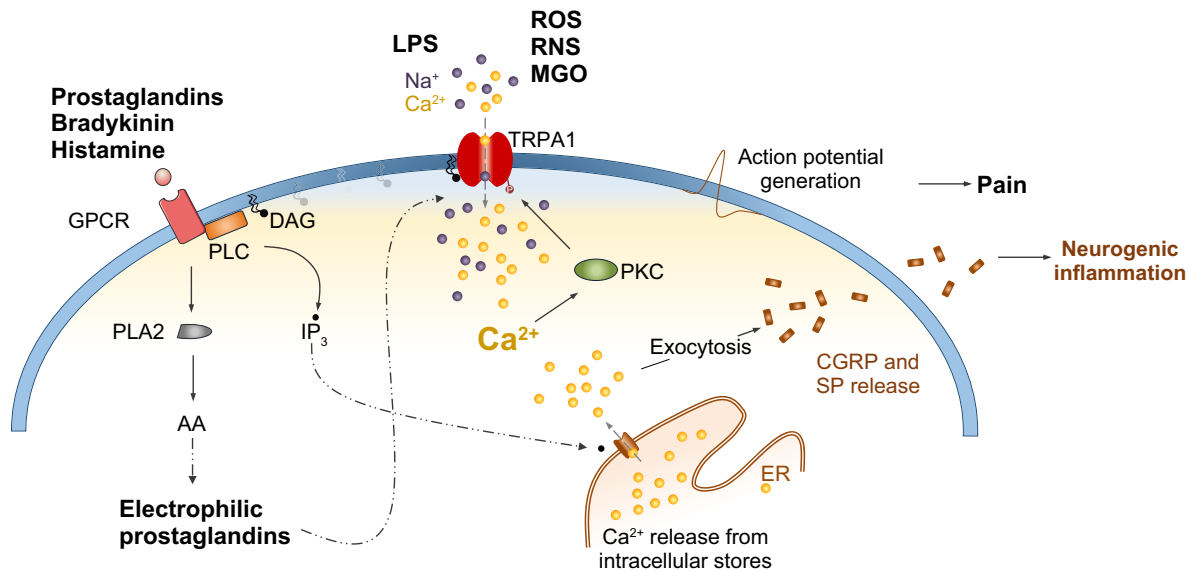


FIGURE 8. Activation and sensitization of TRPA1 by inflammatory mediators. Activation of the G protein-coupled receptor (GPCR) stimulates phospholipase C (PLC), with the ensuing formation of inositol 1,4,5-trisphosphate (IP_3) and diacylglycerol (DAG). IP_3 further activates the IP_3 receptor (IP_3R), inducing the release of Ca^{2+} from the endoplasmic reticulum (ER). The hydrolysis of phosphatidylinositol 4,5-bisphosphate (PIP_2) releases the inhibitory effect over TRPA1. DAG and Ca^{2+} -mediated protein kinase C (PKC) activation sensitizes TRPA1. In parallel, GPCRs activate phospholipase A_2 (PLA_2) leading to the downstream formation of arachidonic acid (AA) metabolites, which are endogenous activators of TRPA1. The rise in intracellular Ca^{2+} triggers the release of inflammatory mediators (i.e., CGRP). In the inflammatory context, TRPA1 is directly activated by reactive oxygen (ROS) and nitrogen (RNS) species, methylglyoxal (MGO), or lipopolysaccharide (LPS) derived from invading gram-negative bacteria. TRPA1 activation leads to the generation of action potentials in sensory neurons, triggering acute pain sensation.

TRPA1 is expressed on astrocytes in the central nervous system (457, 735, 736, 823, 849), where it is involved in the spontaneous peptide hormone exocytosis (800) and its ablation decreases mature oligodendrocyte apoptosis and thereby demyelination and behavioral alterations and morphological changes induced by cuprizone (115, 700). TRPA1 activation may regulate mitogen-activated protein kinase pathways, the transcription factor c-Jun, and a proapoptotic Bcl-2 family member (Bak). Inhibition of TRPA1 may therefore be a strategy to treat multiple sclerosis and to limit demyelination and consequent damage of the central nervous system (700). Furthermore, TRPA1 inhibition decreases myelin damage during ischemia, suggesting this channel as a therapeutic target in white matter ischemia (299). TRPA1 seems to be implicated in neuronal development and oligodendrocyte maturation, thereby regulating emotion, cognition, learning, memory, and social behavior (455). Interestingly, TRPA1 mediates the stimulatory effect of oligomeric forms of the amyloid-beta peptide on intracellular Ca^{2+} in mouse astrocytes and the resulting synaptic dysfunction, which may suggest a role of this channel in Alzheimer's disease (121).

Psoriasisiform dermatitis induced by imiquimod, an immune regulator with antitumoral activity, is increased by ablation of TRPA1 in mice. Although this compound was reported to induce Ca^{2+} influx in TRPA1-expressing cell lines and the channel was found to be expressed in CD4^+ T helper

cells (379), the mechanism underlying the protective role of TRPA1 in this model of psoriasiform dermatitis remains unclear.

3. TRPA1 in osteoarthritis

Osteoarthritis is a common disorder characterized by the progressive loss of articular cartilage and remodeling of the underlying bone. Typically, small joints of the hands and feet, the hips, and the knees are the most affected by the disease. Osteoarthritis induces damage of articular cartilage, formation of new bone in the subchondral region, and formation of new bone and cartilage at the joint margins (154, 504). The chronic low-grade joint inflammation in osteoarthritis highly contributes to disease development and progression (500).

Cartilage degradation is caused by changes in the production of catabolic, anabolic, and inflammatory mediators within the joint, induced by increased expression of matrix-degrading metalloproteinases (MMPs) and proinflammatory mediators such as IL-6 and PGE_2 (370). Monosodium iodoacetate (MIA)-induced arthritis is commonly used as a model of osteoarthritis. MIA-induced inflammation and degenerative cartilage changes and joint pain are decreased in TRPA1-deficient animals (511, 543). TRPA1 is functionally expressed in primary human osteoarthritis chondrocytes and in the human T/C28a2 chondrocyte cell line. This

channel also mediates the production of osteoarthritis-related factors such as MMP-1, MMP-3, MMP-13, IL-6, and PGE₂ as evidenced by pharmacological inhibition and genetic depletion of TRPA1 (608). Pharmacological block of TRPA1 reduces mechanical hypersensitivity in nociceptive neurons of the spinal dorsal horn (511), whereas pain sensation is sustained (629). This suggests a role of TRPA1 in mechanical hypersensitivity rather than in ongoing pain in osteoarthritic animals.

4. TRPA1 in migraine

Migraine is a neurovascular disease characterized by episodic attacks of predominantly unilateral headache, often associated with nausea, vomiting and hypersensitivity to light, sound, and odors. Trigeminovascular system activation, followed by dural neurogenic inflammation and sensitization, is linked to development of migraine attacks. Several mediators, including CGRP and substance P, are important factors in migraine-associated symptoms (201).

TRPA1 and TRPV1 play crucial roles in the migraine pathophysiology (88, 201, 581). The implication of TRPA1 is strongly supported by its sensitivity to exogenous and endogenous agents that are known to trigger migraine and by the reduction of the activity of this channel by antimigraine agents (via inhibition or desensitization) (89, 90). In the migraine context, TRPA1 stimulation may occur at the level of dural afferent fibers (335). This may be via direct chemical stimulation, as reported for umbellulone (231), or via local oxidative, nitrative, carbonylic and electrophilic stress (88, 421, 728). In some cases, TRPA1 was shown to be associated with the release of vasodilating CGRP.

It was recently proposed that migraine is a homeostatic protective response to brain oxidative stress that is mainly detected by TRPA1 (120). It must be noted that the contribution of the modulation of TRPA1 by H₂S and NO to headache generation is not straightforward. Spinal TG neurons are activated by HNO derived from these chemicals, whereas activation of meningeal afferents by HNO may produce reduced spinal TG activity, due to a rise in the electrical threshold caused by TRPA1 activation in afferent fibers (810). Stimulation of TRPA1 leading to migraine may also occur peripherally, through the effect of environmental irritants, such as inhaled acrolein, which acutely increase meningeal blood flow via the trigeminovascular system (433, 434). Of note, inhaled acrolein acts on TRPA1 in the periphery producing lipidic imbalance in trigeminal tissue and enhanced levels of modulators of TRPV1, TRPV3, and TRPV4 (462). The TRPA1-mediated signaling in meningeal afferents was proposed to be increased by obesity (502). Interestingly, the antimigraine effect of parthenolide, a major component of feverfew, is related to partial agonism and desensitization of TRPA1 (506). Similar findings were reported for the major component of butterbur (*Petasites hybridus*), isopetasin (89). Recent studies on the therapeutic

action of extracranially applied botulinum neurotoxins on migraine indicate that their action may be mediated by de-functionalization of TRPA1- and TRPV1-expressing meningeal nociceptors (938). Furthermore, pharmacological evidence obtained with the use of umbellulone and AITC show that TRPA1 activity and CGRP are crucial for the regulation of cortical spreading depression, an underlying cause of migraine aura (356). Migraines are also increased in patients who have experienced infantile colic (689). As TRPA1 is the last from sensory TRPs to be expressed during development (319), it could be speculated that human TRPA1 expression coincides temporally with the emergence of colic symptoms.

5. TRPA1 in oro/facial/dental pain

Odontalgia (toothache) is one of the most common types of pain induced by inflammation of the dental pulp (pulpitis). One of the main features of dental pulp inflammation is hypersensitivity to cold. The oro-dental system is innervated by trigeminal branches expressing multiple receptors that are involved in the detection and transduction of noxious and non-noxious thermal stimuli, including TRPM8 and TRPA1 channels. TRPA1 plays a crucial function in the oro-dental system and increase in the channel expression has been linked with hyperalgesia and allodynia following tooth injury (295). In vivo and in vitro administration of nerve growth factor (NGF) increases TRPA1 mRNA and TRPA1 protein expression in a concentration- and time-dependent fashion in TG neurons (212). These findings indicate that elevated TRPA1 expression induces cold hyperalgesia and bradykinin-induced acute thermal hyperalgesia in the orofacial region.

Enhanced expression of TRPA1 and TRPV1 channels in TG neurons was also described in a model of orofacial pain based on skin incisions on the oral mucosa or the whisker pad skin. This again leads to mechanical allodynia and hyperalgesia to cold (840).

Pain associated with stimulation of a sensitive tooth involves the mechanisms of mechanotransduction. The majority of pulpal afferents express acid-sensing ion channel 3 (ASIC3) and TRPA1, suggesting these channels as targets for the treatment of dentin sensitivity (310).

Odontoblasts are cells of the outermost cellular layer of the dental pulp. These cells have as main function the dentinogenesis, but are also implicated in tooth pain (494). Odontoblasts have cytoplasmic extensions throughout the dentin and locate within the dentinal tubules. They lie in close proximity to sensory unmyelinated nerve fibers that project from the dental pulp to the inner half of the dentin and may sense and transduce external stimuli (136). Odontoblasts express several classes of ion channels, such as L-type Ca²⁺ channels, mechanosensitive K⁺ channels, and voltage-gated Na⁺ channels (18, 495). Human odontoblasts also highly

express TRPA1, TRPV1, and TRPM8, which play crucial roles in the detection of chemical and thermal stimuli. TRPV1 and TRPA1 colocalize with peptidergic sensory neurons expressing substance P and CGRP, indicating the role of TRPA1 and TRPV1 in neurogenic inflammation (235, 236).

Pain and allodynia related to mechanical trauma in the oral mucosa are linked to prostanoid- and PAR2-dependent activation of TRP channels. In oral mucositis, spontaneous pain occurs due to TRPV1 and TRPA1 activation and mechanical allodynia through TRPV4 activity, independently of bacterial infection (345). Moreover, inhibition of TRPA1 alleviated mechanical allodynia associated with the action of bacterial toxins in oral ulcerative mucositis induced by systemic administration of the chemotherapeutic drug 5-fluorouracil (918).

Some tooth-whitening (bleaching) treatments employ hydrogen peroxide at high concentrations (up to 35%) to oxidize chromogens in the dentin. Other whitening systems use carbamide (urea) peroxide. Patients undergoing bleaching procedures often complain of painful sensations that are referred to as “bleaching sensitivity” (BS) (311, 463). Bleaching products quickly diffuse into the dental tissues reaching and activating TRPA1 channels in the intradental nerves, causing BS pain (503). Post-bleaching sensitivity is also related to the morphological changes that presumably enhance the tissue permeability, resulting in transient sensitivity after the whitening procedure.

In the search for novel pharmaceutical TRPA1 analgesics compound, AZ465, was identified as a reducer of CGRP release from human dental pulp (609). Administration of the TRPA1 inhibitor HC-030031 attenuated mechanical allodynia and cold hyperalgesia in a model of orofacial pain based on skin incisions on the oral mucosa or the whisker pad skin (840). HC-030031 also decreases spontaneous guarding pain behavior elicited by skin and deep tissue incision in mice (781), a phenomenon that seems to be related to the strong response of muscle tissue to H_2O_2 (780).

6. TRPA1 in bacterial infections

Bacterial infections, as well as injections of the complete form of the Freund's adjuvant mycobacteria mixture (CFA), are well known to induce acute pain, mechanical allodynia, tissue swelling, and chronic inflammation (248). Numerous reports demonstrate that the CFA-induced inflammatory pain is attenuated by administration of TRPA1 inhibitors and in *Trpa1* KO mice (328, 511, 656).

Recent studies have shown that LPS activates sensory TRP channels (117), including several TRPA1 isoforms: mouse, human (523), and *Drosophila melanogaster* (754). LPS activates TRPA1 in mouse sensory neurons in a TLR4-independent manner, and genetic ablation of this channel re-

duces mouse pain and inflammatory responses triggered by LPS (523). In fruitflies, TRPA1 expressed in bitter-sensing neurons mediates gustatory avoidance to LPS (754). Of note, LPS also activates TRPV4 channels in airway epithelial cells (21), and with lower potency TRPV1, TRPM3, and TRPM8 (118). The mechanism of activation of TRPA1 by LPS seems to be related to the ability of this molecule to induce mechanical perturbations in the plasma membrane (765). Moreover, it was demonstrated that LPSs with diverse conformations of lipid A, such as LPS from *E. coli* or *S. minnesota*, have different potencies in inducing changes in the membrane order that relate to their ability to activate TRPA1 (765).

The pathophysiological mechanisms underlying low back pain and enhanced pain in diabetes was linked with low-grade bacterially induced inflammation (13). Furthermore, patients with chronic low back pain triggered by lumbar disk herniation display elevated plasma levels of methylglyoxal (480), an agonist of TRPA1 expressed in human mechano-insensitive C fibers (223), which suggests this channel as a potential treatment target in these patients.

Interestingly, there seems to be a link between TRPA1 and viral infections, and the expression of this channel is up-regulated dIMR-32 neuronal cells by soluble factors released during HRV replication in the Wi-38 lung cells (1).

C. TRPA1 and Itch

Itch (or pruritus) is an unpleasant sensation provoking scratching behavior (909), leading to decreased life quality (7, 78, 478). Itch is related to eczema, psoriasis, urticaria, renal failure, cholestasis, lymphoma, and chronic liver diseases (909). The scratching reflex is associated with sensitization of nerves sensing exogenous pruritogens and endogenous chemicals produced by epithelial and immune cells.

Itch is related to TRPA1 activation, but the underlying sensation is different from pain (395, 514). Mas-related G protein-coupled receptors (Mrgpr) constitute a class of histamine-independent receptors activated by mast cell mediators and chloroquine (285, 478). MrgprA3 and MrgprC11 act upstream of TRPA1, and their expression overlaps with that of TRPA1 and TRPV1 (342, 894). Of note, MrgprD is required for cold allodynia in neuropathic pain induced by chronic constriction injury via a PKA-TRPA1 pathway, but the itching behavior is unaltered in this experimental model (865). TRPA1-deficient mice display reduced scratching behavior in response to chloroquine and BAM8-22, implicating TRPA1 in histamine-independent pruritus (894). Although BAM activates TRPA1 and TRPV1, only TRPA1 is required for BAM-evoked itch-related behaviors. BAM-induced TRPA1 activation is linked to PLC activity, but inhibition of PLC does not alter chloroquine-evoked activation of TRPA1 (894). Mice lacking PLC β respond to chloro-

quine with normal itch behavior related to TRPA1 but not to TRPV1 activation (342). It was recently reported that chloroquine-evoked scratching responses in TRPA1-deficient mice do not differ from those of wild-type (WT) animals. Instead, evidence was provided for a role of TMEM16a Ca^{2+} -activated Cl^- channels in chloroquine-induced increases in sensory neuron excitability and scratching (692).

Serotonin (5-hydroxytryptamine, 5-HT) is released by mast cells and is associated with atopic dermatitis, cholestasis, and psoriasis (752). Serotonin receptor (HTR7) stimulation leads to TRPA1 activation, triggering itch behaviors (559). Mice lacking HTR7 or TRPA1 display reduced scratching and skin lesion severity, highlighting their role in acute and chronic itch (559). HTR7-linked activation of TRPA1 requires functional adenylate cyclase (AC), and TRPA1 is sensitized by AC and cAMP in sensory neurons (559, 717). Remarkably, TRPA1 does not couple to all HTRs. HTR2 activation by α -methyl-5-HT triggers itch and pain behaviors that are independent of TRPA1 (478, 717, 894).

As mentioned above, TRPA1 inhibition reduces the pain induced by endothelin-1. However, TRPA1 inhibition exacerbates the scratching behavior of mice triggered by intradermal injection of this peptide, but the mechanism underlying this effect remains unknown (470).

In the context of cholestatic hepatitis and other obstructive liver disease (464), it was found that activation of G protein-coupled bile acids receptor 1 (TGR5) expressed in primary sensory neurons sensitizes TRPA1 via a $\text{G}\beta\gamma$ - and PKC-dependent mechanism (472). TRPA1 inhibition prevents bile acid-stimulated release of the pruritogenic neuropeptide gastrin-releasing peptide (GRP), and decreases bile ac-

id-induced scratching. Furthermore, spontaneous scratching induced by endogenous bile acids in mice overexpressing TGR5 is reduced by TRPA1 inhibition (472).

Atopic dermatitis (AD) is a chronic itch and skin inflammatory disorder characterized by intolerable and incurable itch (307, 483, 762, 952). The PAR2 plays an important role in the production of the cytokine thymic stromal lymphopoietin (TSLP), which triggers AD (545, 952). TSLP is expressed in human cutaneous epithelial cells in AD and bronchial epithelial cells in asthma (352). TRPA1 is required for TSLP-evoked sensory neuron activation, triggering robust itch-evoked scratching, involving PLC signaling (895) (**FIGURE 9**).

IL-13 is one of the critical mediators of AD (620). This Th2 cytokine enhances the growth of dermal neuropeptide-secreting afferent nerves and enhances the expression of TRPA1 in dermal sensory nerve fibers, DRG neurons, and mast cells. TRPA1 ablation decreases scratching behaviors (620). Another cytokine, IL-31, is implicated in inflammatory and lymphoma-associated itch (113, 182) and induces itch by activating IL-31RA in the skin sensory nerves expressing TRPA1 and TRPV1. IL-31-induced itch is highly reduced in TRPV1- and TRPA1-deficient mice but not in c-kit or proteinase-activated receptor 2 mice (148). Thus IL-31RA signaling links neuroimmune crosstalk between T cells and sensory nerves.

The lymphoma-associated itch is suppressed by a miRNA-711 inhibitor and a blocking peptide that disrupts miRNA-711/TRPA1 interaction. Extracellular miRNA directly activates TRPA1 channels inducing TRPA1-dependent itch, confirming miRNAs role in pruritogenesis (300, 729).

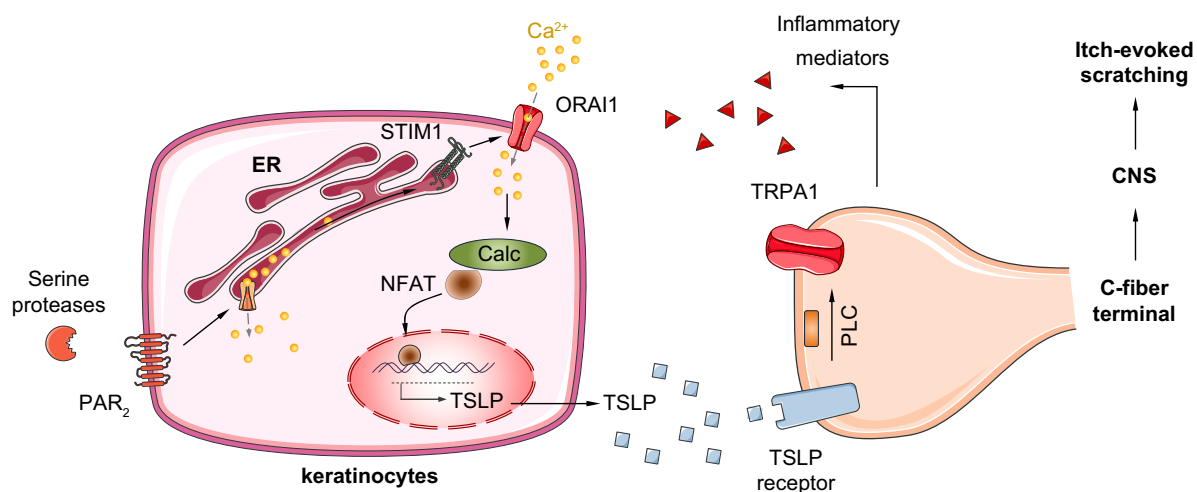


FIGURE 9. Schematic diagram depicting the thymic stromal lymphopoietin (TSLP)-mediated sensitization of TRPA1 during chronic itch. Orai1/NFAT-dependent release of TSLP activates the TSLP receptor in innervating cutaneous sensory neurons. Keratinocyte-secreted TSLP depolarizes a subset of C-fibers thereby producing itch in a TRPA1-dependent manner. Activation of TRPA1 in sensory neurons leads to neurogenic inflammation through the release of neuropeptides in the skin. CNS, central nervous system; ER, endoplasmic reticulum; PAR2, proteinase-activated receptor 2; PLC, phospholipase C.

TRPA1 is a major integrator of histamine-independent inflammatory and pruritogenic signals in oxazolone-induced contact dermatitis in mice. TRPA1 deletion reduces the levels of the proinflammatory cytokines IL-4, IL-6, and CXCL-2. TRPA1-deficient animals display reduced skin edema, keratinocyte hyperplasia, nerve growth, leukocyte infiltration, and antihistamine-resistant scratching behavior once exposed to the haptens oxazolone and urushiol (478). Topical application of xylene and toluene in mouse induces edematogenic and nociceptive responses that are prevented by HC-030031 and by the genetic deletion of TRPA1. TRPA1 activation may be therefore implicated in some of the symptoms of irritant-mediated contact dermatitis (602).

Phthalates are found in many consumer and industrial products (366), but increase the risk of developing different allergies (407, 449) and asthma (95). It was proposed that exposure to di-isononyl phthalate enhances expression and/or activation of TRPA1 via NF- κ B signaling. This induces higher levels of IL-6 and Th2 cytokines, stimulating the development of ADC in mice (366). TRPA1 activation correlates with the enhancing effect of phthalates used in industry and as components of mosquito repellents in a fluorescein isothiocyanate model of contact hypersensitivity (442–444, 732, 733). In a follow-up study, it was found that esters with glycerol and a short-chain fatty acid, dibutyrin and tributyrin, activate TRPA1 and enhanced skin sensitization to fluorescein isothiocyanate (724).

TRPA1 mediates the itch-related mouse behaviors elicited by cheek injection of lysophosphatidic acid. This compound activates both TRPA1 (and TRPV1) when applied intracellularly requiring the amino acid pairs K672-K673 and K977-R978 (396).

Taken together, the evidence gathered so far in animal models strongly indicates for the implication of TRPA1 in the trigger and/or maintenance of pain and inflammatory conditions. However, clinical data remain scarce. Due to its accessibility and its relative isolation from the internal tissues of the body, the skin is probably the most tractable organ for the study of the pathophysiological roles of TRPA1 in humans. The available evidence for the implication of TRPA1 in itch suggests this condition as a convenient model for the development of therapies targeted to TRPA1-mediated diseases, in particular those featuring pain and neurogenic inflammation. It must be noted, however, that the most popular agonists used so far to induce human TRPA1 activation in vivo (AITC and cinnamaldehyde) are far from acting solely on this channel (20, 26, 124, 140, 241, 269, 554, 555, 811), which urgently prompts for the development of more specific agonists.

D. TRPA1 in Cardiovascular Diseases

The application of TRPA1 agonists causes dilation of several arteries, and these responses are smaller when the ves-

sels are treated with TRPA1 blockers or in preparations isolated from *Trpa1*-deficient mice (76, 227, 282, 313, 429, 432, 669, 676, 783).

The peripheral vasodilating effects of the TRPA1 agonist cinnamaldehyde (61) were first reported over 40 yr ago (303). This effect is accompanied by a release of catecholamines, suggesting that the effects of this compound are dependent on sensory nerve-mediated mechanisms (302, 347). TRPA1 is expressed in the adventitial nerves in rat femoral, mesenteric, and meningeal arteries and in chemoreceptor afferents of chicken aorta (76, 669, 913). Another prominent TRPA1 agonist, allicin (492), is proposed to protect against coronary endothelial dysfunction via its vasodilatory action on vascular endothelium (62, 459, 786). TRPA1 is also expressed in endothelial cells from mouse and rat cerebral arteries, mouse and rat mesenteric arteries, rat femoral arteries, and rat vascular smooth muscle cells (VSMC) (227, 783, 920). These findings raise the possibility that not only mechanisms mediated by sensory nerves are responsible for TRPA1-dependent vasorelaxation. To date, two mechanisms have been proposed to explain the TRPA1-dependent vasodilation: the nerve-evoked vasodilation and the endothelium-dependent vasodilation (110, 225) (FIGURE 10).

In this context, it was suggested that TRPA1 activation in sensory nerves induces the release of CGRP, which then binds to its G protein-coupled receptor expressed on the VSMC membrane, causing myocyte hyperpolarization and relaxation (76, 224, 228, 282, 283, 432). Alternatively, the Ca^{2+} influx via TRPA1 activation in endothelial cells may lead to VSMC relaxation and vasodilation (225). In this regard, it is notable that TRPA1 expression is abundant in the endothelial cell plasma membrane that is in proximal contact with VSMC. These junctions host the cellular signaling players necessary for endothelium-dependent VSMC hyperpolarization and vasodilation, such as Ca^{2+} -activated K^+ channels (K_{Ca}) and myoendothelial gap junctions (291, 454, 711). The activation of TRPA1 (by AITC) in endothelial cells induces Ca^{2+} influx and vasodilation of pressurized rat cranial vessels. AITC-induced vasodilation is abrogated by the inhibition of TRPA1 by HC-030031, the disruption of the endothelium, and the inhibition of K_{Ca} channels (227, 291, 676, 783).

Notably, TRPA1-mediated vasodilation in mouse mesenteric artery rings occurs in an endothelial and neuropeptide-independent manner (669). This gave rise to the notion of another TRPA1-induced vasodilation mechanism, presumably working through direct actions on VSMC. However, it is difficult to translate the outcomes of this study since vasodilation was also seen in *Trpa1* KO mice upon administration of higher doses of cinnamaldehyde. These results suggest that cinnamaldehyde vasodilatory actions may involve TRPA1-independent pathways. In this sense, contro-

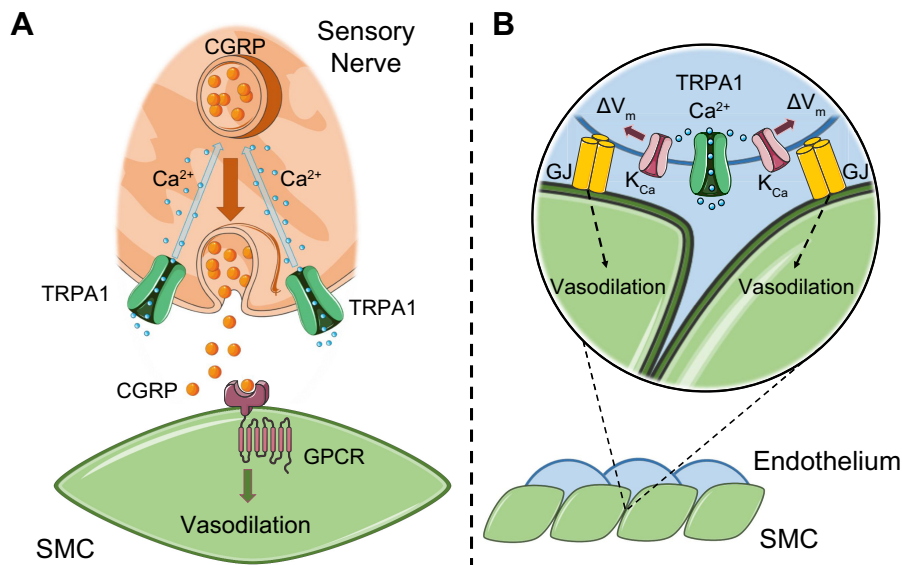


FIGURE 10. Two proposed mechanisms of TRPA1-dependent vasodilation. *A*: the activation of TRPA1 channel in perivascular sensory nerves leads to a Ca^{2+} influx that subsequently causes the release of calcitonin gene-related peptide (CGRP) from the sensory nerves that innervate the vascular wall. CGRP binds its G protein-coupled receptor (GPCR) on smooth muscle cells (SMC) and causes membrane hyperpolarization, myocyte relaxation, and arterial dilation. *B*: TRPA1 agonists in the bloodstream may activate the channel in endothelial cells resulting in Ca^{2+} influx. High intracellular Ca^{2+} leads to hyperpolarization of the endothelial cells via the stimulation of the Ca^{2+} -activated K^+ channels (K_{Ca}). This hyperpolarization is conducted to adjacent SMC, through myoendothelial gap junctions (GJ), hyperpolarizing the vascular SMC leading to vasodilation.

versy can also arise from a study reporting an endothelial-independent relaxation of VSMC after stimulation with cinnamaldehyde (920). Moreover, it should be noted that cinnamaldehyde is a potent blocker of the L-type Ca^{2+} channel blocker expressed in VSMC (26). The reduced Ca^{2+} influx in VSMC, through the inhibition of the L-type Ca^{2+} currents, may partly account for cinnamaldehyde-induced vasodilation.

TRPA1 agonists cause the release of adrenaline from the adrenal cortex (302, 303, 347), which can lead to systemic cardiovascular changes and to a variety of vascular actions in dissimilar vascular beds. A cooperative action of H_2S and NO is proposed to be required for the TRPA1-CGRP signaling pathway and the regulation of the vascular tone (50, 200, 224, 228, 297, 429, 663). Intriguingly, mesenteric arterioles from *Trpa1* KO mice have significantly less ability to relax in response to NO compared with arteries from WT animals (111). Another cooperative mechanism that involves the contribution of TRPA1 in peripheral vasoactivity is the cold-mediated vasoconstriction/vasodilation. This cyclic physiological response to long cold exposures is characterized by a transient initial vasoconstriction due to the activation of sympathetic nerves that is followed by vasodilation mediated by sympathetic nerve inhibition (465, 730). In these responses, TRPA1 and TRPM8 can act as vascular cold sensors mediating mice hindpaw vasoconstriction. The cold-induced contractions of the vessels are only prevented by the simultaneous application of TRPA1 and TRPM8 blockers (637, 638). Remarkably, the relaxation of the blood vessel wall is accompanied by increased CGRP and NO levels. More importantly, a TRPA1-TRPM8 synergy does not seem to be necessary in this mechanism since the vasodilation in response to cold exposures is abolished by the separate use of a TRPA1 or a TRPM8 antagonist (636).

It is suggested that TRPA1-expressing sensory neurons may be involved in the vascular component of neurogenic inflammation. The activation of TRPA1 in sensory fibers by toxic metabolites (such as HNE, *N*-acetyl-*p*-benzoquinonimine, and acrolein) caused edema in rat hindpaws and tracheal plasma extravasation in mice, rats, and guinea pigs. TRPA1 activation leads to a release of pro-inflammatory neuropeptides, such as CGRP and substance P (40, 580, 829). The edema and the tracheal plasma extravasation are attenuated by the blockade of TRPA1 and in mice lacking the *Trpa1* gene. Moreover, various TRPA1 agonists may contribute to the formation and propagation of pro-inflammatory factors, for example, ROS. In endothelial cells of cerebral arteries, the TRPA1 activation by ROS, and the consequent Ca^{2+} entry, induces dilation of cerebral arteries from WT but not from *Trpa1* KO mice (783). ROS production is exacerbated in cerebral arteries in pathological conditions such as hypoxia (664) and hypertension (640). For example, hypoxia induces activation of TRPA1 in the cerebral endothelium leading to vasodilation. This effect constitutes an adaptive response that could help reduce the damage in ischemic stroke (664).

ROS-evoked TRPA1 sensitization was shown in another experimental model. Peripheral postischemic dysesthesia is an abnormal sensation that is often accompanied by vascular impairment. The ligation of mice hindlimbs induces transient ischemia. Reperfusion of the hindlimb, by releasing the ligature, elicits dysesthesia-like behaviors manifested as hindpaw licking responses in mice. These responses are inhibited by ROS scavengers, by TRPA1 antagonists, and by *Trpa1* deficiency. Likewise, intraplantar injection of H_2O_2 produced similar paw licking responses. These *in vivo* results correlate with *in vitro* findings in human TRPA1-expressing cells and mouse DRG neuron cultures where H_2O_2 -evoked TRPA1 responses are increased

after hypoxia pretreatment (712, 750). In this respect, TRPA1 has been proposed to take part also in chronic post-ischemic pain in rat hindpaw. Chronic ischemia increases mechanical and cold allodynia in rat paws as well as HNE and TNF- α and the protein levels of TRPA1 (397). Also, in a rat model of femoral artery occlusion (FAO), protein levels in the DRG neurons innervating the femoral arteries (occluded and control) were assessed following 72 h of occlusion. FAO increased the protein levels of TNF- α , PAR2, and TRPA1, particularly in C-fibers. PAR2 activation and AITC produced an increment in TRPA1 currents amplitude in the DRG neurons isolated from occluded femoral arteries rats that was higher than in control femoral arteries. FAO also increases the sympathetic nerve activity in response to AITC, which is attenuated by the blockade of TRPA1 by HC-030031 or by the suppression of TNF- α (910–913). Similar mechanisms are expected during peripheral atherosclerosis. Actually, TRPA1 expression is increased in macrophage-foam cells in mouse atherosclerotic aortas (940).

TRPA1 is functionally expressed throughout the endocardium, myocardium, and epicardium of mouse hearts. Its activation by AITC induces intracellular Ca^{2+} transients through the activation of the Ca^{2+} /calmodulin-dependent kinase II. The Ca^{2+} transients are abolished by HC-030031 and absent in *Trpa1* KO mice (42, 43). However, the real physiological contribution of TRPA1 to the cardiac function remains unknown. Nonetheless, an altered TRPA1 activity during pathophysiological conditions could be relevant for the heart, for instance, in the cases of ischemia-reperfusion injury, oxidative stress, and diabetic cardiomyopathy (171, 488, 619, 877). Recently, a role for TRPA1 in cardiac pathophysiology was related to the exposure to high levels of toxic substances, such as tobacco smoke, diesel exhaust, and airborne pollutants. In this sense, gaseous inhalation of high doses of acrolein and ozone induces changes in heart rate, electrocardiogram, arrhythmias, blood pressure, and breathing rate in WT mice. In contrast, neither acrolein nor ozone produced variations of these parameters in *Trpa1* null mice. Accordingly, WT mice treated with HC-030031 were largely protected from acrolein-induced mortality (3, 172, 438, 439). In line with these findings, spontaneous hypertensive rats exposed to diesel exhaust or AITC presented higher sensitivity to trigger arrhythmias. When pretreated with the TRPA1 antagonist ruthenium red, the heightened sensitivity to develop arrhythmic events was prevented (306, 327).

In summary, TRPA1 is involved in the regulation of the vascular tone via its functional expression in the perivascular sensory innervation, in the endothelium, and in VSMC. Due to its high sensitivity to endogenous signals released upon tissue damage (e.g., ROS), this channel seems to be a crucial player in ischemia and reperfusion and, together with TRPV1, is a plausible candidate for the initiation of

cardiac pain. However, further translational research is needed to determine whether TRPA1 is a potential therapeutic target for the prevention or treatment of cardiovascular diseases.

E. TRPA1 in Gastrointestinal Tract Diseases

The enteric nervous system (ENS) is a complex cellular network that consists, among various cell types, of intrinsic and extrinsic afferent nerve fibers. The latter are deriving from nodose ganglia (NG) and DRG that innervate different regions of the gastrointestinal (GI) tract. Extrinsic afferent nerve fibers contribute to the detection of luminal stimuli and convey visceral inputs to the central nervous system (CNS) (202, 246, 261). TRPA1 has been proposed to sense various environmental factors that enter in contact with the digestive system (107, 323, 594). Transcripts and functional expression of TRPA1 have been found in NG and DRG neurons innervating mouse and rat stomach and colon (128, 417, 822), mouse jejunum and ileum (128, 144, 162, 501, 654), and guinea pig esophagus (482, 931, 932). TRPA1 has been also detected in intestinal tissue samples of mouse, guinea pig, and dog (71, 214, 654), as well as in rat and human colonic epithelial cells (362, 363) and enterochromaffin cells (EC), mouse duodenal and colonic enterocytes (255), and mouse and human mucosa (81, 214, 604, 675). Moreover, TRPA1 is expressed in mouse enteroendocrine cells (EEC), commonly in the duodenum and jejunum, and in human enteric glial cells cultures, where TRPA1 transcripts are upregulated by *E. coli* LPS incubation (159, 476). More importantly, TRPA1 has been found in mouse colonic (but not duodenum, ileum, and jejunum) and human intestinal myenteric and motor neurons (29, 527, 668). These enteric neurons, by interacting with other intestinal cell types, control various GI functions, such as motility, epithelial barrier function, and ion secretion (138, 261). Therefore, the expression of TRPA1 in intrinsic, neuronal, and non-neuronal cells of the intestines may contribute to physiological functions of this channel distinct from its somatosensory role in gut-projecting sensory neurons in normal tissue (108, 128). In fact, a role for TRPA1 in rat and human colonic mucosa ion secretion has been proposed. The application of the channel agonist AITC induces mucosal Cl^- and HCO_3^- secretion in a concentration-dependent manner. The anion secretion in the mucosa layer is abolished by the use of HC-030031. This TRPA1-related anion secretion in response to colonic luminal stimuli is proposed to be mediated by direct activation of the channel or by a PGE_2 synthesis mechanism independent of neural pathways in the colon (362, 363). TRPA1 is also proposed to contribute to intestinal motility and gastric accommodation responses. TRPA1 agonists accelerate the colonic transit in mouse and may be effective in the treatment of constipation (413). Furthermore, intracolonic treatment of AITC increases colonic motor responses and defecation patterns in conscious dogs. TRPA1-mediated cholinergic

and serotonergic neuronal pathways are important for the AITC-induced colonic motility (in enteric neurons) and the generation and propagation of giant migrating contractions (in extrinsic nerves fibers) (214, 709, 756). Similarly, gastric accommodation and emptying are decreased by treatment with TRPA1 agonists, which increases the gastric tone via TRPA1-mediated cholinergic and serotonergic neuronal pathways (215, 423). Additionally, a decreased adrenal sympathetic nerve activity upon gastric application of β -eudesmol is only observed in WT but not in *Trpa1* KO rats (621, 623). In this context, garlic powder, containing the TRPA1 agonist allicin, induces gastric relaxation and epigastric symptoms of pressure and warmth in healthy subjects (257). Nonetheless, the TRPA1 contribution to the regulation of the GI motility is not restricted to ENS intrinsic and extrinsic afferent pathways. 5-HT is abundantly stored in EC, which are mucosal endocrine cells present in the GI tract length. TRPA1 channels in EC not only chemosense the gut environment (81), but mediate the stimulation of 5-HT release from the EC to the intrinsic and extrinsic afferents in the ENS. Thus secreted 5-HT promotes the excitation of neuronal pathways and, subsequently, physiological and pathophysiological responses (216, 238, 604, 834). It was later concluded, however, that 5-HT does not mediate the motor effect of AITC in the guinea pig small intestine (710). The TRPA1 agonist methylglyoxal induces 5-HT secretion in RIN-14B cells derived from a rat pancreatic islet tumor (a model for EC cells), and this is reduced by TRPA1 inhibition (789).

TRPA1 has been implicated in digestion and satiety mechanisms. TRPA1 is coexpressed with cholecystokinin (CCK), an endogenous brain-gut peptide, and 5-HT in duodenal and jejunal EEC. It was proposed that TRPA1 agonists in garlic, cinnamon, and ginger help digestion by facilitating the CCK and 5-HT secretion from the EEC in a TRPA1 Ca^{2+} influx-dependent manner (159, 255, 569, 675, 923). On the other hand, in cranial visceral CCK-expressing nodose neurons, AITC evokes inward currents that are abolished by HC-030031. Since cranial visceral afferent pathways probably have an effect of satiety, the action of spiced diets, such as garlic and cinnamon, may contribute to reduction of the food intake and be associated with satiety sensations (160). In addition to this, oral treatment with AITC suppresses food intake and increases emesis in mice. These reflexes are abolished by pretreatment with RR, an unspecific TRP channel blocker, and are proposed to act in a CCK-TRPA1 fashion (819, 947).

The GI tract is continuously exposed to a variety of gut microorganisms that are implicated in several physiological functions such as food digestion, nutrition, and immune integrity. Environmental factors, including stress, smoking, and diet changes (e.g., pungent compounds in food), can deregulate the immunity system of the GI tract and its epi-

thelial barrier function (17, 81, 324). This unbalanced state leads to chronic inflammatory disorders of the GI tract, i.e., idiopathic inflammatory bowel disease (IBD), ulcerative colitis, and Crohn's disease (725, 906). These disorders are characterized by an increased visceral hypersensitivity (VH) due to lower perceptual thresholds for pain and discomfort (852, 893). The mechanisms involving lower GI tract pain and VH are associated with a peripheral sensitization of neuropeptidergic visceral afferent fibers (841, 848). TRPA1 colocalizes with substance P, CGRP, tyrosine kinase A (TrkA), and TRPV1 in the mechanosensitive afferents nerve fibers that project to the mucosal and serosal/mesenteric layers (128, 144, 417). In this sense, TRPA1 is proposed to contribute to GI inflammation and pain responses mechanisms since its activity in gut afferent fibers regulates the release of neuropeptides during inflammatory conditions (60, 448, 933). Activation and sensitization of TRPA1 in these fibers contribute to the release of substance P and CGRP, which may induce and maintain GI inflammatory states in mice and rats and increase visceral muscle contractions in *D. melanogaster* (60, 92, 239, 887, 916, 933). Notably, it has been shown that TRPA1 has a protective role in a T cell-mediated colitis model by inhibiting TRPV1 activity in CD4 T cells (96).

Current experimental methods to study inflammatory states in the GI tract include the exposure of intestine sections to irritant chemicals, such as AITC (392, 530), which lead to experimental IBD. After the IBD is induced, visceral pain is assessed by electromyography (EMG) recordings that estimate the visceromotor response (VMR) to mechanical colorectal distension (CRD) (381, 544). Induced colitis, by administration of trinitrobenzene sulfonic acid (TNBS), provokes a TRPA1-mediated increment in the VMR to CRD and promotes the secretion of substance P and CGRP by colonic DRG neurons (107, 239, 851). Hypersensitive VMR, and the release of substance P and CGRP, are reduced in *Trpa1* KO mice and in mice with reduced TRPA1 expression on the DRG neurons that innervate the colon. Additionally, the TNBS- or AITC-induced VH is accompanied by higher TRPA1 mRNA, protein levels, and currents in colonic DRG neurons of mouse and rat (144, 162, 239, 392, 418, 468, 501, 560, 922). Moreover, TNBS-induced colonic inflammation in rats increases the H_2O_2 levels and VH. These effects are reverted by HC-030031, suggesting a TRPA1 role in visceral mechanosensation (405). Accordingly, gastric distension induces visceral pain via the activation of extracellular signal-regulated protein kinase 1/2 (ERK1/2) and mitogen-activated protein kinase (MAPK) in the TRPA1-expressing NG and DRG afferents that innervate rat stomach (417, 419). Likewise, mechanosensitivity is increased in mice colonic and guinea pig esophageal sensory afferents upon stimulation of the adenosine A_2 receptor or by the application of the inflammatory mediator bradykinin (128, 131, 482, 931, 932).

Inflammatory agents can directly activate (e.g., HNE) or indirectly sensitize (e.g., agonists of PAR2) TRPA1, thereby inducing pain and hyperalgesia in the GI (144, 239). Histamine-mediated sensitization of TRPA1 most likely contributes to the increased visceral pain perception in irritable bowel syndrome (IBS) patients (59). Also, the stimulation of PAR2 in mast cells leads to TRPA1 activation, and it is proposed as a mechanism for TRPA1-mediated hyperalgesia (144). PAR2 was found to colocalize with TRPA1 in CGRP-positive colonic nerve fibers in an inflammatory TNBS rat model (147). However, *Trpa1*-null mice show no changes in mechanosensory functions after PAR2 activation (128). Although TRPA1 is suggested to be involved in the pathogenesis of IBD, its role in nociception during inflammation (i.e., the involvement of PAR2) remains controversial. TRPA1 is upregulated in inflamed human and mouse colon samples and has protective roles through the reduction of the expression of multiple proinflammatory neuropeptides, cytokines, and chemokines (431). In the same way, TRPA1 activators, such as carvacrol, carvacryl, and ASP7663, reduce the production of proinflammatory cytokines and ROS in a mouse jejunal mucosal inflammation model and the VMR to CRD in rat constipation and pain model (23, 24, 414). Of note, the downstream factors of TRPA1 pathways, substance P and CGRP release, play opposite roles in IBD. Substance P is implicated in the pathogenesis of IBD, and CGRP is proposed to exert a protective role in colitis (239, 841). Furthermore, *Trpa1*

KO mice are resistant to experimental colitis, have no differences compared with control mice, or present more prominent inflammatory extents in the intestines (144, 239). Several factors, including species differences, diversity of genetic backgrounds, and variations on the GI tract regions and their corresponding innervations (e.g., colon-splanchnic nerve or esophagus-vagal nerve) may explain the disparity of these results (112, 239). In spite of this, TRPA1 is considered a potential target in the treatment of inflammatory pain and mechanosensitivity in GI disorders (60, 80, 953) (FIGURE 11).

Stressful life events can have striking influences on visceral perception in IBS patients. TRPA1 is involved in the stress-induced visceral hyperalgesia in rats probably due to an upregulation of the channel in the colonic afferent DRG. This suggests that TRPA1 could be a target to treat the stress-induced visceral hyperalgesia in IBS (934). Interestingly, a novel TRPA1 variant (c.2755C>T), found in a parent-son pair, selectively cosegregated with cramp-fasciculation syndrome symptoms such as IBS and GI refluxes (598).

In summary, it is clear that TRPA1 plays pathophysiological roles in the GI tract. Yet, a better understanding of its contributions to gut-related syndromes is required to treat patients with GI complaints.

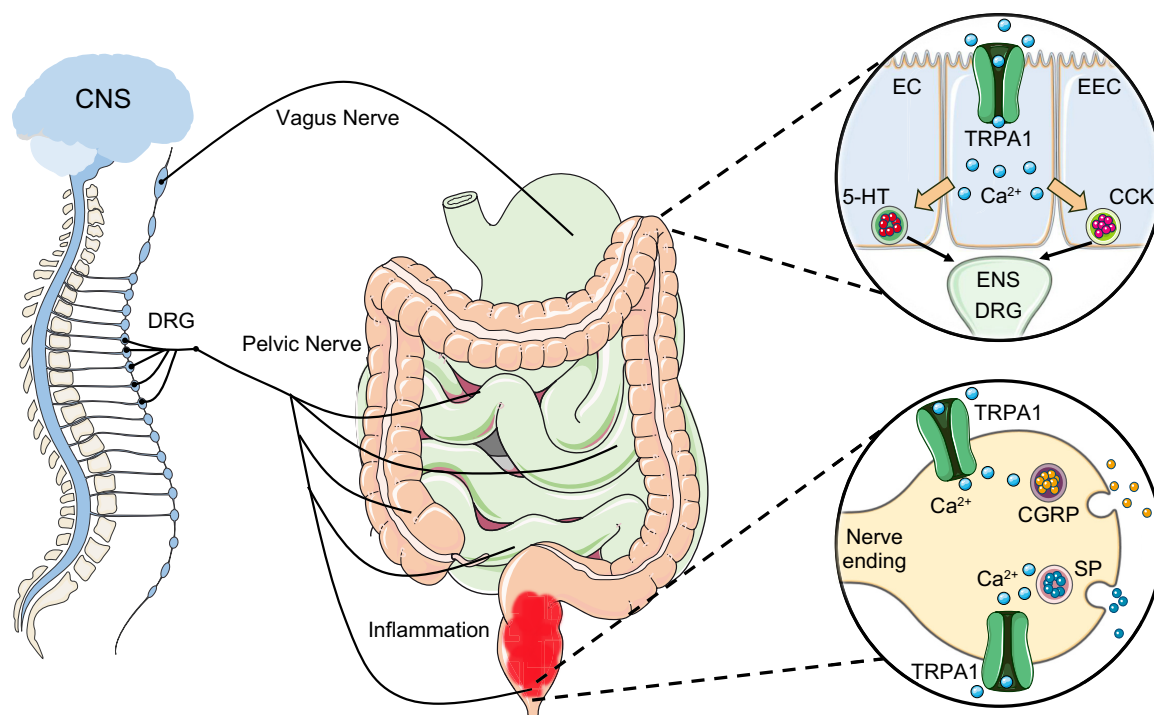


FIGURE 11. TRPA1 agonists may modulate gastrointestinal motility. Following its activation in enterochromaffin cells (EC) and enteroendocrine cells (EEC), the TRPA1-induced intracellular Ca²⁺ increase gives rise to increased secretion of serotonin (5-HT) and cholecystokinin (CCK), which eventually trigger activity of the myenteric and dorsal root ganglion (DRG) neurons embedded in the myenteric plexus. In addition, the TRPA1-expressing sensory nerves in the GI tract (via Ca²⁺ influxes) mediate the release of neuropeptides [such as substance P (SP) and calcitonin gene-related peptide (CGRP)] in response to GI inflammatory conditions.

F. TRPA1 in Diabetes, Obesity, and Pancreatitis

Diabetes mellitus is the most common metabolic syndrome in humans, and it is characterized by increased blood glucose concentrations. Type I diabetes occurs in response to idiopathic or immune-mediated destruction of pancreatic β -cells that results in a failure of the pancreas to secrete enough insulin. On the other hand, type II diabetes (T2DM) is due to resistance to insulin in the target tissues. Nonetheless, as T2DM progresses, it is common that a dysfunction of the pancreatic β -cells follows the pre-existent condition leading to hypoinsulinemia rather than a peripheral insulin resistance (592, 658, 957). The TRPA1 agonist cinnamaldehyde has been long used in traditional medicine as antidiabetic (6, 592, 658, 755). However, the mechanism underlying the effects of this compound on lowering hyperglycemia is presumably TRPA1-independent. TRPA1 is functionally expressed in EEC in primary mouse intestinal cultures. Agonists of the channel promote the glucagon-like peptide-1 (GLP-1) secretion, a blood glucose-lowering hormone, from EEC (237, 947). The GLP-1 secretion, however, is not observed in primary intestinal cultures derived from *Trpa1*-null mice (947). TRPA1 is expressed in rat pancreatic β -cells where, upon application of its agonist (like HNE or AITC), it contributes to increase intracellular Ca^{2+} levels leading to insulin release. On the contrary, the TRPA1 antagonist HC-030031 inhibits glucose-induced insulin release, suggesting a possible role for this channel in the release of insulin (139, 607).

Current therapies for diabetic patients include oral treatment with sulfonylureas, such as glibenclamide. Sulfonylureas block K_{ATP} channels, increasing the insulin release from pancreatic β -cells. Despite this, sulfonylureas are ineffective for long-term treatments and have associated undesirable effects (607, 658). In pancreatic β -cells, sulfonylureas can activate TRPA1, which can potentially contribute to the sulfonylurea-induced toxicity due to the cooperative insulinotropic effects that result from both the inhibition of K_{ATP} channels and the activation of TRPA1 (209).

Diabetes is a chronic disease that aggravates with time. A complication of diabetes is the peripheral diabetic neuropathy (PDN). In this scenario, the TRPA1 activation in small-diameter fiber endings is proposed as a mechanism that contributes to PDN in the early phase of diabetes (411). Diabetes induces endoplasmic reticulum (ER) stress. This condition can be experimentally modeled by tunicamycin, which produces ER stress and neuropathic pain in rats (343). In addition to ER stress, the hyperglycemic conditions stimulate the release of a cocktail of TRPA1 agonist (such as HNE, methylglyoxal, and ROS) causing mechanical allodynia in diabetic rodents. Both ER stress-induced and TRPA1-induced mechanical hypersensitivity are attenuated by the use of TRPA1 blockers and in *Trpa1* KO mice (33, 70, 230, 288, 312, 336, 343, 409, 411, 490, 624, 684,

879, 880). In line with these findings, TRPA1 blockers delay the loss of substance P immunoreactivity-expressing small-diameter endings in diabetic animals (409). Curiously, TRPA1 is involved in PND sensitivity to cold but not in the mechanical hyposensitivity in later phases of diabetes (318). It should be mentioned, though, that streptozotocin (STZ) directly activates TRPA1 (31). Thus PND models should be redefined, because the direct potentiation of TRPA1 by STZ complicates the interpretation of some of the previous cited results.

Apart from PND, diabetes also carries other life-threatening problems, for example, cardiovascular complications. In human cardiac fibroblasts, methylglyoxal induces a Ca^{2+} influx that is inhibited by HC-030031. The use of siRNA to knock down TRPA1 in cardiac fibroblasts reduces the methylglyoxal-evoked Ca^{2+} entry (619). This study suggests that a TRPA1-mediated mechanism is involved in the development of diabetic cardiomyopathy, which represents a major cause of morbidity and mortality in T2DM.

Yet, obesity is, undoubtedly, the most common risk factor for this metabolic disease (658). As mentioned above, TRPA1 is proposed to play a role in food intake and satiety (159, 160, 569, 675, 819, 903, 904, 923, 947). Thus TRPA1 has been considered as a therapeutic target for appetite suppression in obesity (957). In this context, some flavonoids [from citrus origin and with therapeutic value in the prevention and treatment of cardiovascular diseases (27, 296)] have been used as TRPA1 activators to stimulate the release of CCK in STC-1 cells (an EEC cell line model). The flavonoids naringin and hesperidin, as well as their aglycones naringenin and hesperetin, respectively, increase intracellular Ca^{2+} via TRPA1 potentiation. Therefore, a dose-dependent stimulated CCK secretion is observed, which could contribute to appetite regulation and food intake (390, 643). A similar approach is proposed with cinnamaldehyde, which by triggering TRPA1 activity in EEC reduces the levels of secreted ghrelin, “the hunger hormone” (137, 951). Furthermore, in an adipocyte cell line, TRPA1 activation reduces lipid accumulation (471). Interestingly, TRPA1 agonists induce adrenaline secretion via CNS, which prevents fat accumulation and obesity in mouse (347, 505, 625, 804, 878). Altogether, this evidence suggests that these natural compounds are candidate biomolecules for satiety control in obesity in T2DM.

Chronic pancreatitis (CP) is a devastating inflammatory disease that exacerbates over time and leads to permanent damage of the pancreas. It is characterized by persistent abdominal pain and ultimately impairs the ability to digest food and the production of pancreatic hormones. Pancreatic inflammation involves central and peripheral sensitization (551). Pancreatic peripheral innervation consists of NG and DRG sensory neuron projections that are likely to innervate the duodenum (466, 720). TRPA1 channels, ex-

pressed in these neurons, are involved in pancreatic inflammation by sensing the physical and chemical injuries in the pancreas. As in GI inflammation models, TNBS is also used to induce CP in mice. The inflammation and pain induced in this model are significantly lower in *Trpa1* KO mice (143). In a more common CP mouse model, induced by cerulein, an increment in DRG TRPA1 immunoreactive fibers and transcripts is observed. TRPA1 antagonists, or the deletion of *Trpa1*, attenuate the cerulein-induced pancreatic inflammation (146, 720, 721). Moreover, inflammatory mediators that activate TRPA1 (like HNE, PGE₂, or H₂S) increase pancreatic inflammation and pain in these mice (146, 812). Additionally, the TRPA1 activation mediates PAR2 stimulation, which contributes to pancreatic nociceptor excitation (813). The evidence suggests that TRPA1 contributes to sensing pancreatic inflammation, although a synergistic interaction of TRPA1 with other proteins, such as TRPV channels, is needed (146, 551, 720, 721). A novel TRPA1 and TRPV blocker inhibits inflammation and pain-associated behavior in a mouse model of acute pancreatitis (368). Consequently, TRPA1 is an interesting drug target for the treatment of CP, although more profound investigations are required.

G. TRPA1 in Respiratory Disease

TRPA1 plays a fundamental role not only for the normal airway function (556), but becomes particularly important for respiratory diseases characterized by hypersensitivity, for instance, asthma, rhinitis, chronic obstructive pulmonary disease (COPD), and chronic cough (272, 273, 484, 863, 921). Chemosensory airway reflexes can provoke severe complications in patients affected by inflammatory airway conditions (899).

The whole respiratory tract is innervated by primary sensory afferent nerves, which can be activated by mechanical and chemical stimuli via activation of TRPA1 on these vagal sensory afferents (85, 326, 564, 577, 777). TRPA1 is expressed in the nasal mucosa (582), and it is primarily expressed in small-diameter nociceptive neurons, where its activation contributes to the perception of noxious stimuli in the airways (686). TRPA1 expression is found also in immune cells, smooth muscle cells, and airway epithelial cells (133, 369, 561, 564, 584, 824). Airway sensory nerves may respond via TRPA1 to exogenous (AITC, cigarette smoke, chlorine, tear gas) and endogenous irritants (40, 41, 99, 130, 383, 928). TRPA1 activation in the nasal cavity induces significant bradypnea and a decrease of the respiratory frequency (344). Nasal trigeminal nerve endings are particularly sensitive to oxidants formed in polluted air and during oxidative stress as well as to chlorine, which is frequently released in industrial and domestic accidents (97, 98, 100).

Cough is a defense mechanism mediated by the vagus nerve that protects the airway by clearing the respiratory tract of

potentially harmful irritants and pathogens. Broncho-pulmonary vagal fibers respond to a variety of TRPA1 agonists (132, 562, 577, 686) and trigger cough in animals and humans (106, 805). Paroxysms of coughing are worsened by numerous TRPA1 agonists in chronic cough patients (557). For example, recent data demonstrate an interaction between diesel exhaust particles (DEP), which can contain electrophilic and nonelectrophilic TRPA1 agonists (194) and airway C-fiber activation. Components of DEP lead to mitochondrial ROS production, which is known to activate TRPA1 on nociceptive C-fibers (686, 805). In addition, TRPA1 is activated by compounds found in wood/biomass smoke particulate materials (195). TRPA1 antagonists or the antioxidant *N*-acetylcysteine attenuate IL-8 expression in human lung cells exposed to combustion particles (350). Also, compounds within cigarette smoke such as acrolein and crotonaldehyde cause cough through activation of TRPA1 (75, 76, 106, 727), suggesting a role for TRPA1 in cigarette smoke-induced cough, which is a key disease driver of COPD. TRPA1 is activated by the chemical sensitizer toluene diisocyanate, a potent sensory irritant known to cause chemical-induced occupational asthma (206). TRPA1 is also involved in the induction of mouse lower airway hyperreactivity to methacholine by exposure to chlorine and ovalbumin, an experimental nonallergic model that may be relevant for cleaning workers and competitive swimmers (331).

The airways of asthmatic patients contain increased levels of PGE₂ and bradykinin, which activate TRPA1 and TRPV1 and trigger coughing (280, 497). However, activation of TRPA1 inhibits the airway smooth muscle cell proliferative phenotype, a key contributing factor to asthma (937). A strong association was found between TRPA1 gene variants, childhood asthma, and total IgE concentrations (263).

Nasal hyperreactivity is a common feature in patients suffering from allergic and nonallergic rhinitis (846). These symptoms have been postulated to arise from hypersensitivity of the trigeminal fibers innervating the nasal mucosa (274). Recently, it was shown that patients suffering rhinitis exhibit a decreased threshold to the TRPA1 agonist AITC (845, 846), which correlates strongly with total symptom scores and is resolved after chemical destruction of the nasal sensory nerves (845, 846). Repeated applications of azelastine hydrochloride and fluticasone propionate (MP29-02) desensitized mouse sensory neurons expressing the transient receptor potential channels TRPA1 and TRPV1 (422). This effect may contribute to the therapeutic action (reduction of inflammatory mediators and nasal hyperreactivity) of this formulation in allergic rhinitis.

TRPA1 is expressed in non-neuronal cells in the respiratory tract such as human lung fibroblast cells and pulmonary alveolar epithelial cells. Activation of TRPA1 potentially

induces the release of chemokines in inflamed airways (561), and conversely, inflammatory cytokines can enhance TRPA1 translocation to the membrane (791). An *in vitro* model of respiratory virus infection of human bronchial epithelial cells showed that cytokines released in the supernatants lead to increased TRPA1 expression (631). The TRPA1-mediated responses to cigarette smoke may result not only from the activation of the channel in sensory nerve endings but also in airway epithelial cells (474, 868). This can be secondary to ROS production and result in IL-8 production via the activation of the MAPKs/NF- κ B pathway (474). Additionally, the exposure to cigarette smoke extract increases TRPA1 expression in airway epithelial cells (590). In human bronchial epithelial cells, TRPA1-dependent Ca^{2+} influx following cigarette smoke exposure is secondary to ROS production and results in IL-8 production via the activation of the MAPKs/NF- κ B pathway. Nevertheless, a study on high-level acrolein-induced toxicity suggests that TRPA1 plays an important protective role in both acute and post-exposure processes (172). The pseudostratified columnar epithelium of the bronchial lumen in cystic fibrosis coexpresses IL-8 and TRPA1. Here, the inhibition of TRPA1 expression results in a relevant reduction of release of several cytokines, including IL-8 and the pro-inflammatory cytokines IL-1 β and TNF- α . This suggests that TRPA1 mediates the severity of airway inflammation driven by cystic fibrosis bronchial epithelial cells (672). A nasal challenge with AITC induces the release of the mucin 5 subtype (MUC5B), a mucin specifically associated with inflammatory airway disease (16).

Thus the lungs and in particular the upper airways constitute a primary target of environmental pollutants that can activate TRPA1 and thereby induce well-known protective reflexes, such as mucus secretion, sneezing, and cough. The ventilatory function of the airways can be assessed by monitoring ventilatory parameters recorded with plethysmography. As up until now (100, 801), future studies can employ this relatively simple and noninvasive technique to continue deciphering TRPA1 airway pathophysiology and therapeutics. A particularly intriguing issue is to determine the implication of TRPA1 in the induction and maintenance of airway hypersensitivity conditions, especially as a sensor of thermal changes and chemical irritants, which are still known as unspecific environmental stimuli in the clinical research community. Other advances in this direction may be achieved with the use of nasal mucosal potential measurements (403, 524, 816). This methodology allows monitoring the responses of sensory nerves to chemical stimuli in humans, in a rather noninvasive and simple manner, and has already served to test the effects of AITC in human nasal mucosa in pathological conditions (845).

H. TRPA1 in the Urogenital System

The lower urinary tract (LUT) contains the bladder and urethra, which are involved in the involuntary storage and

voluntary expulsion of urine. Overactive bladder (OAB) is a common LUT pathology associated with detrusor muscle overactivity. In OAB, the properties of bladder afferent pathways are affected, leading to bladder storage dysfunctions and, subsequently, to an increase in urinary urgency, frequency, and pain (36, 747). TRPA1 is proposed to contribute to the bladder function since TRPA1 proteins and transcripts are found on mechanosensory lumbosacral fibers that innervate mouse, rat, guinea pig, and human urinary bladder mucosa (urothelial and suburothelial spaces), pig ureter, and human urethra (169, 221, 249, 284, 528, 564, 589, 778). TRPA1 mRNA and proteins are found in the bladder wall, particularly in rat and human urothelial cells and rat and human smooth muscle cells. TRPA1 is also found in blood vessels throughout rat bladder, pig, and human interstitial cells and human prostate (221, 242, 284, 359, 430, 768, 778, 885).

TRPA1 may play an important role in the regulation of bladder contraction. Agonists of TRPA1 (AITC and CA) increase the bladder contraction, stretch sensitivity, hyperalgesia, and micturition in rats and guinea pigs (37, 193, 222, 589). Moreover, in various OAB models, TRPA1 protein and mRNA expression levels are increased in the bladder and in the DRG neurons innervating it (38, 193, 221, 376, 499, 539, 634, 862). This upregulated TRPA1 expression may increase sensory transductions and induce the OAB symptoms. Furthermore, the TRPA1 antagonist HC-030031 diminished OAB symptoms and decreased the micturition reflexes (38, 156, 222, 522). In cyclophosphamide-induced OAB models, the toxic metabolite and TRPA1 agonist acrolein is produced. Acrolein can be responsible for the bladder inflammatory responses in cyclophosphamide-induced cystitis through direct activation of the channel (249, 271, 525). Additionally, in the rat model, PAR2 expression is enhanced. The blockade of PAR2 reduces the TRPA1 signaling pathway and attenuates OAB symptoms (149). Similarly, another compound produced in inflammatory states, nitro-oleic acid (OA-NO₂), enhances the contractile activity of rat bladder strips by TRPA1 activation in afferent nerves (45). In the same way, the administration of H₂S, a bacterial metabolite synthesized during inflammation, induces urodynamic parameter changes, presumably via TRPA1 activation in the DRG neurons innervating the bladder (467, 525, 645, 778). TRPA1 coexpresses with TRPV1, substance P, and CGRP in the bladder nerve endings (51, 222, 284, 589, 778), hence the notion of TRPA1 as an important player in inflammatory bladder (cystitis) conditions. Bladder contractions and painlike behavior in response to TRPA1 agonists (or inflammatory mediators, e.g., LPS) are accompanied by increased levels of substance P, CGRP, and PGE₂ (37, 359, 364, 659, 886). This suggests that the TRPA1-mediated contraction of the detrusor muscle involves the stimulation and secretion of neuropeptides and prostanoids from sensory afferents (747). Remarkably, TRPA1 agonists have no effect on spontaneous contrac-

tions of isolated human urethral strips (284, 886). Conversely, after phenylephrine-induced contraction, TRPA1 agonists induce a dose-dependent relaxation of urethral strips, probably in a TRPV1-dependent mechanism (284, 886). However, this does not exclude that TRPA1 is involved in the initiation of afferent activity in pathophysiological states (36, 364). It should be noted, though, that the role of TRPA1 in the LUT has been mostly assessed using agonists and antagonists of the channel. Some studies should be carefully interpreted since the TRPA1 agonist AITC can also activate TRPV1 (241). On the other hand, only two studies involved *Trpa1* KO mice, but showed a reduced number of voids and painlike behaviors in *Trpa1* KO mice compared with WT after ROS- or LPS-induced cystitis (634).

The data indicate that TRPA1 contributes to urinary bladder function in disease. This organ is another very good model to further investigate the pathophysiological roles of TRPA1. Indeed, the use of cystometry (842) allows tracking robust reflexes of the bladder wall upon intravesicular administration of chemical irritants in anesthetized or awake animals, and can therefore be used in the development of novel TRPA1 antagonists. Furthermore, voiding reflexes triggered by stimulation of other parts of the body (for instance in the skin) can be used to dissect the roles of sensory TRP channels (e.g., TRPM8 vs. TRPA1) in the detection of environmental stimuli such as cold (843).

VIII. CONCLUSIONS

TRPA1, primarily a chemosensor channel with more recently established roles in thermosensation and mechanosensation, is an extremely interesting ion channel, not only for its importance as a drug target, but also for its unique structural and biophysical properties. About 7 years ago we pinpointed a set of issues that were required to be clarified for a better understanding of the function of this intriguing channel (594). Although immense advances have been made, especially with the appearance of the first high-resolution models of the channel structure (651), we realize that the main challenges remain virtually the same. For instance, we still have very little knowledge about how the interaction of electrophilic agonists with nucleophilic amino acid residues translates into channel opening. Similarly, the mechanisms of Ca^{2+} -dependent potentiation and inactivation, the activation by nonelectrophilic compounds, and modulation by cold, heat, reactive species, and many others factors remain to be fully elucidated. On the other hand, it is becoming clear that noxious TRPA1 agonists activate the channel via their specific chemical properties, such as electrophilicity, oxidative power, or the ability to acidify the intracellular milieu, rather than by specific molecular structures. This underscores the idea that TRPA1 functions not as a refined sensor, but as a broadly-tuned detector of almost every potentially injurious external stim-

ulus and damage-associated endogenous signal. Thus TRPA1 seems to be a channel that when activated informs the sensory systems that something goes wrong, with the mere functions of triggering immediate protective behavioral responses and putting in motion the machinery of tissue repair.

Data obtained in multiple animal models of disease strongly indicate that TRPA1 is implicated in pain and inflammation and that it plays important roles in the initiation, progression, and maintenance of chronic inflammatory diseases and tissue injury, including asthma, diabetes, arthritis, and skin diseases. This prompts for the investigation of cell- and tissue-specific TRPA1 properties that could be of help in the design of specific treatments. It is imperative to notice that the basic knowledge on TRPA1 biophysics, pharmacology, and regulation is required to understand the complex pathophysiology of TRPA1, and is particularly essential for the rational design of modulators that may be used as therapeutic agents. For instance, it is important to distinguish between possible beneficial effects due to channel block versus beneficial effects of channel activation. This is because depending on its extent and rate of occurrence, TRPA1 activation may lead to cell excitation or to quiescence via slow depolarization and consequent inactivation of voltage-gated Na^+ and Ca^{2+} channels. The discovery of the functional expression of TRPA1 beyond sensory neurons, i.e., epithelial and smooth muscle cells, fibroblasts, oligodendrocytes, enteroendocrine cells, etc., in which the channel's function is still poorly understood, represents a complication for the development of therapeutic strategies targeting this channel. TRPA1 seems to behave as a molecular sensor with a multidimensional operating point, being tightly regulated by pH, oxidation, O_2 levels, electrophiles and nucleophiles, hydroxylation, temperature, membrane lipids, voltage, intracellular Ca^{2+} , trace heavy metals, etc. Thus it is likely that every cell type in which it is functionally expressed provides a unique scenario for yet unknown amazing features of this intriguingly irritating channel.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

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